



Human-Aware Robot Navigation: From Comprehensive and Realistic Benchmarks to Versatile and Efficient Solutions

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28/11/2025

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Table of Contents

01

Context

02

Assessment Methods

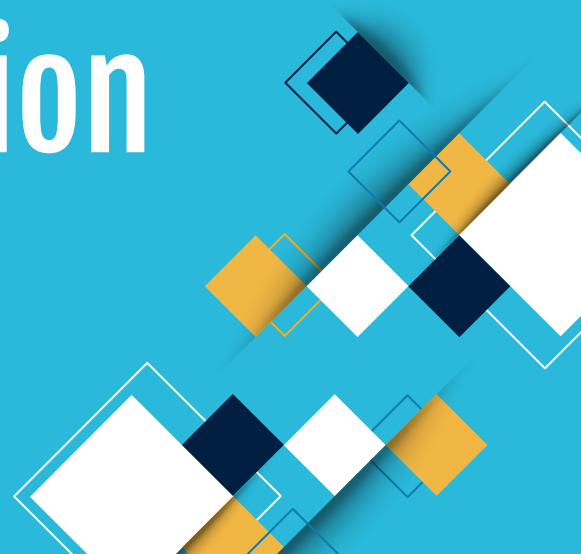
03

Our Solutions

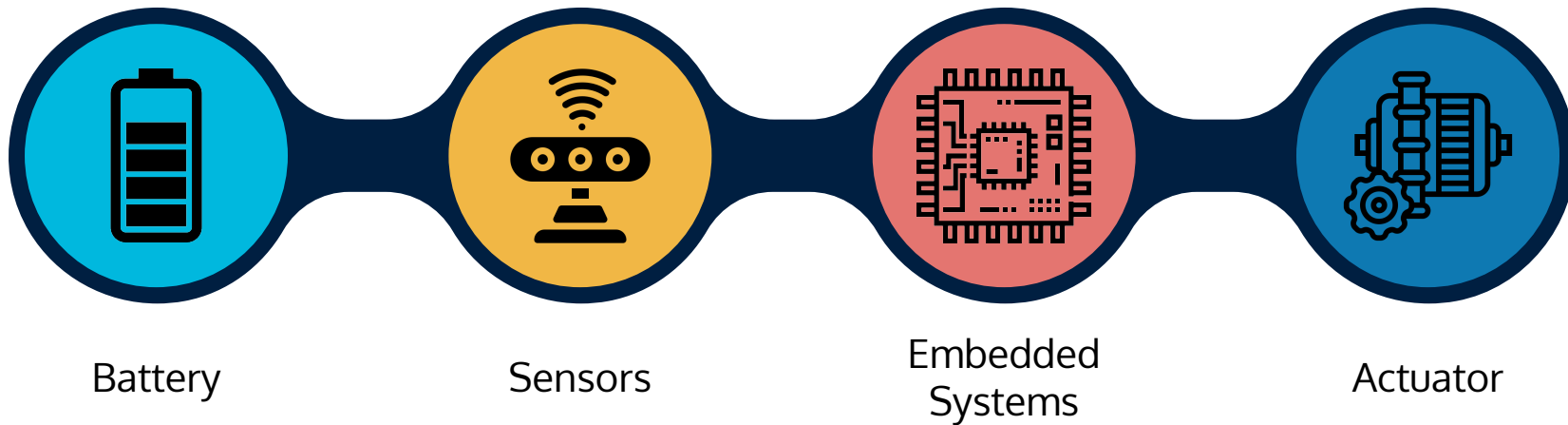
04

Experiments

Context: Mobile Robot and Human-Aware Navigation



Mobile Robot



Mobile Service Robot



Cue



Mirokai



PuduBot 2



SkyPod



Super Market



Restaurant



Airport

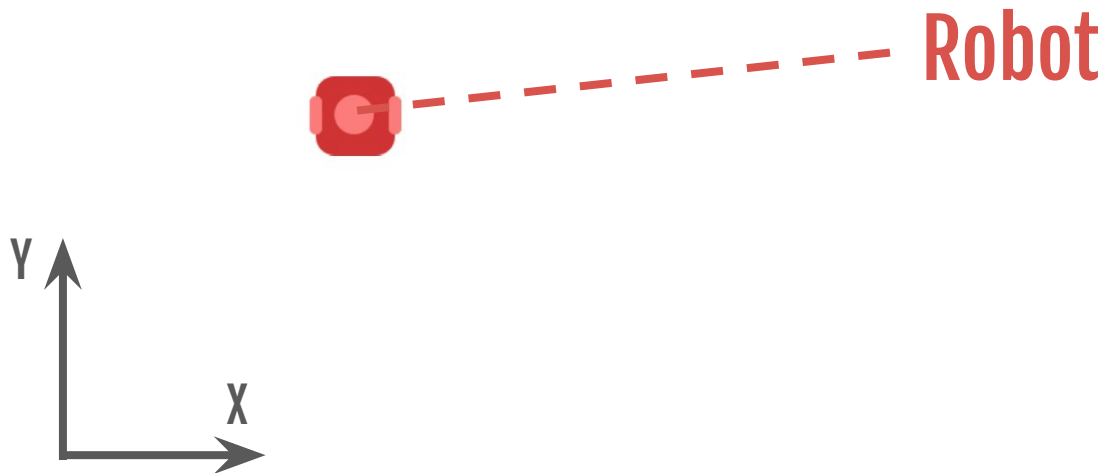


Hospital

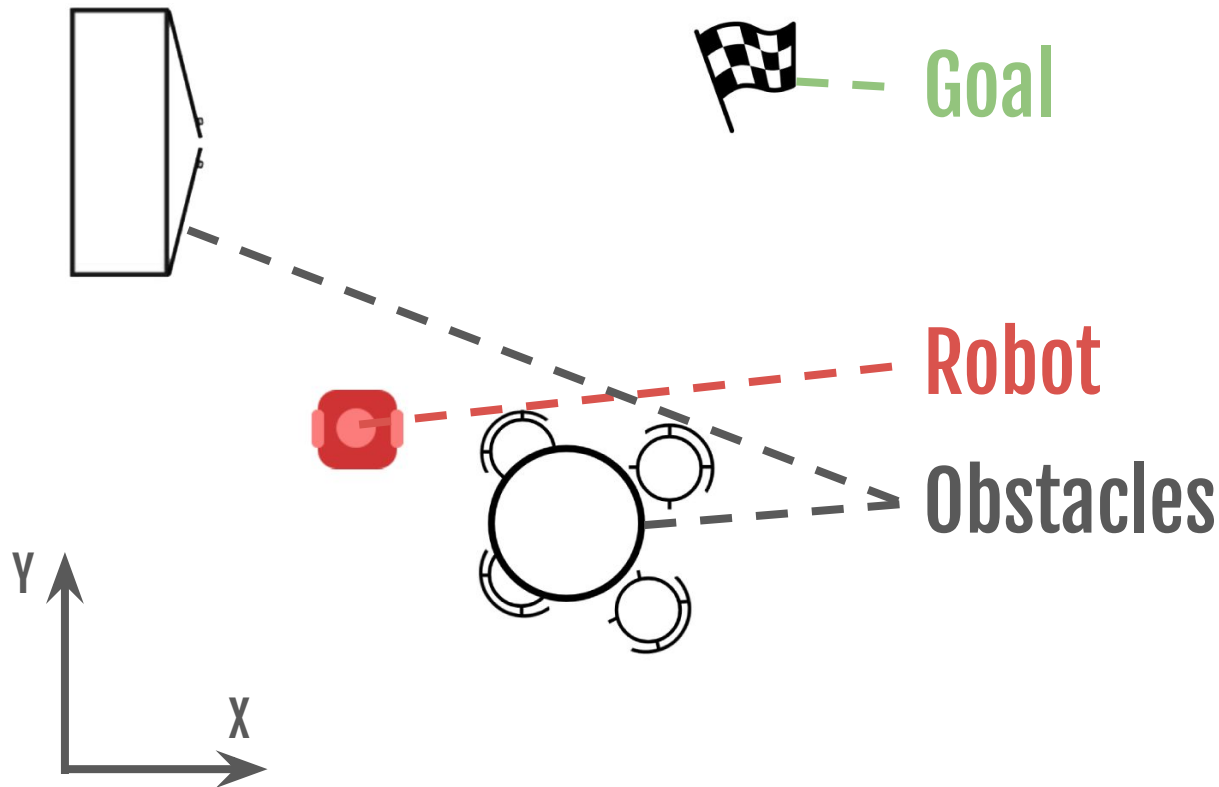


Transport/Delivery

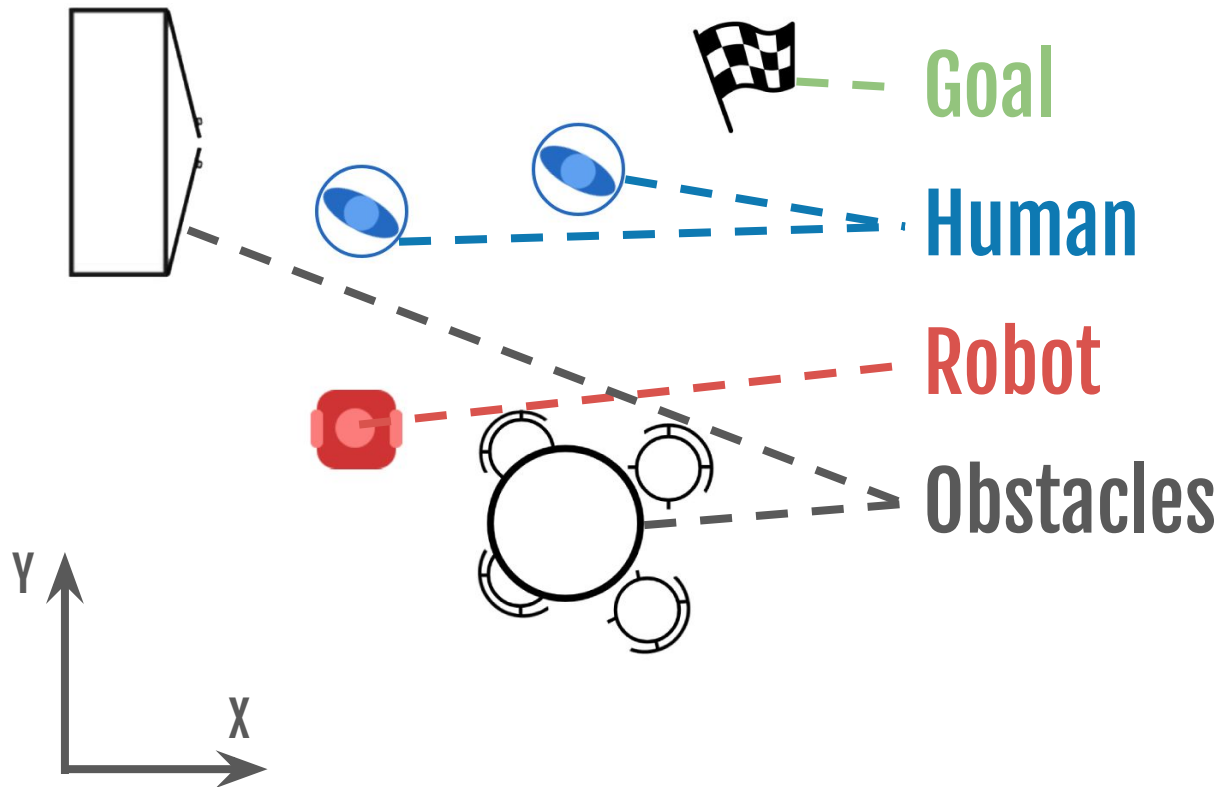
Mobile Robot Environment



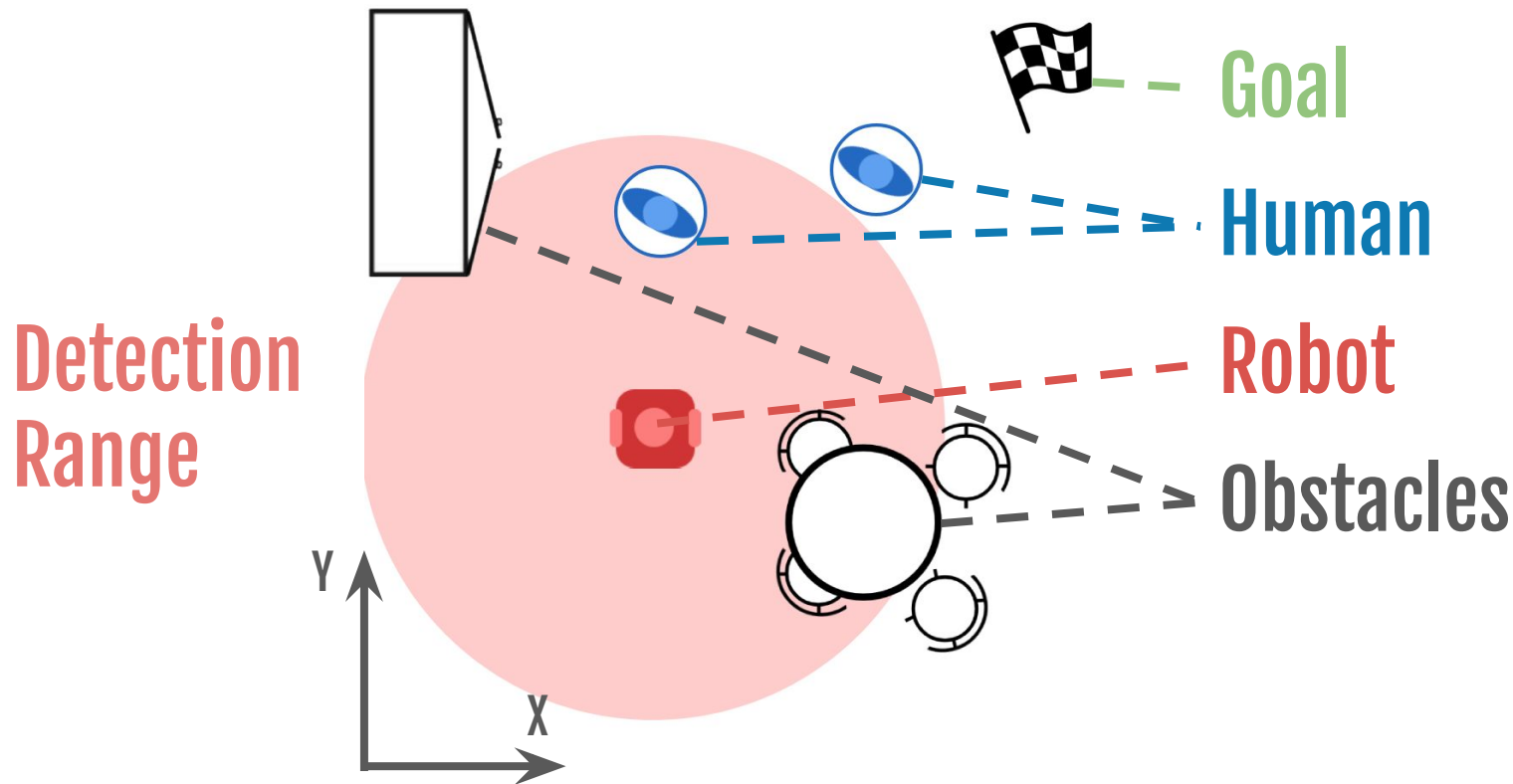
Mobile Robot Environment



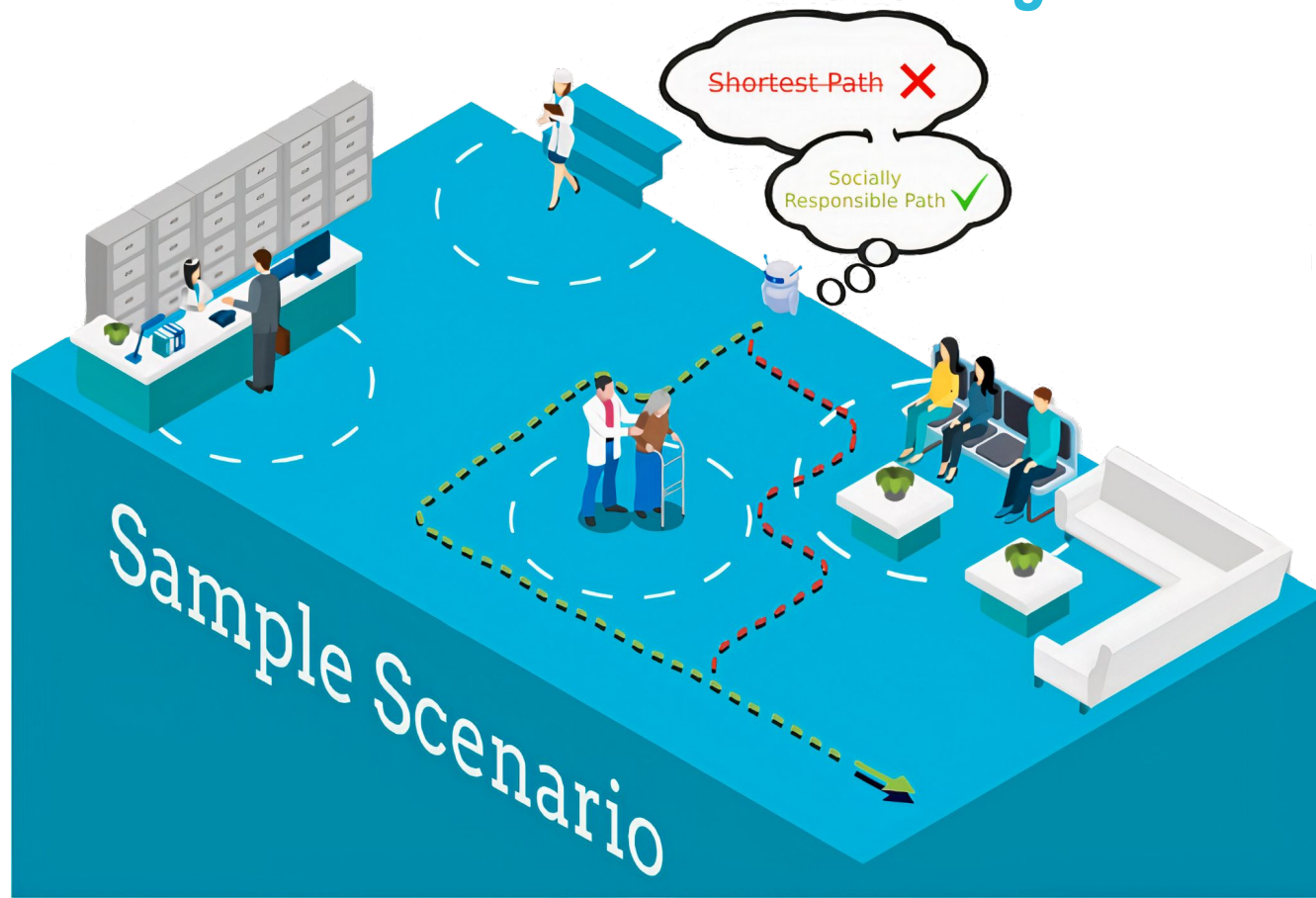
Mobile Robot Environment



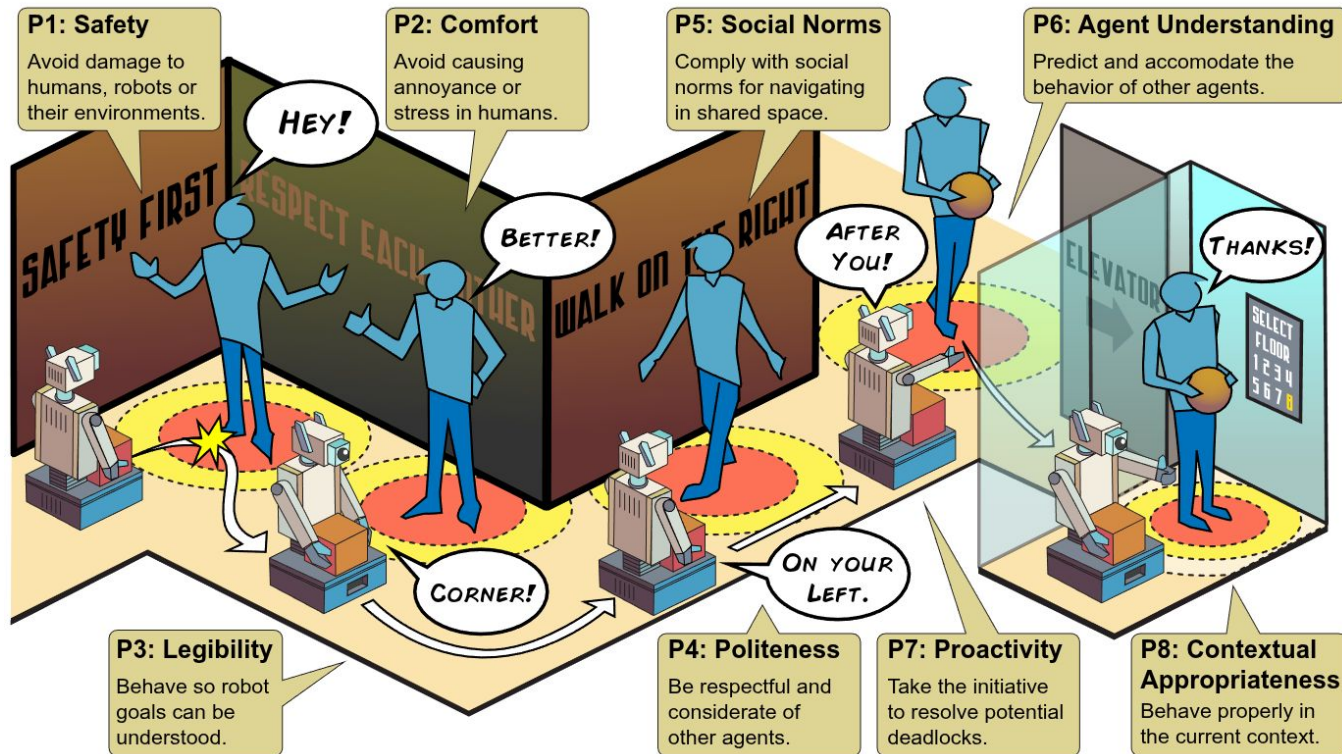
Mobile Robot Environment



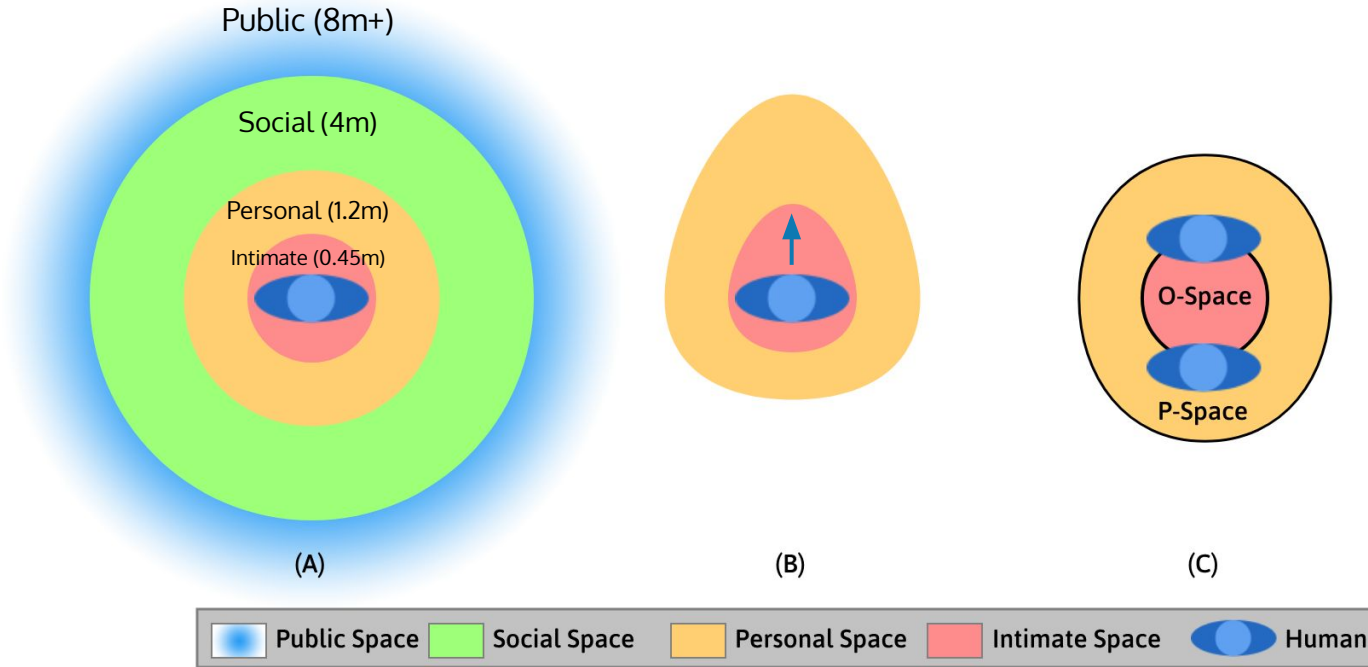
Social Robot Navigation



8 Principles for Social Robot Navigation^[1]

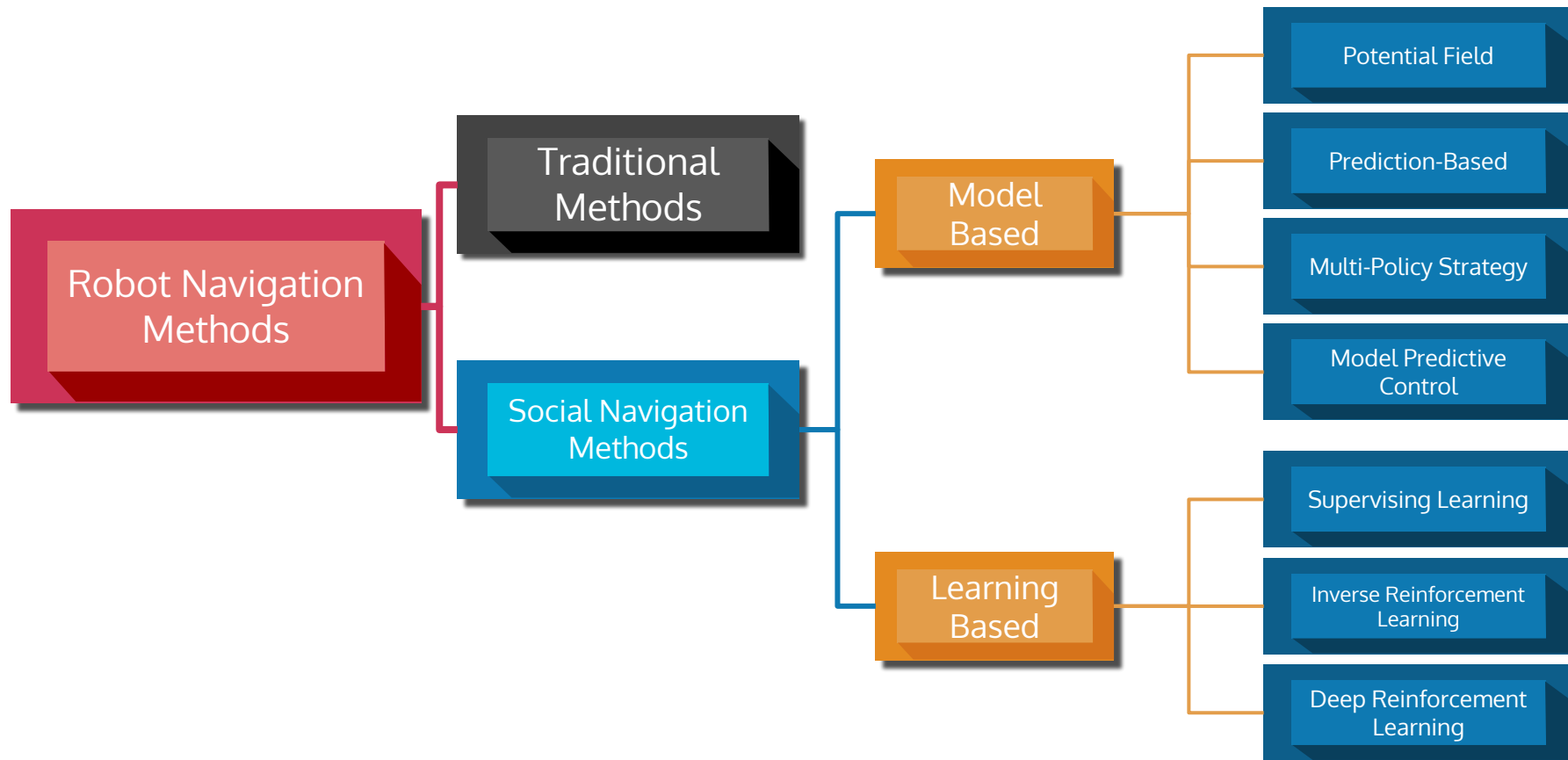


Personal Space Classification^[1]

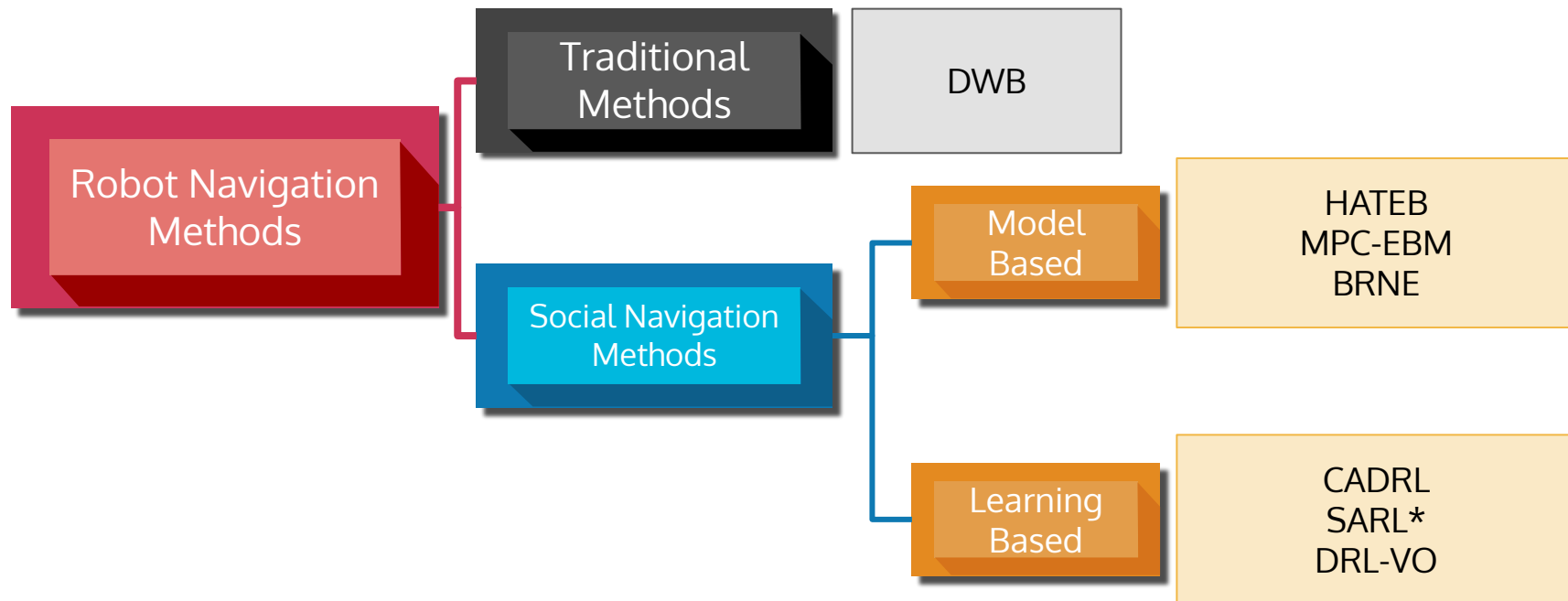


[1] Hall, E.T., The Hidden Dimension: Man's Use of Space in Public and Private, 1966.

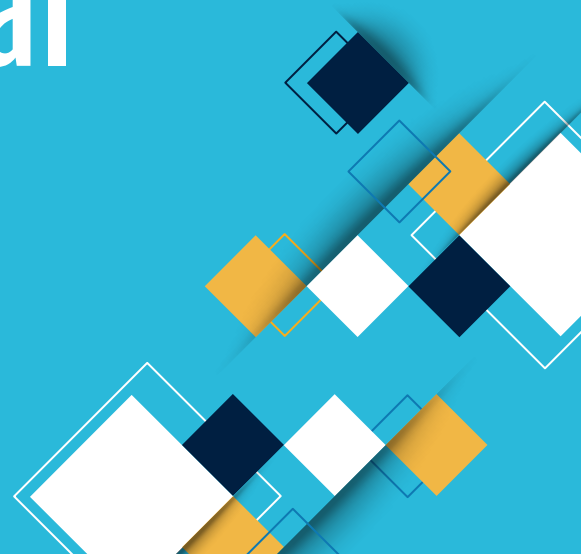
Our Taxonomy



Selected Off-the-Shelf Solutions



Assessment Methods: How to evaluate a Social Navigation Method ?

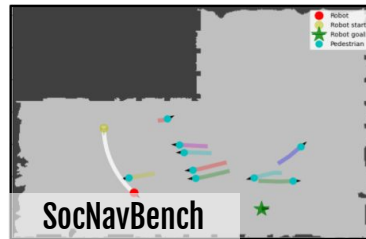


Benchmark for Social Navigation



Why ?

- Automatic / Reproducible Evaluation
- Realistic / Versatile
- Sim2Real / Limited Perception & Control



Benchmark for Social Navigation

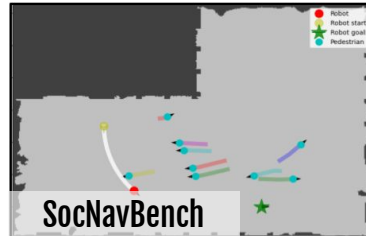


Why ?

- Automatic / Reproducible Evaluation
- Realistic / Versatile
- Sim2Real / Limited Perception & Control

Limitations of Existing Solutions:

- Very few different scenarios
- Very few metrics
- Rarely Deployable on Real Robot





Key points of the Contribution

- ❖ **Realistic** simulation
- ❖ **ROS/ROS2** Compatible
- ❖ **Open Source**^[1]
- ❖ Several **scenarios** implemented with **HouseExpo dataset**.
- ❖ Performance and social **metrics** for evaluation implemented.

[1] https://github.com/agouguet/social_benchmark_robotic_navigation.git

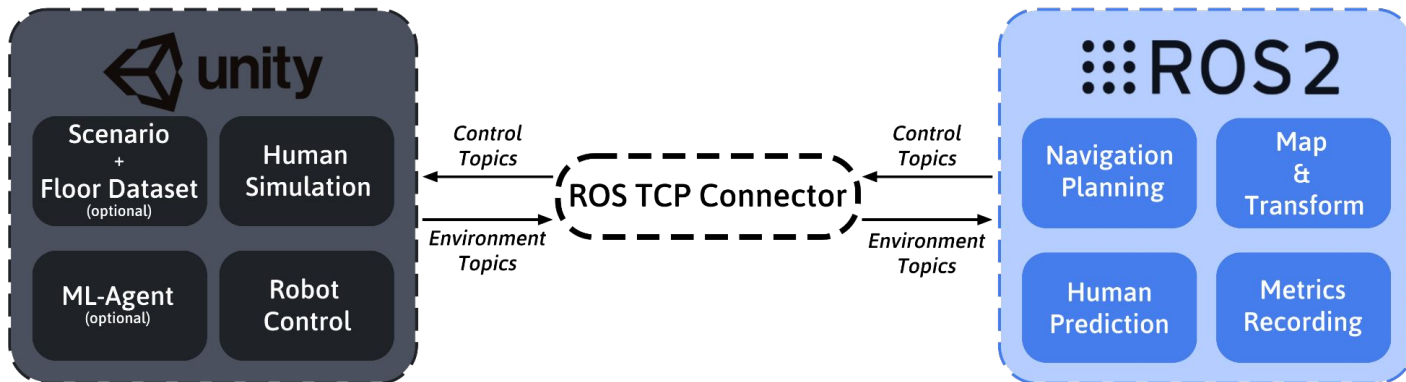


Our Robot Social Navigation Assessment Platform

Key points of the Contribution

- ❖ Realistic simulation
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Architecture



[1] https://github.com/agouguet/social_benchmark_robotic_navigation.git

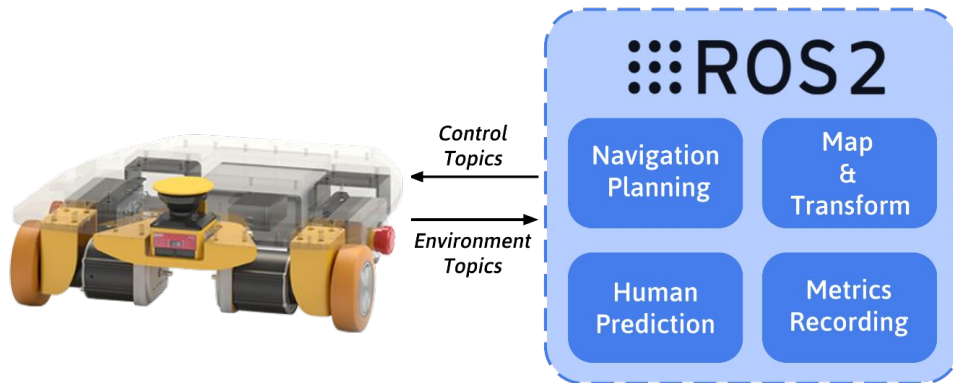


Our Robot Social Navigation Assessment Platform

Key points of the Contribution

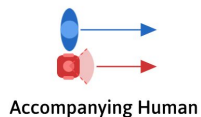
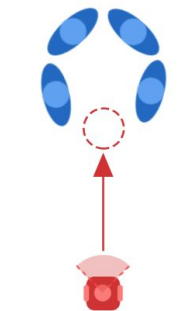
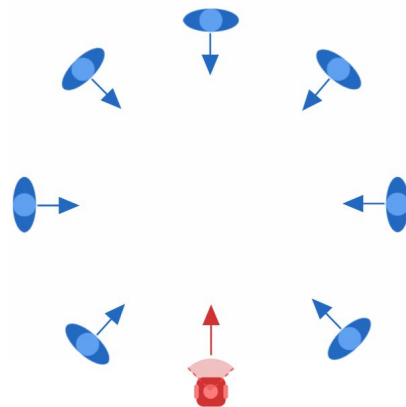
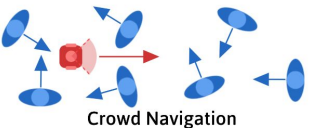
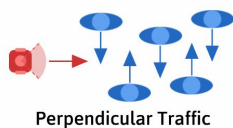
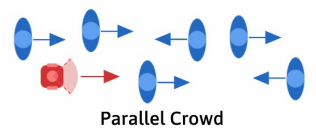
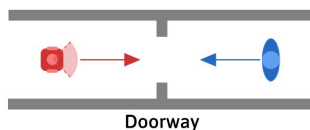
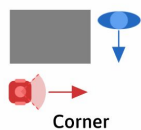
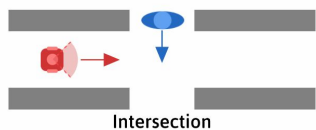
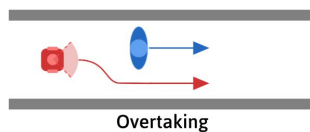
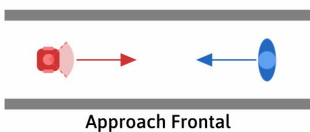
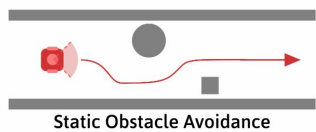
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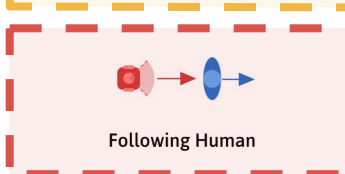
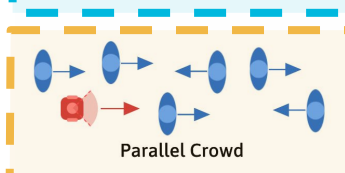
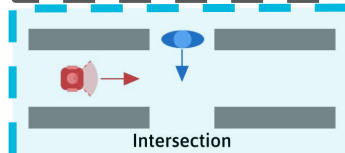
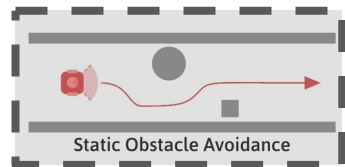
State-of-the-art Scenarios



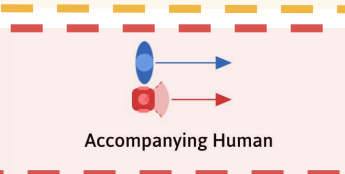
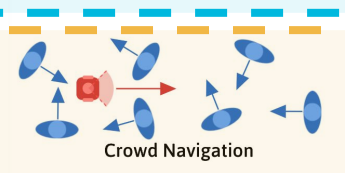
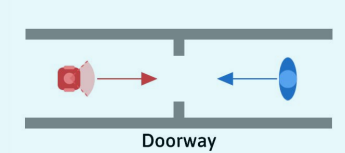
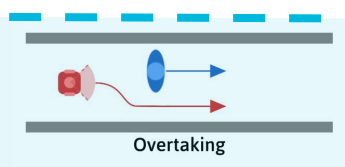
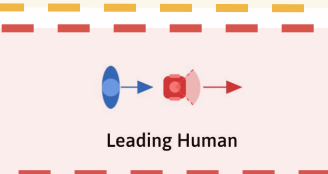
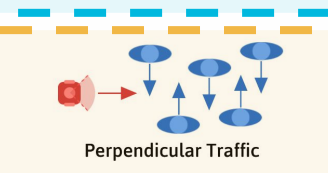
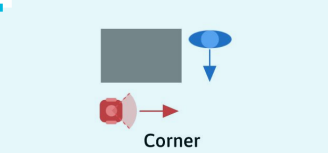
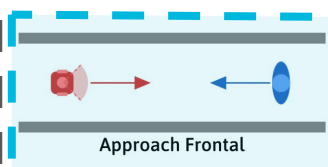
State-of-the-art Scenarios



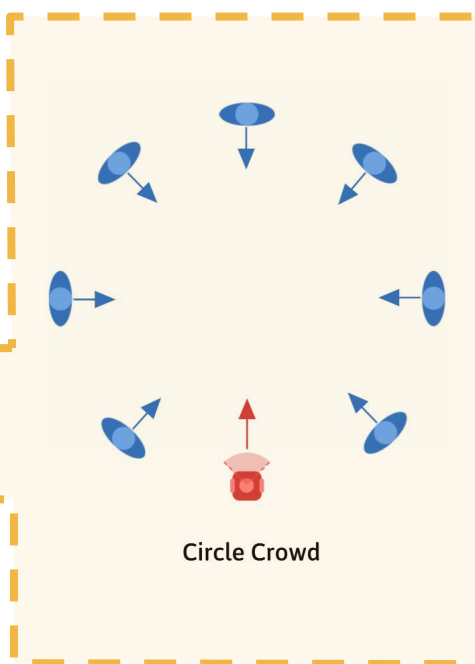
Traditional



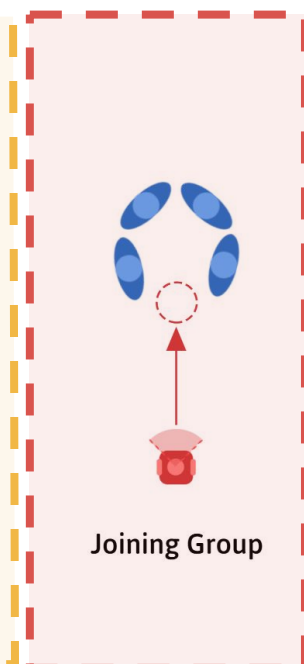
Sparse



Crowd



Interpersonal



State-of-the-art Scenarios

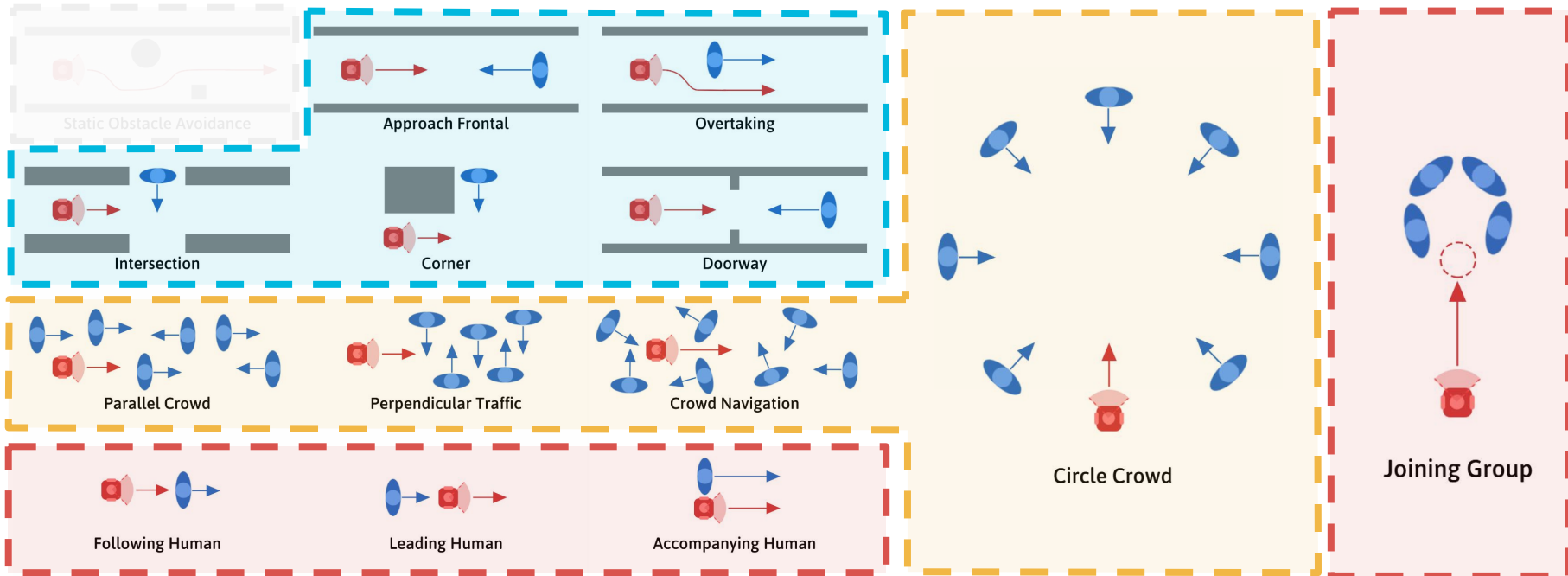


Traditional

Sparse

Crowd

Interpersonal



State-of-the-art Scenarios

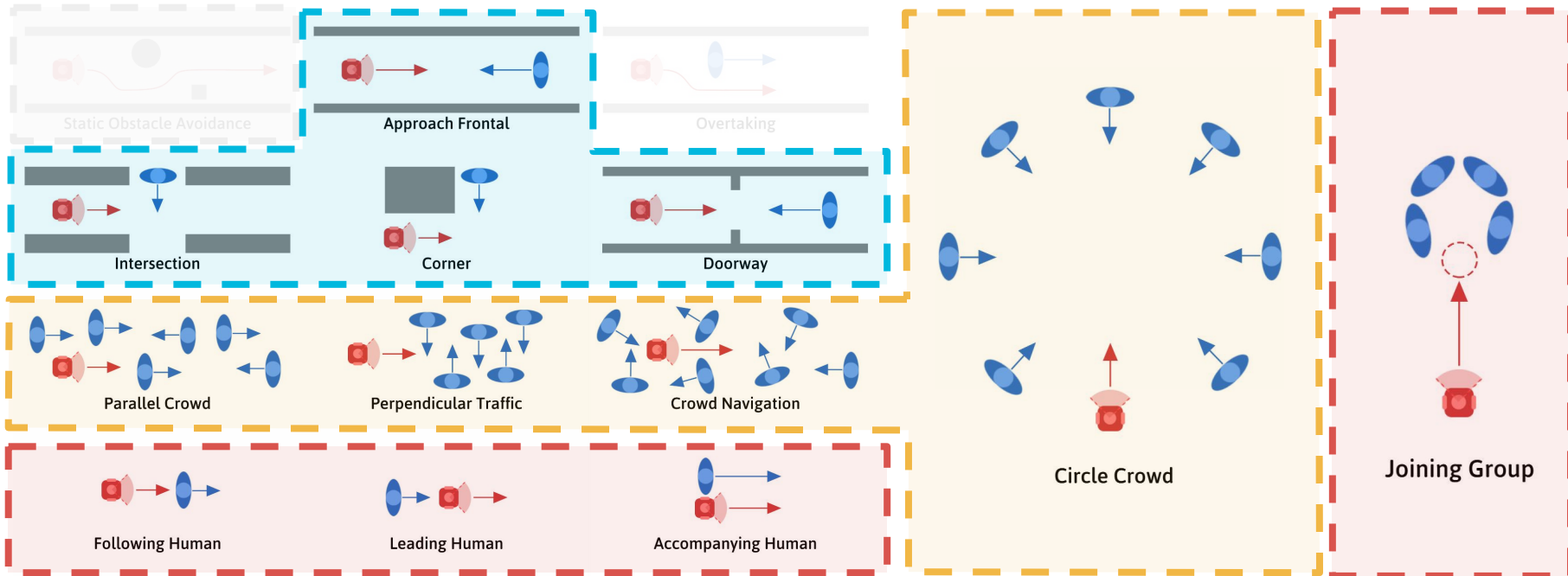


Traditional

Sparse

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Interpersonal



State-of-the-art Scenarios

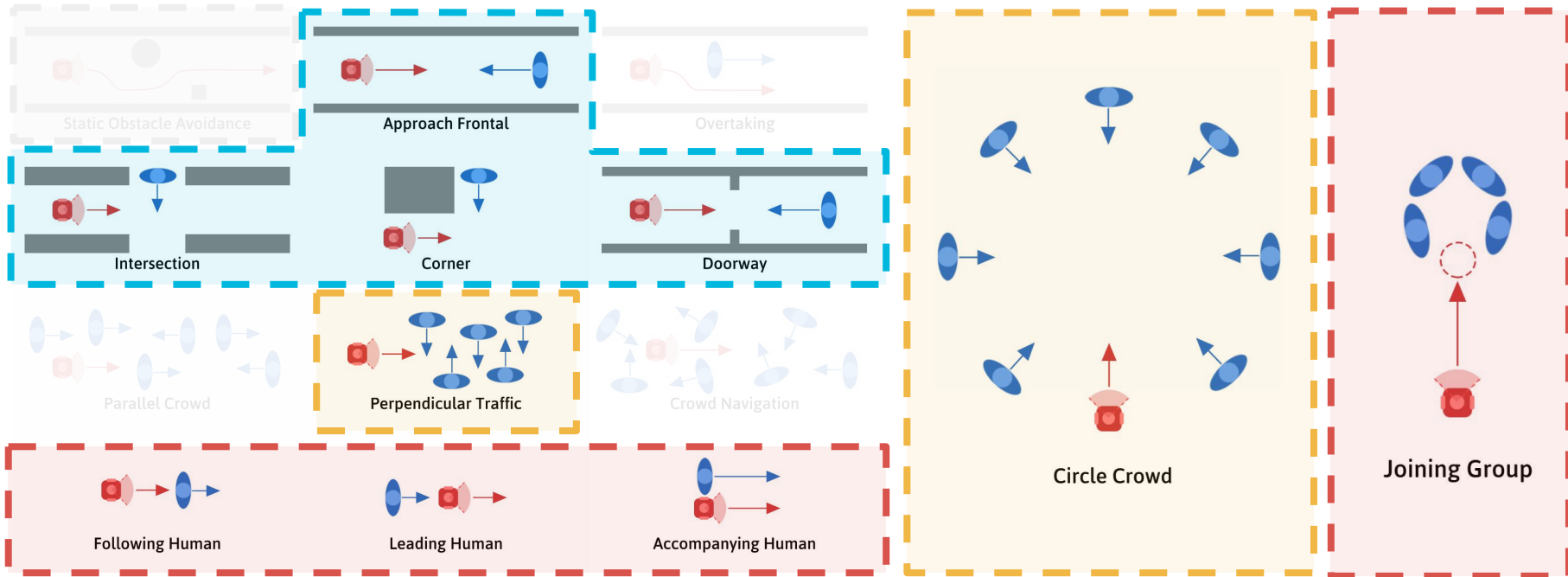


Traditional

Sparse

Crowd

Interpersonal



State-of-the-art Scenarios

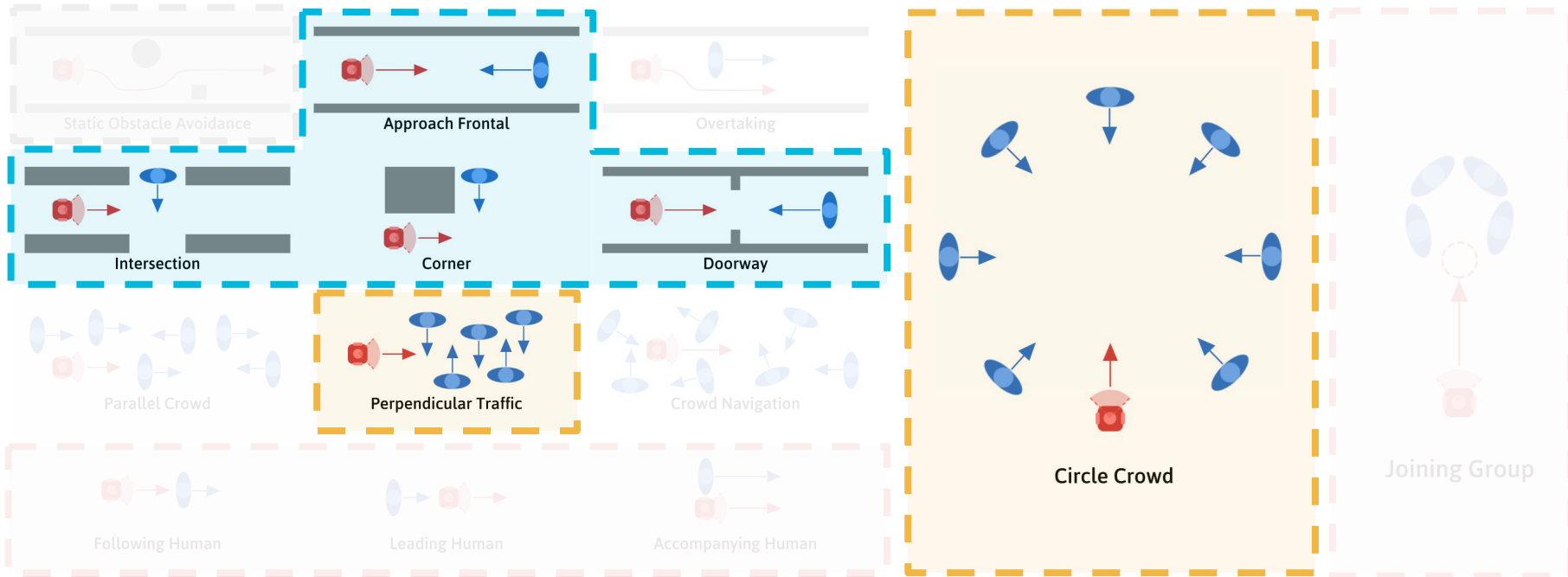


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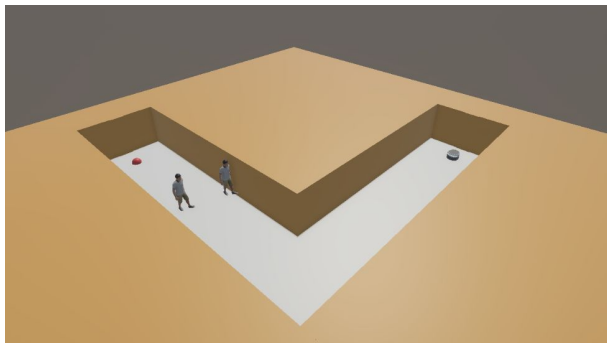
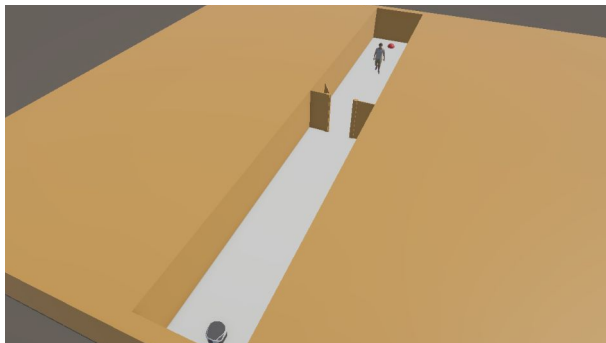
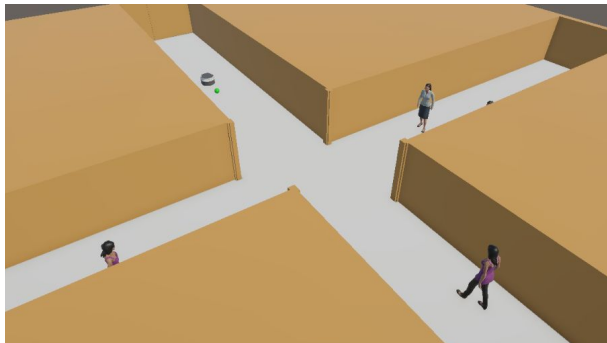
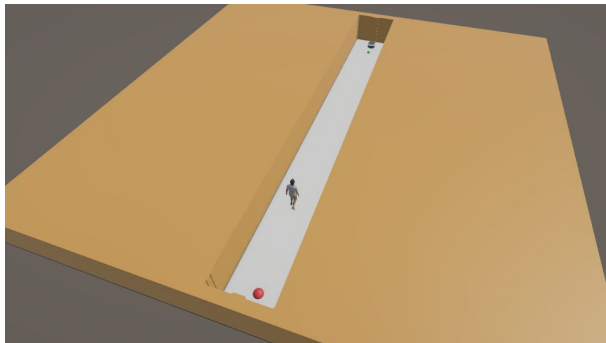
Sparse

Crowd

Interpersonal



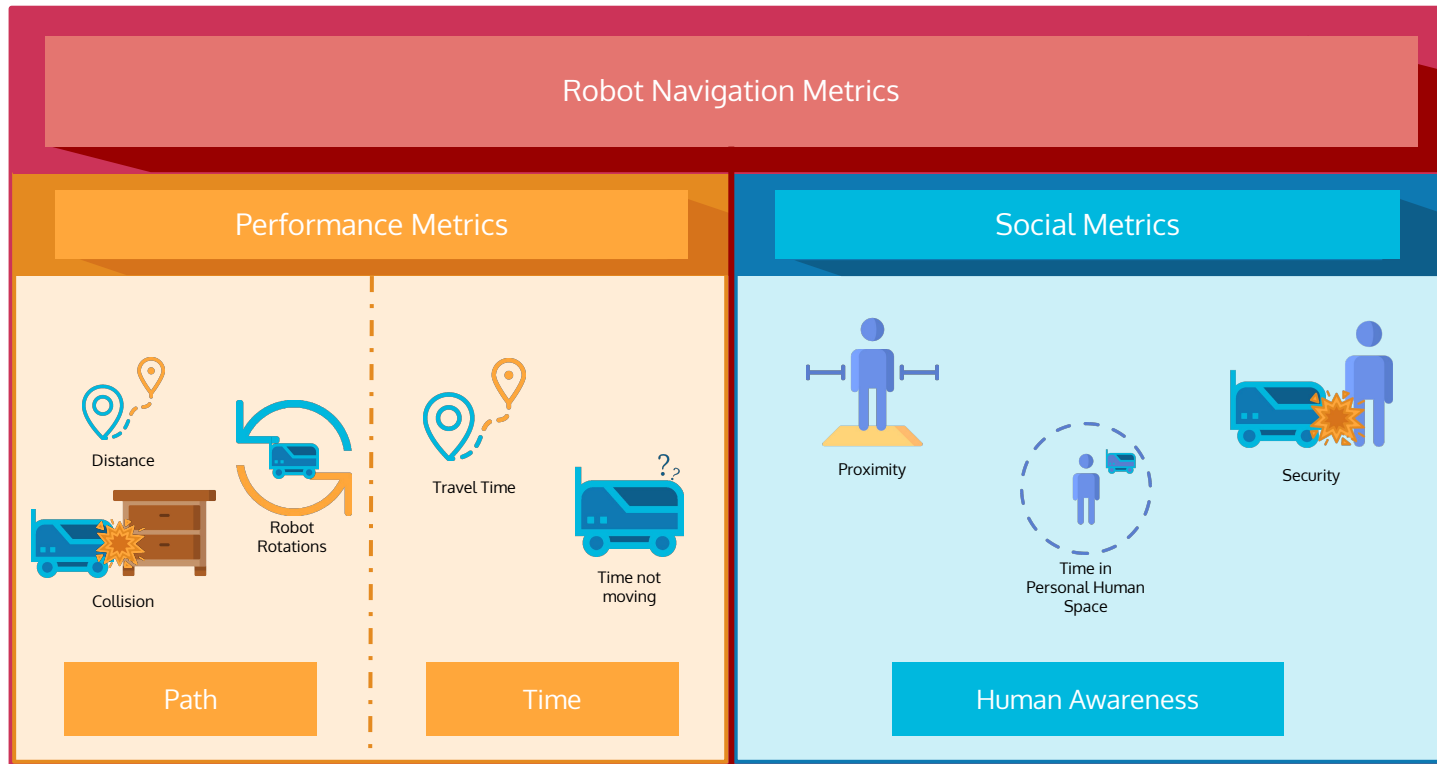
Selected Scenario in RobotSNAP



Sparse

Crowd

How to benchmark ?





How to benchmark ?

9 Selected Metrics:

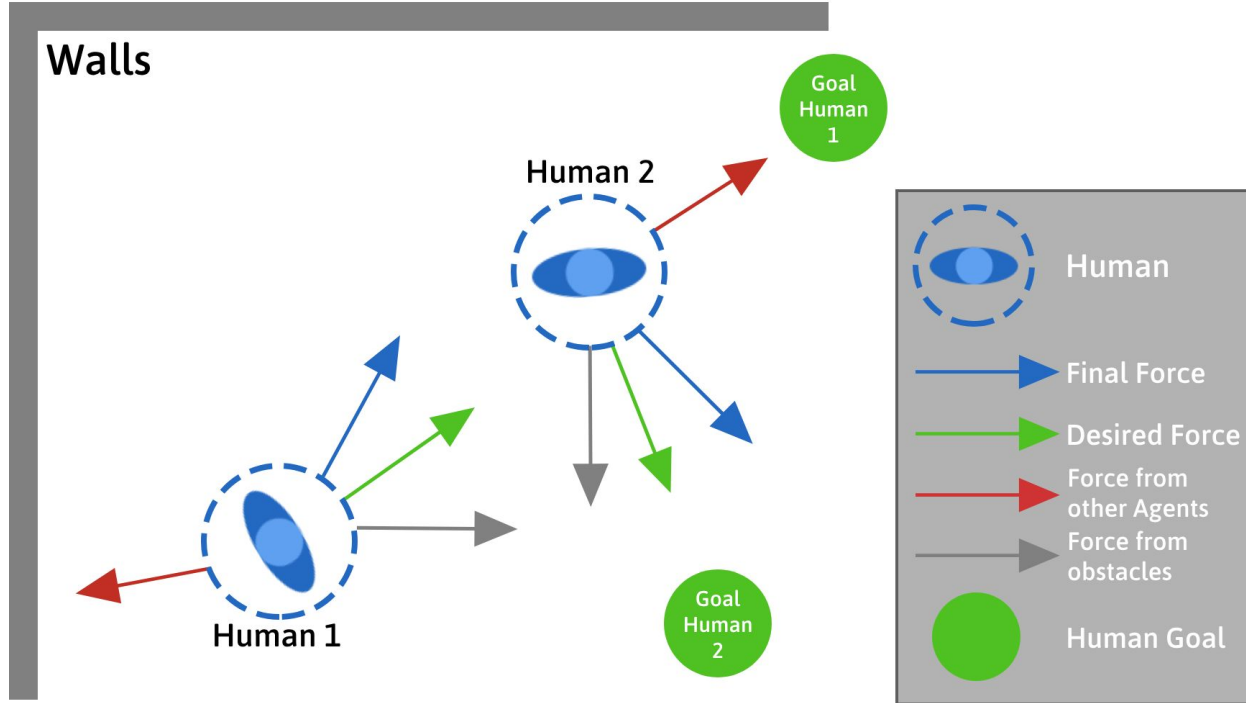
❖ 6 Performance Metrics:

- Success Rate
- Collision with static obstacles
- Path Length (m)
- Time to Reach (s)
- Target Error (m)
- Jerk (m/s^3)

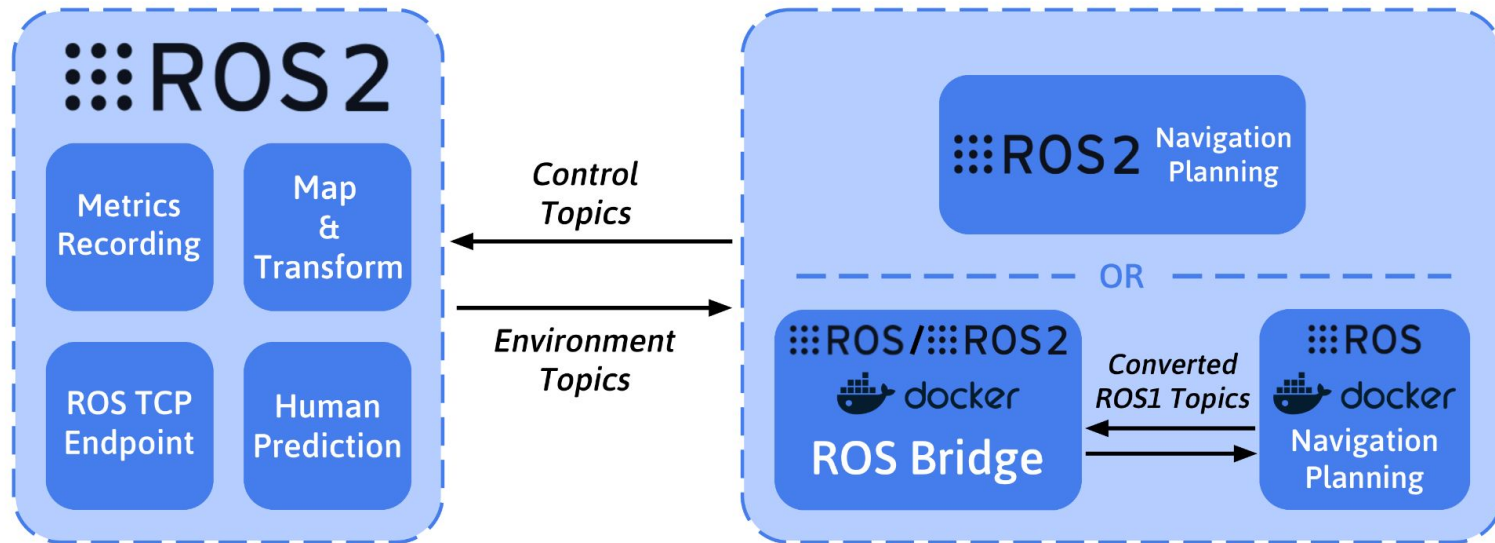
❖ 3 Social Metrics:

- Human Collision
- Minimum Distance to Human (m)
- Minimum Time-to-Collision (s)

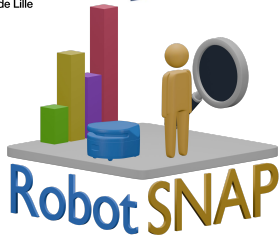
Human Simulation: Social Force Model^[1]



Implemented Off-the-Shelf Solutions^[1]



[1] <https://hub.docker.com/repository/docker/agouguet/benchmark-social-navigation>

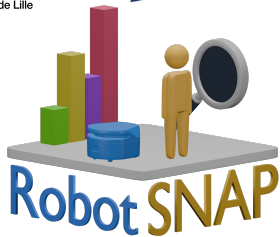


State-of-the-Art Results



			Safety	Social	Performance		
Baseline	Model-based	DWB	■	■	■	Sparse Scenario	Crowd Scenario
		HATEB	■	■	■	■	■
		MPC-EBM	■	■	■	■	■
		BRNE	■	■	■	■	■
	Learning-Based	CADRL	■	■	■	■	■
		SARL*	■	■	■	■	■
		DRL-VO	■	■	■	■	■

■ Weak
 ■ Medium
 ■ Strong



State-of-the-Art Results



			Safety	Social	Performance		Sparse Scenario	Crowd Scenario
Baseline	Model-based	DWB	■	■	■		■	■
		HATEB	■	■	■		■	■
		MPC-EBM	■	■	■		■	■
	Learning-Based	BRNE	■	■	■		■	■
		CADRL	■	■	■		■	■
		SARL*	■	■	■		■	■
		DRL-VO	■	■	■		■	■

■ Weak

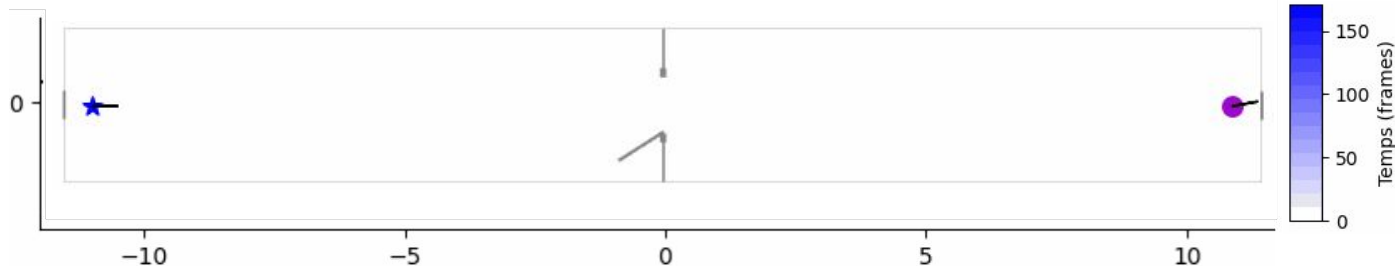
■ Medium

■ Strong

Narrow Passage

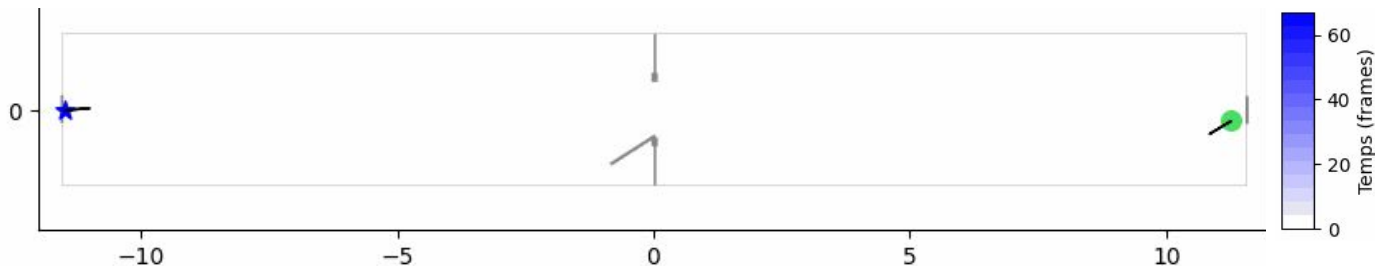


HATEB
(Model-Based)



Robot with orientation

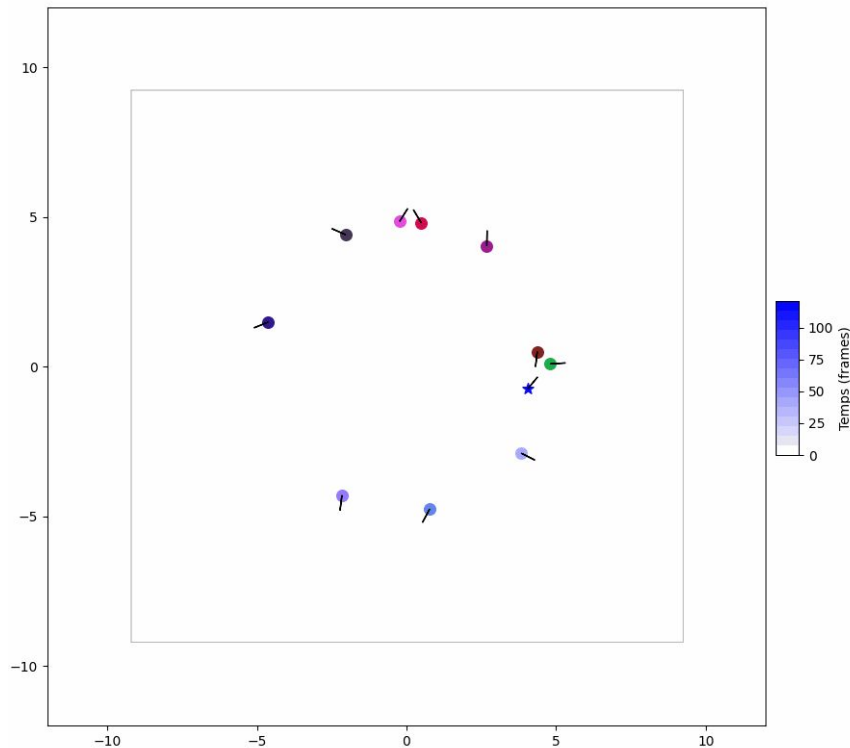
DRL-VO
(Learning-Based)



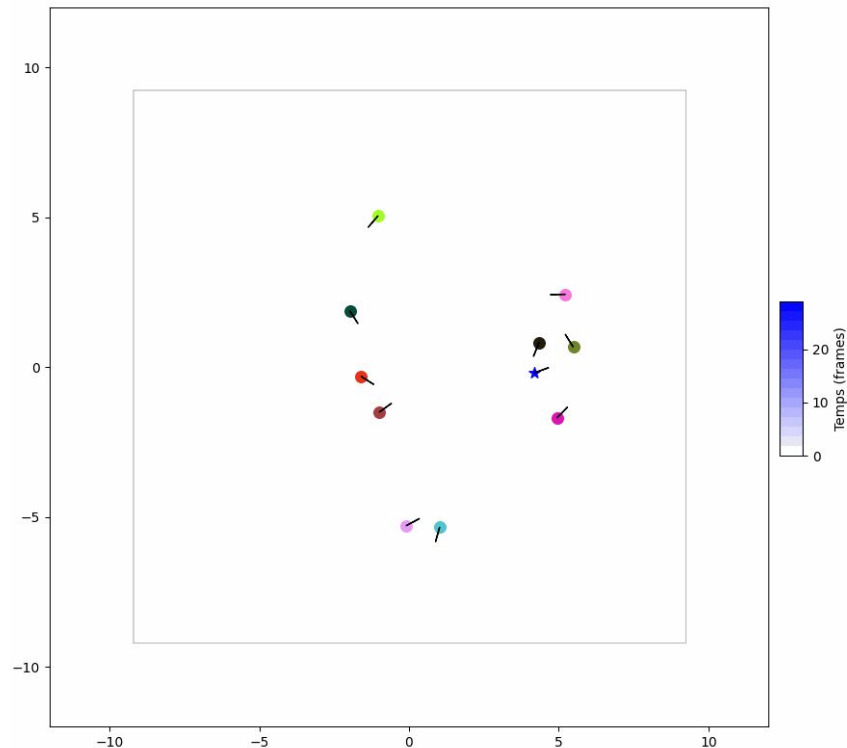
Human with orientation



Circle Crowd

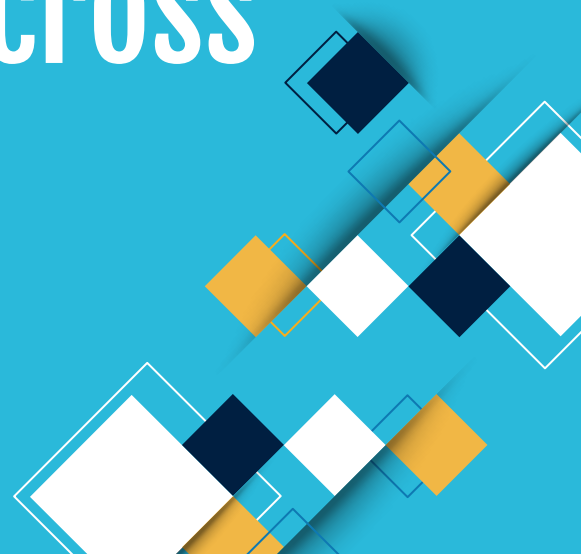


HATEB
(Model-Based)



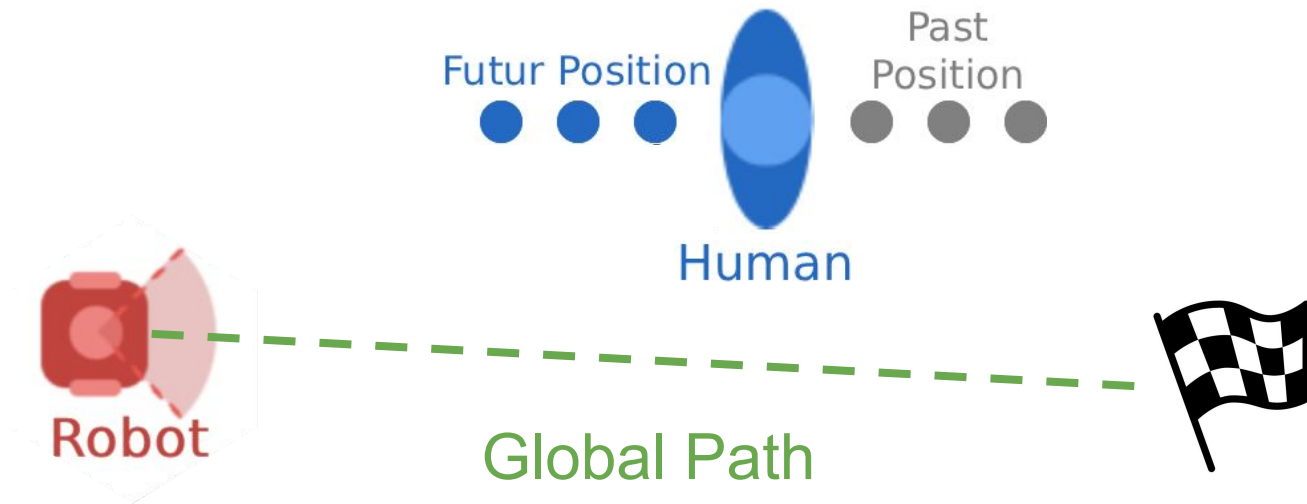
DRL-VO
(Learning-Based)

Our solutions: Improving the versatility of methods across different environments

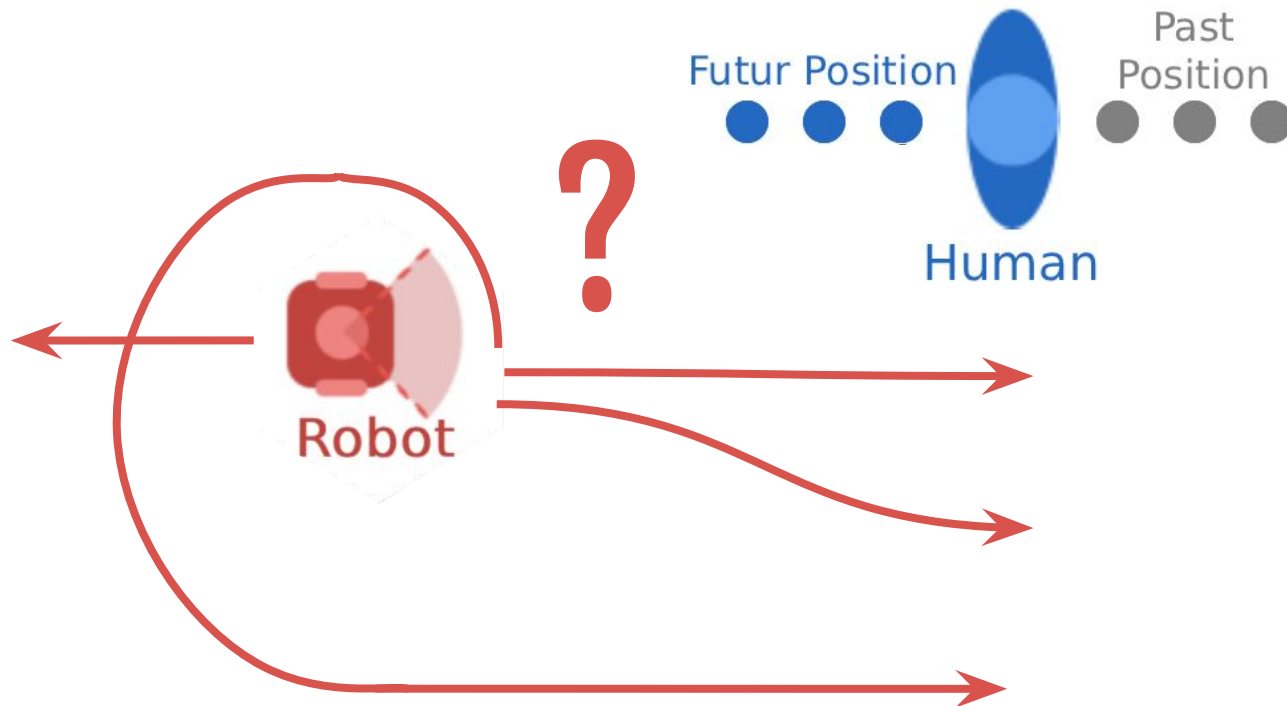




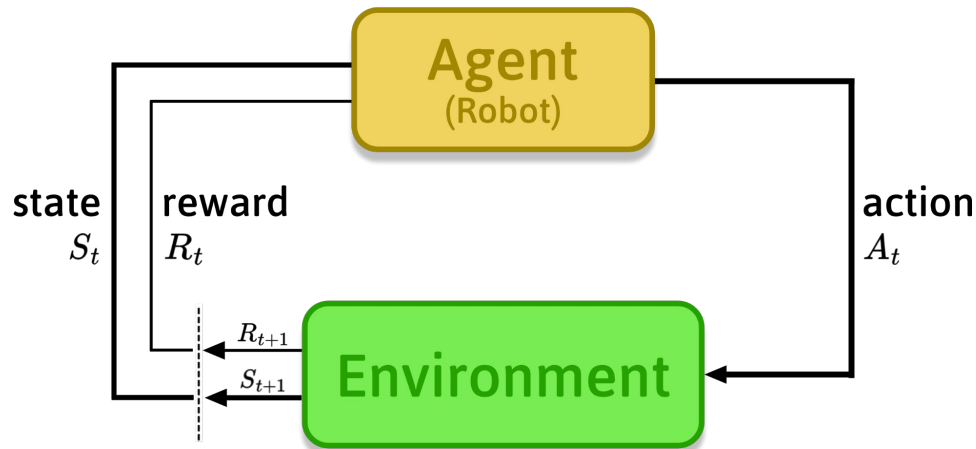
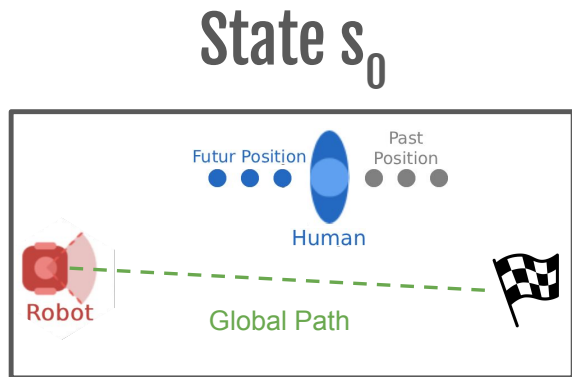
Robot Environment



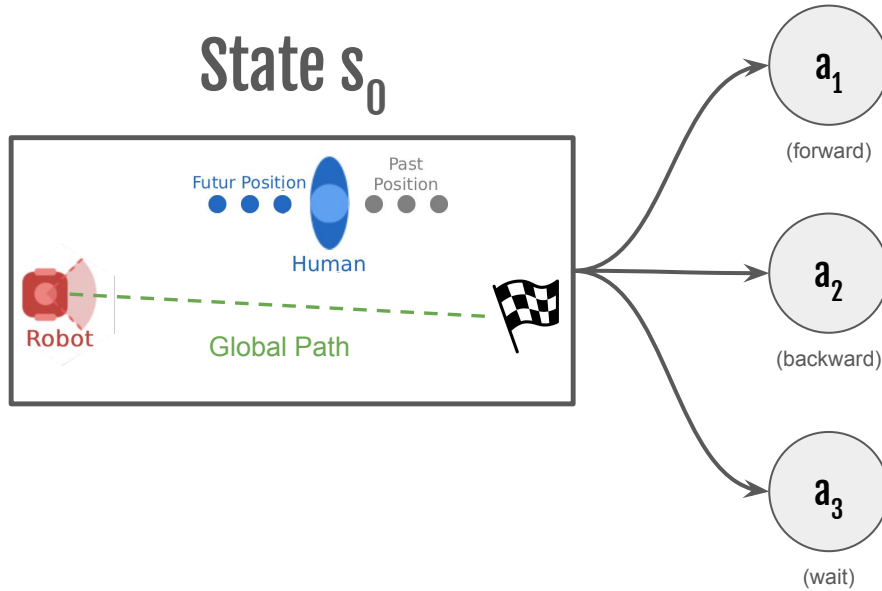
Robot Environment



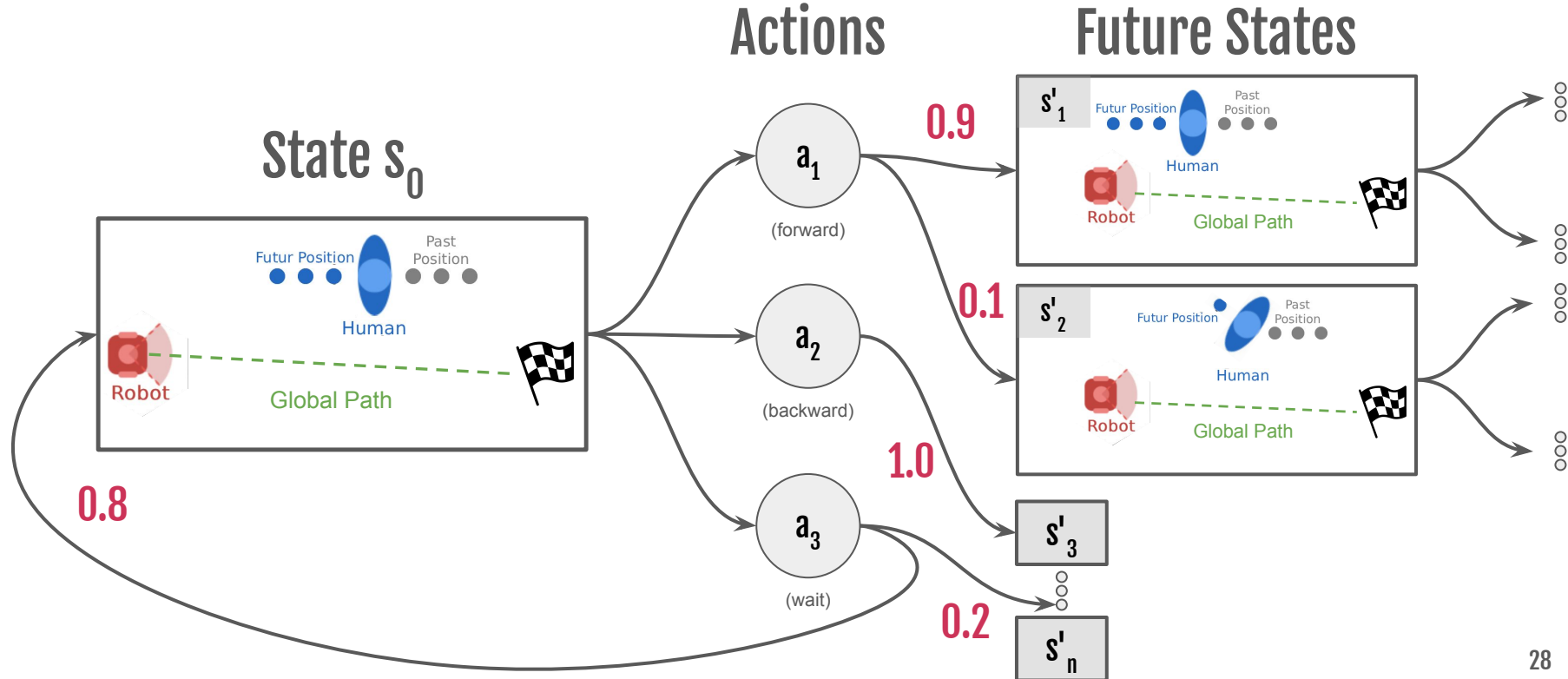
Robot Environment as a Markov Decision Process (MDP)



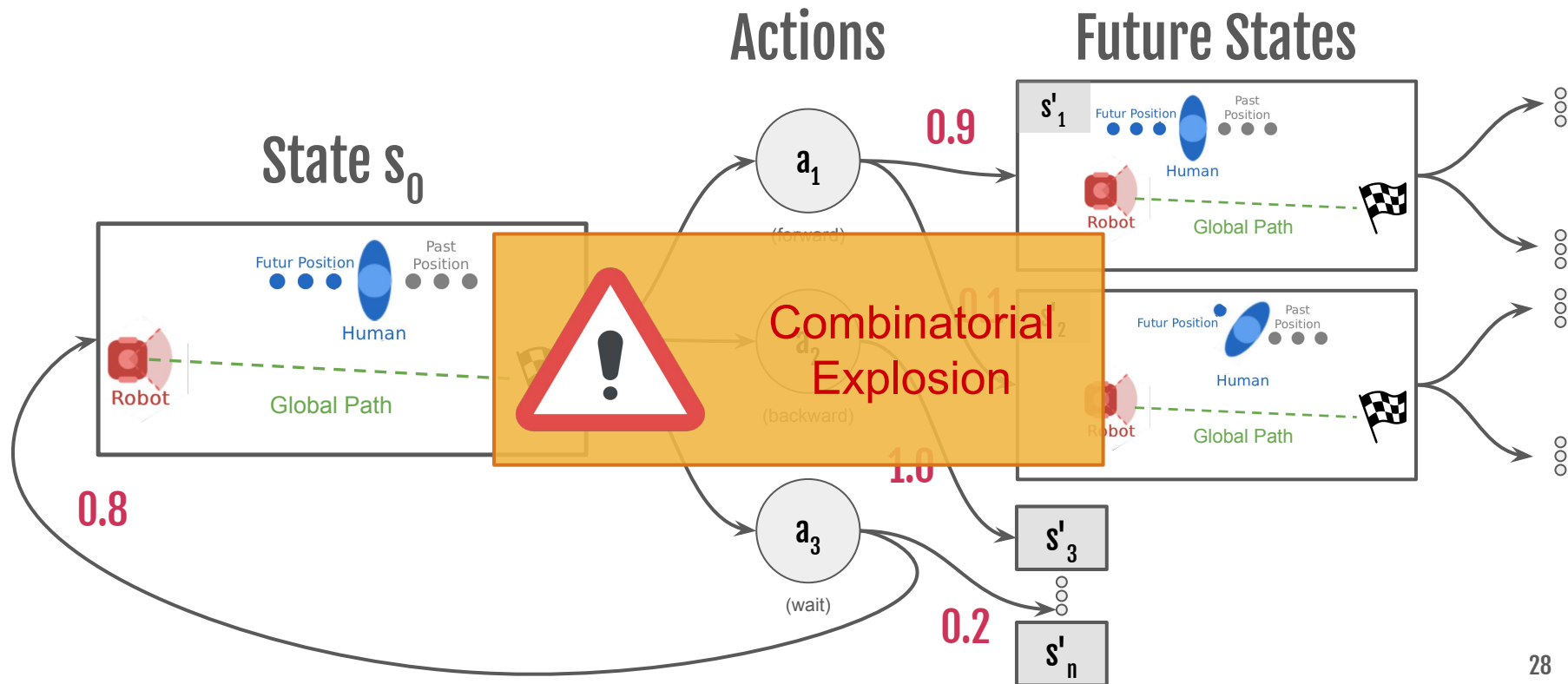
Robot Environment as a Markov Decision Process (MDP)



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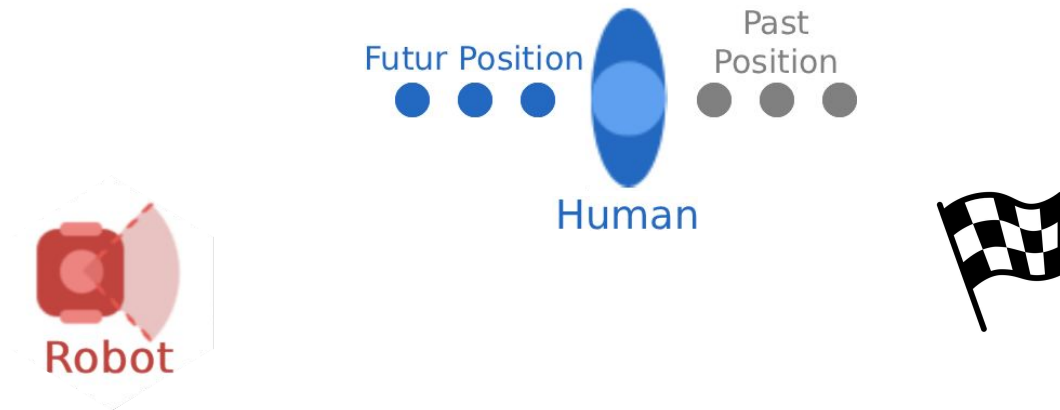
Robot Environment as a Markov Decision Process (MDP)



Heuristic-Based Social Navigation (HBSN)



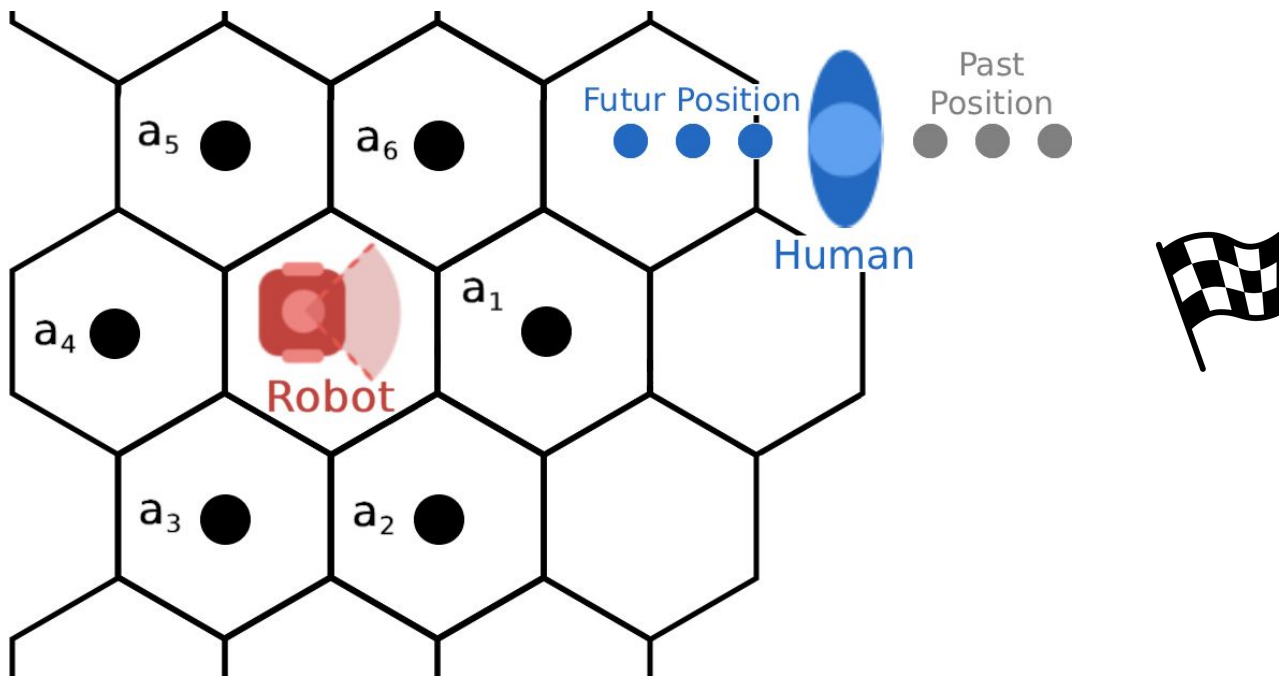
$$\text{social_heuristic} = \underset{a \in A}{\operatorname{argmax}} \left(\overline{\text{score}_{\text{weighted}_a}} - k * \sigma_{\text{weighted}_a} \right)$$



Heuristic-Based Social Navigation (HBSN)



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Scores:

$\text{score}_{\text{move}}$ reducing movement effort.

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Heuristic-Based Social Navigation (HBSN)



$$\text{social_heuristic} = \underset{a \in A}{\operatorname{argmax}} \left(\overline{\text{score}_{\text{weighted}_a}} - k * \sigma_{\text{weighted}_a} \right)$$

Scores:

$\text{score}_{\text{move}}$ reducing movement effort.

$\text{score}_{\text{goal}}$ achieving the goal.

$\text{score}_{\text{near}}$ respecting social space.

$\text{score}_{\text{side}}$ prioritize moving to the right or left in symmetrical situations.

Heuristic-Based Social Navigation (HBSN)



$$\text{social_heuristic} = \underset{a \in A}{\operatorname{argmax}} \left(\overline{\text{score}_{\text{weighted}_a}} - \boxed{k * \sigma_{\text{weighted}_a}} \right)$$

Promotes
consistency/stability

Scores:

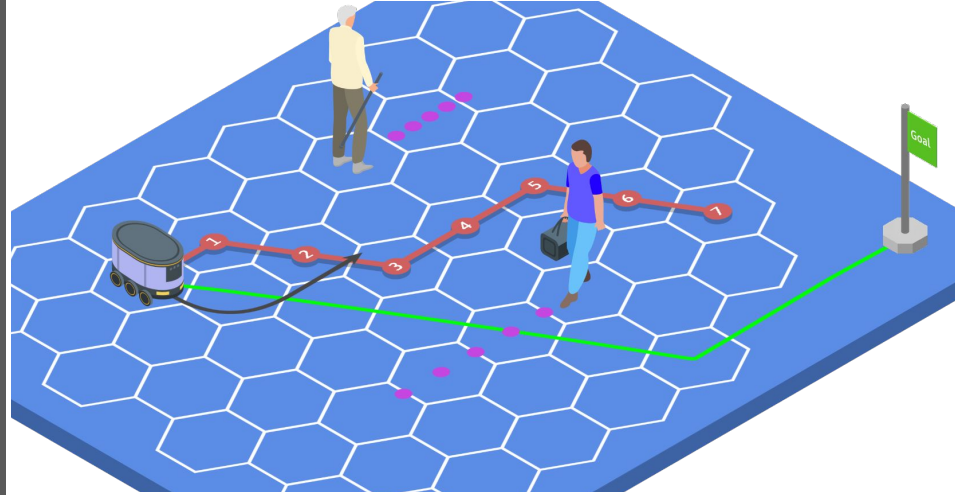
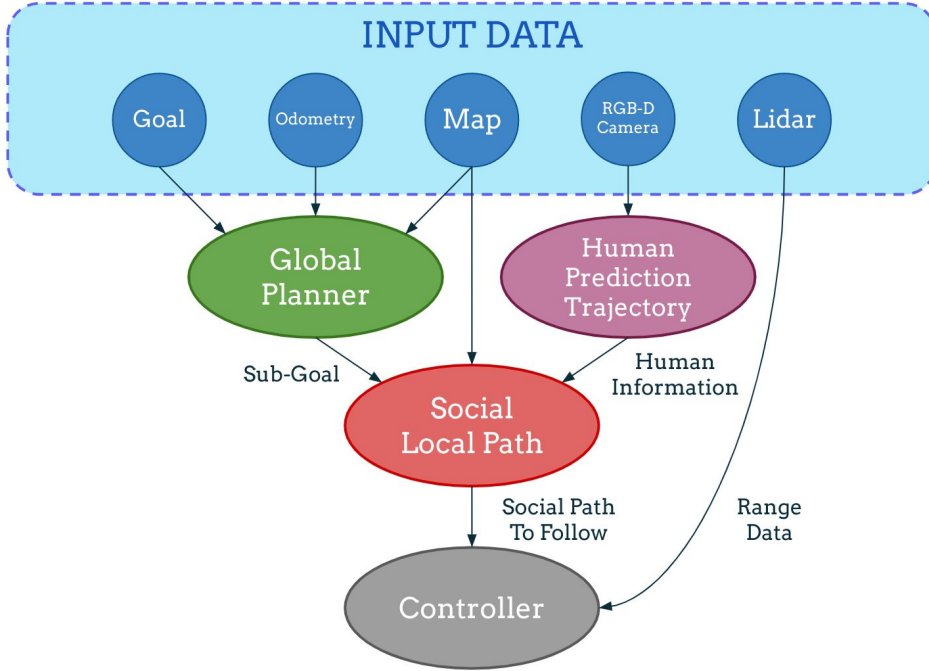
$\text{score}_{\text{move}}$ reducing movement effort.

$\text{score}_{\text{goal}}$ achieving the goal.

$\text{score}_{\text{near}}$ respecting social space.

$\text{score}_{\text{side}}$ prioritize moving to the right or left in symmetrical situations.

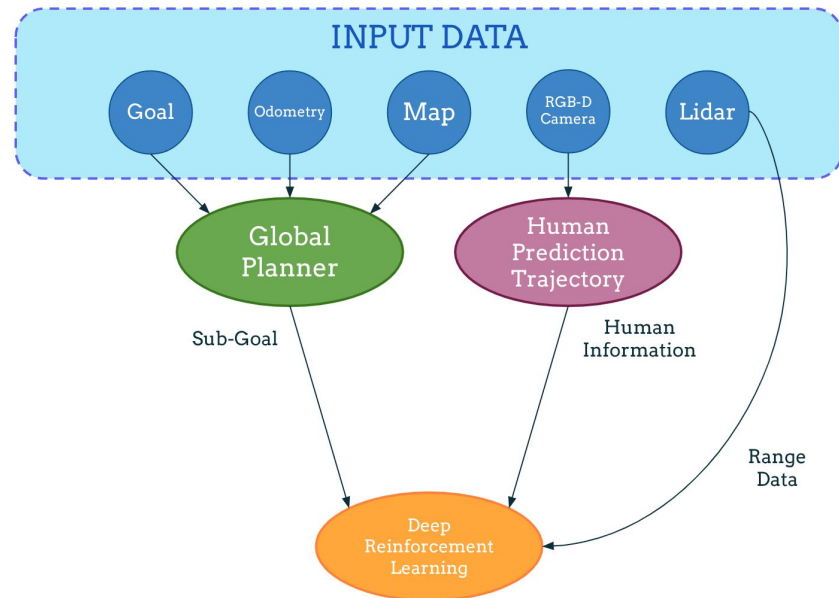
Heuristic-Based Social Navigation (HBSN)



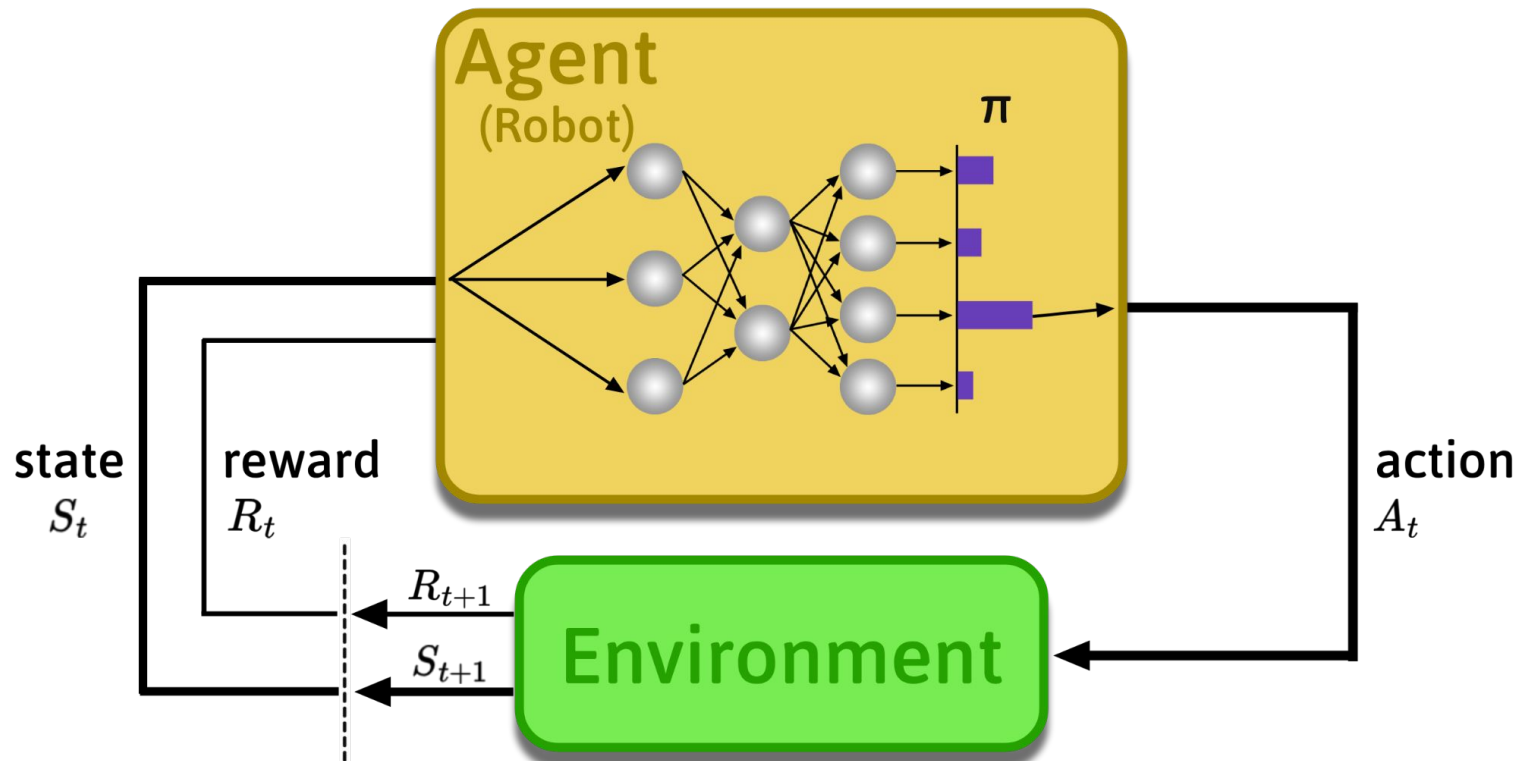
Motivation for Learning-Based Solution

Why ?

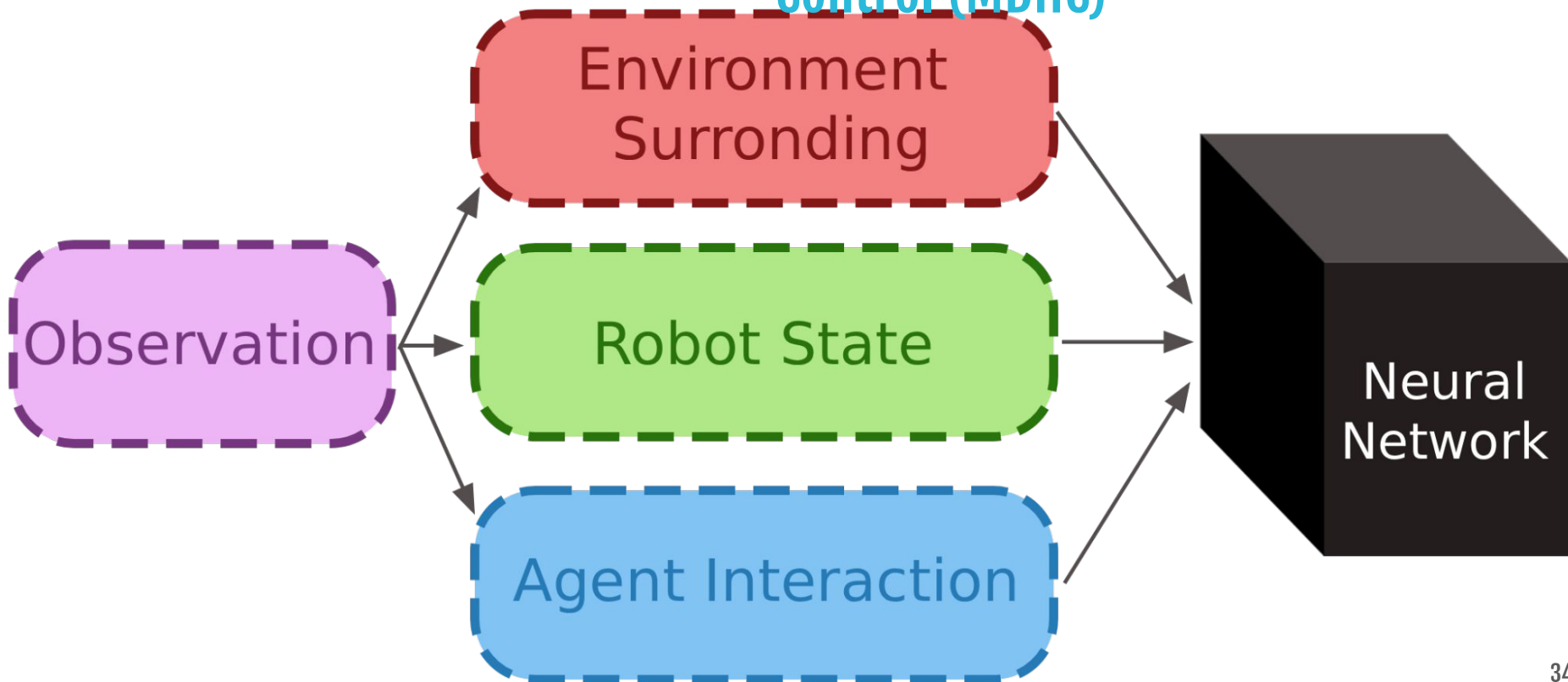
- Model-based rules cannot capture full human variability.
- Better alignment with pedestrian behavior.
- Optimize implicitly for comfort, smoothness, and legibility.



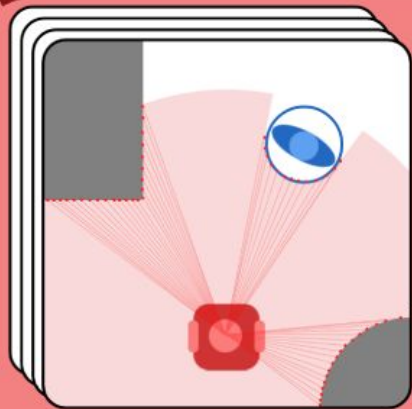
Deep Reinforcement Learning



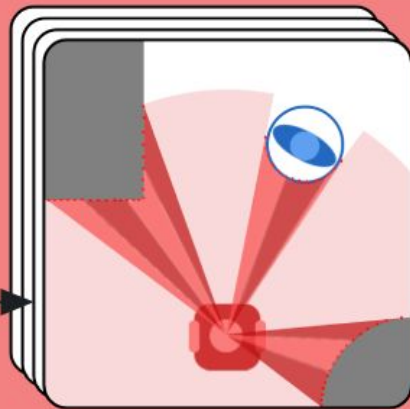
Multi-branch Deep Reinforcement-learning for Human-aware Control (MDHC)



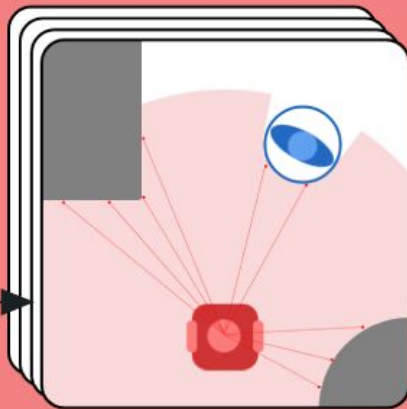
Surrounding Environment



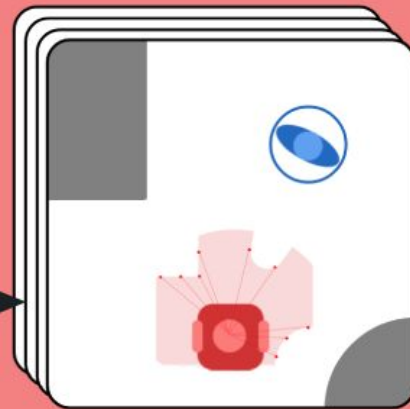
Stacking



Slicing

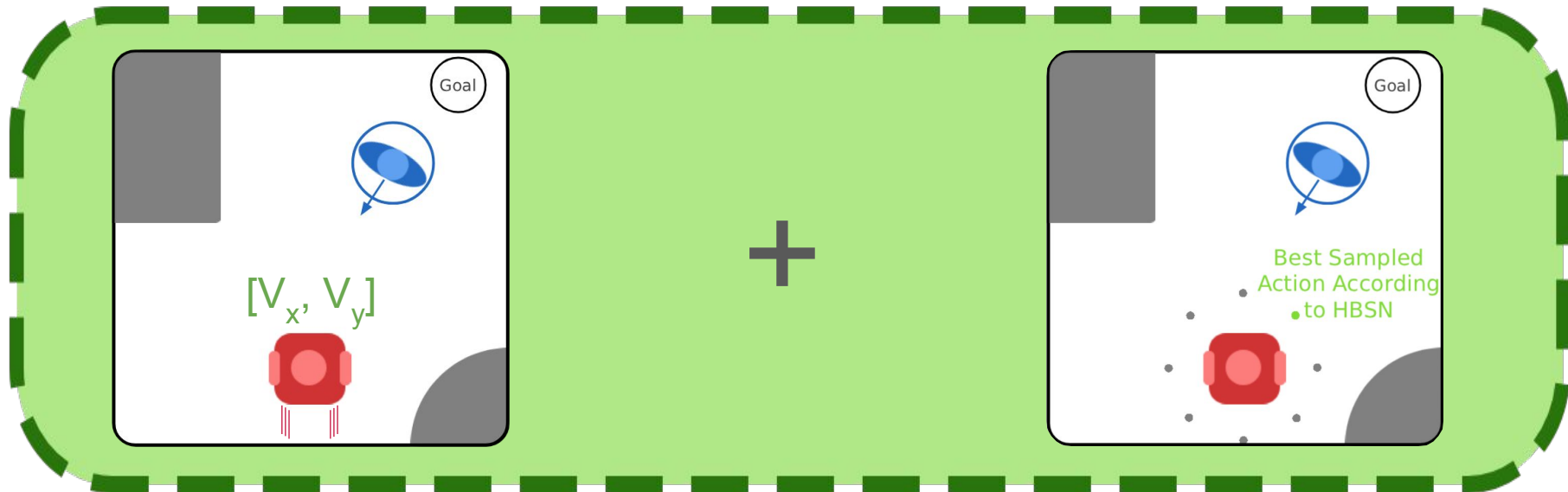


Min Pooling

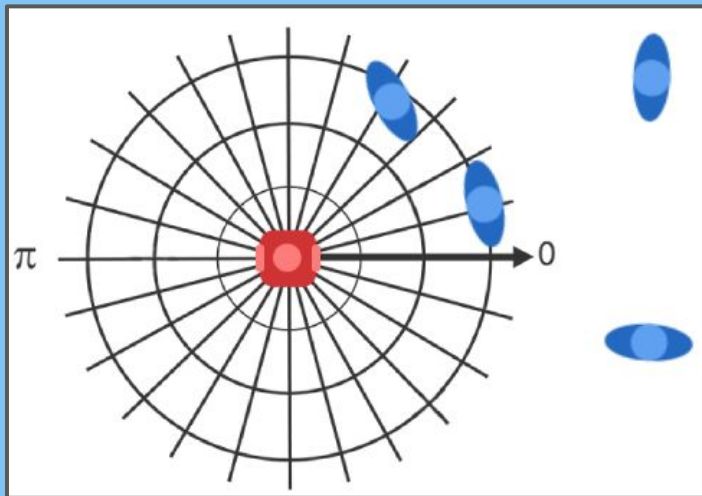


Normalization

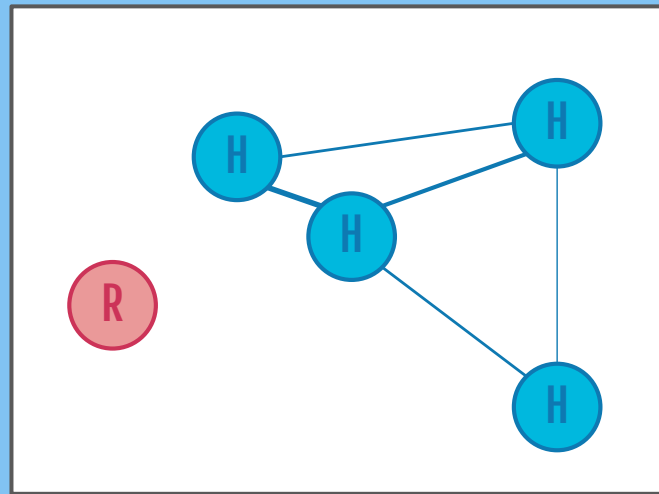
Robot State



Agent Interaction

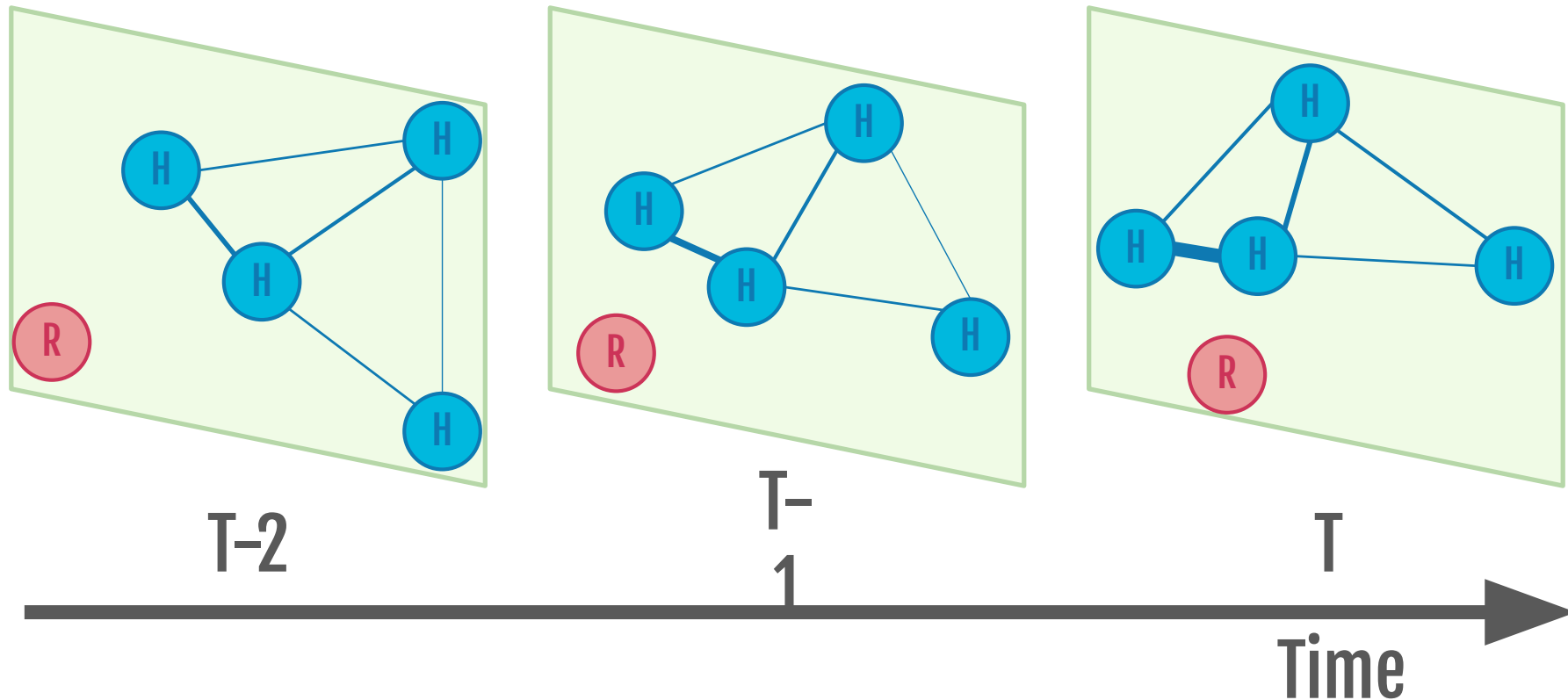


**Instant Human-Robot
Interaction**

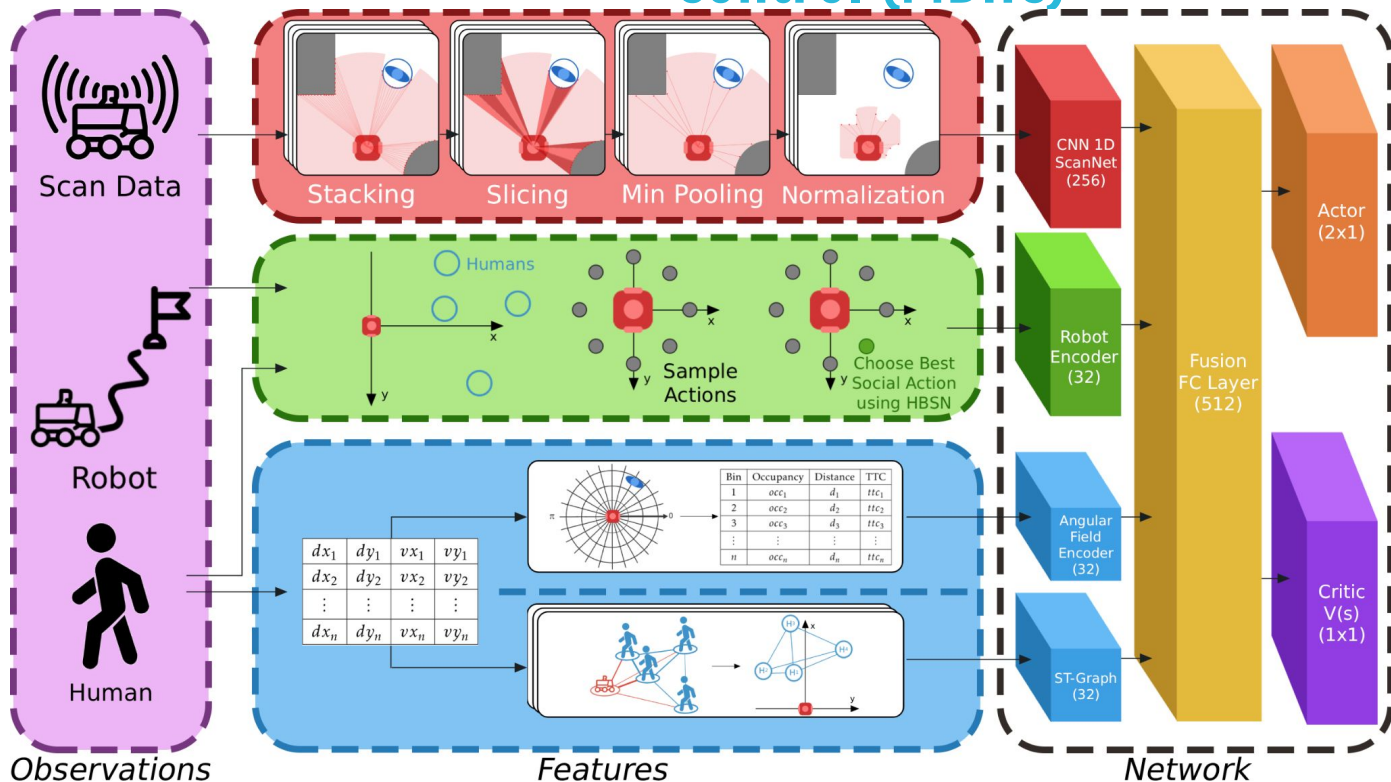


**Human-Human
Interaction over time**

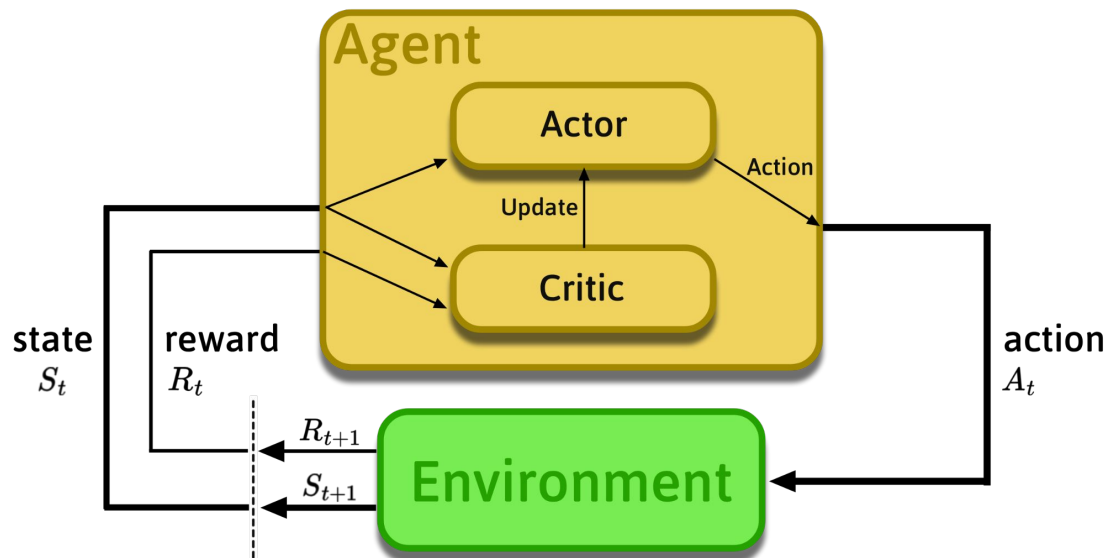
Spatio-temporal Graph



Multi-branch Deep Reinforcement-learning for Human-aware Control (MDHC)



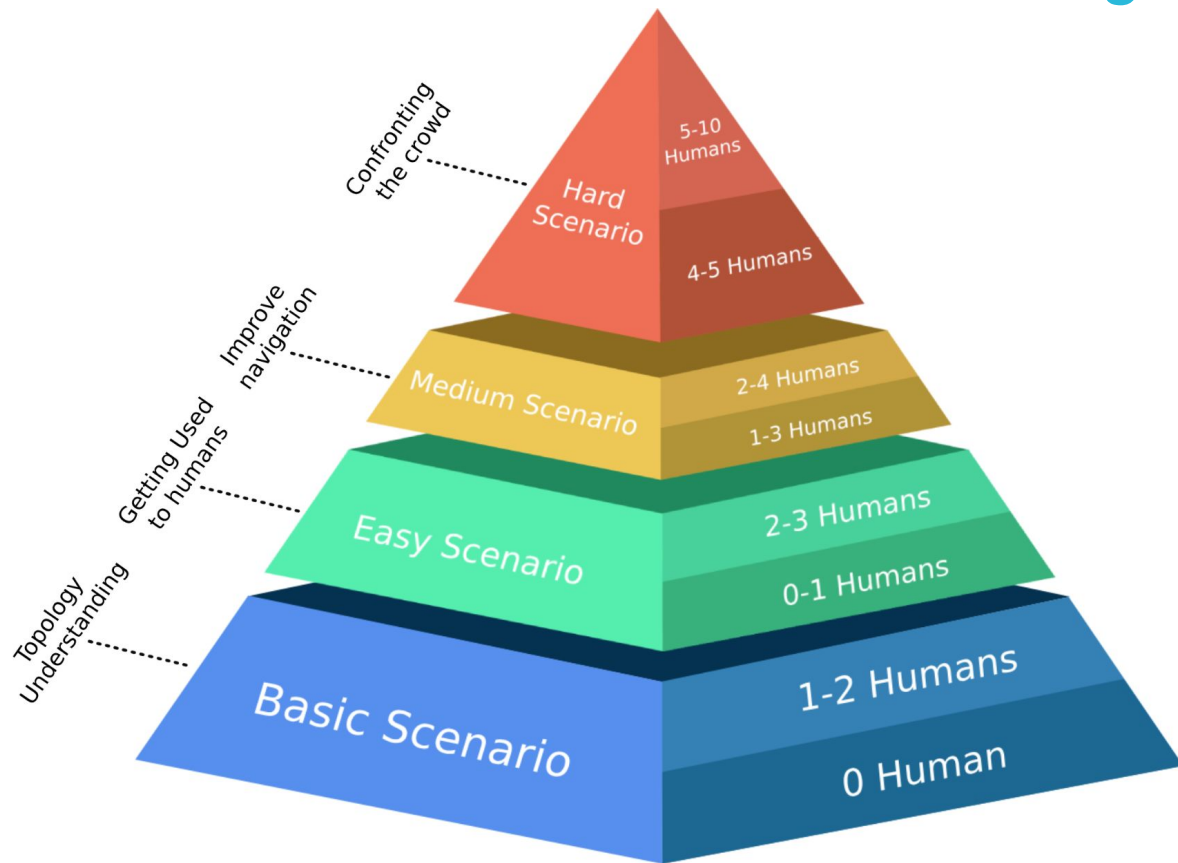
Proximal Policy Optimization (PPO)



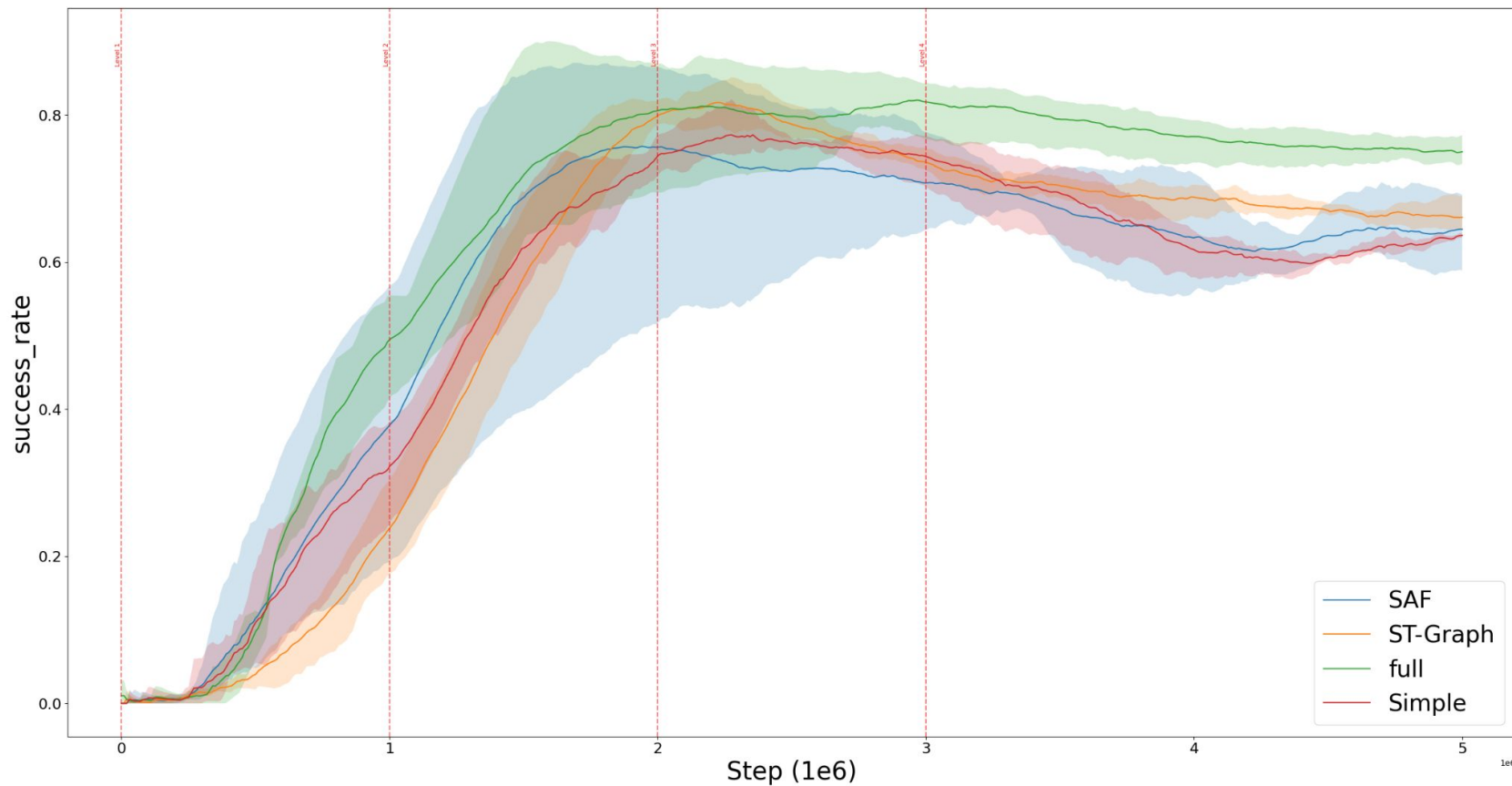
Stable Baseline 3



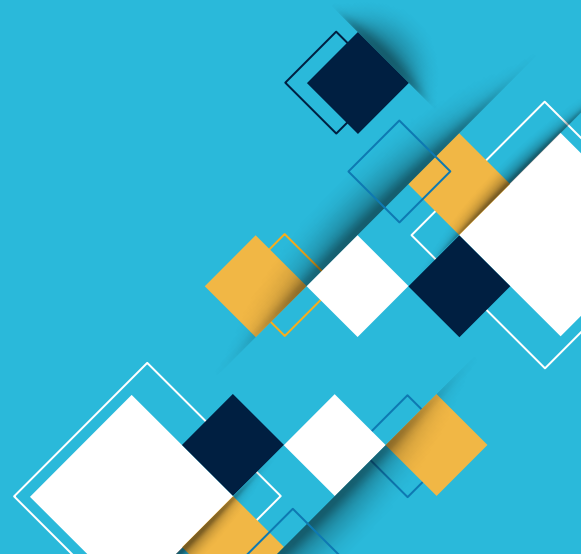
Curriculum Learning



Training Comparison of Neural Network Variants

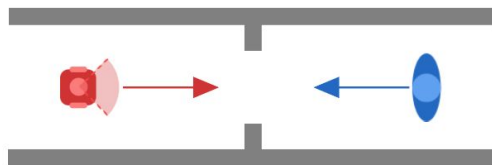
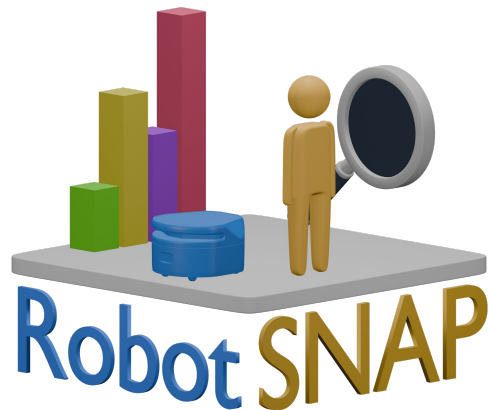


Experiments

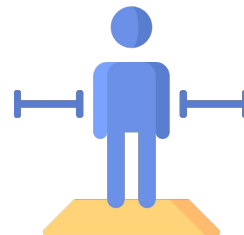




Experiments Setup



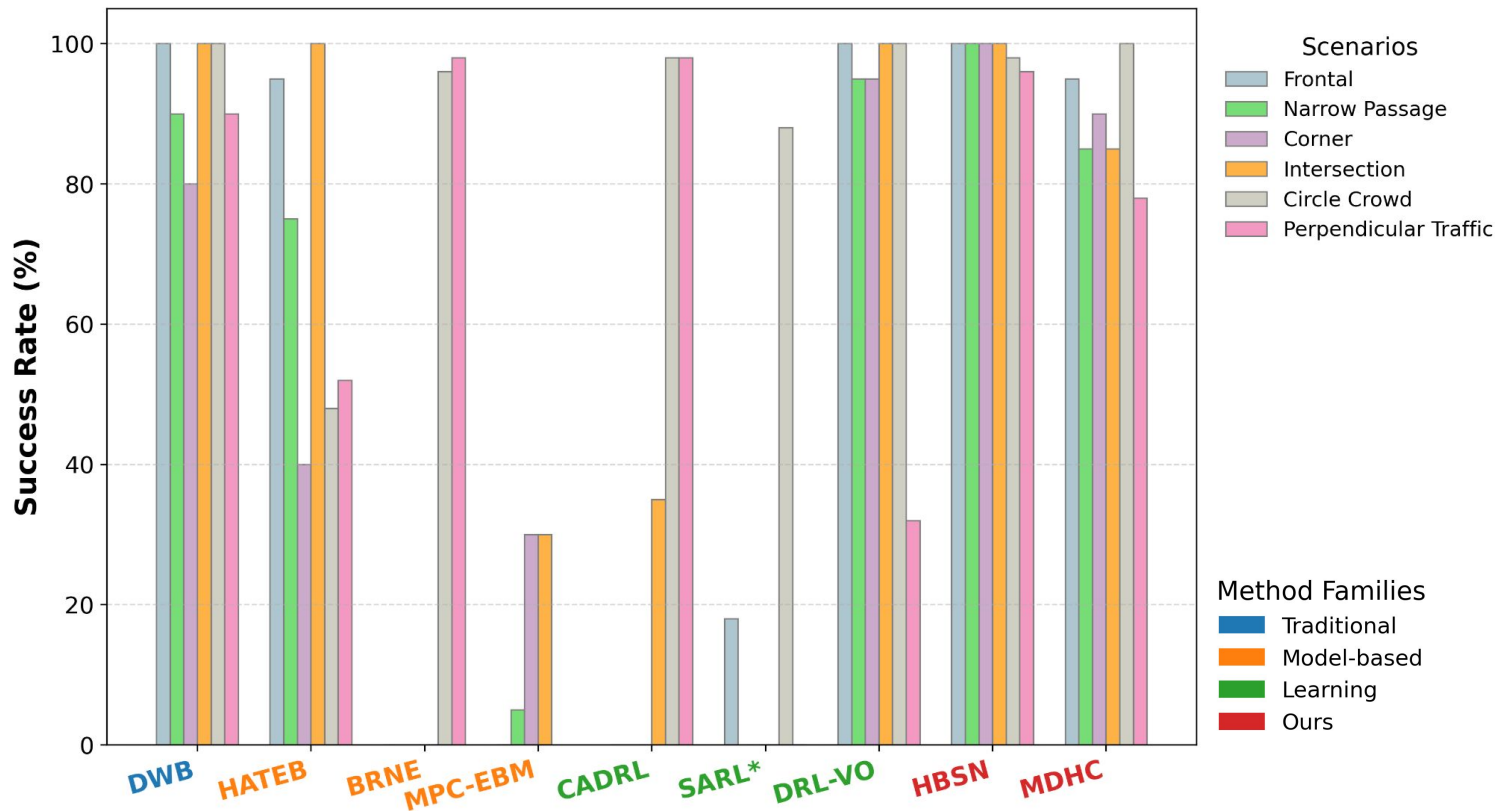
6 Scenarios



9 Metrics

Performance Metrics: Success Rate

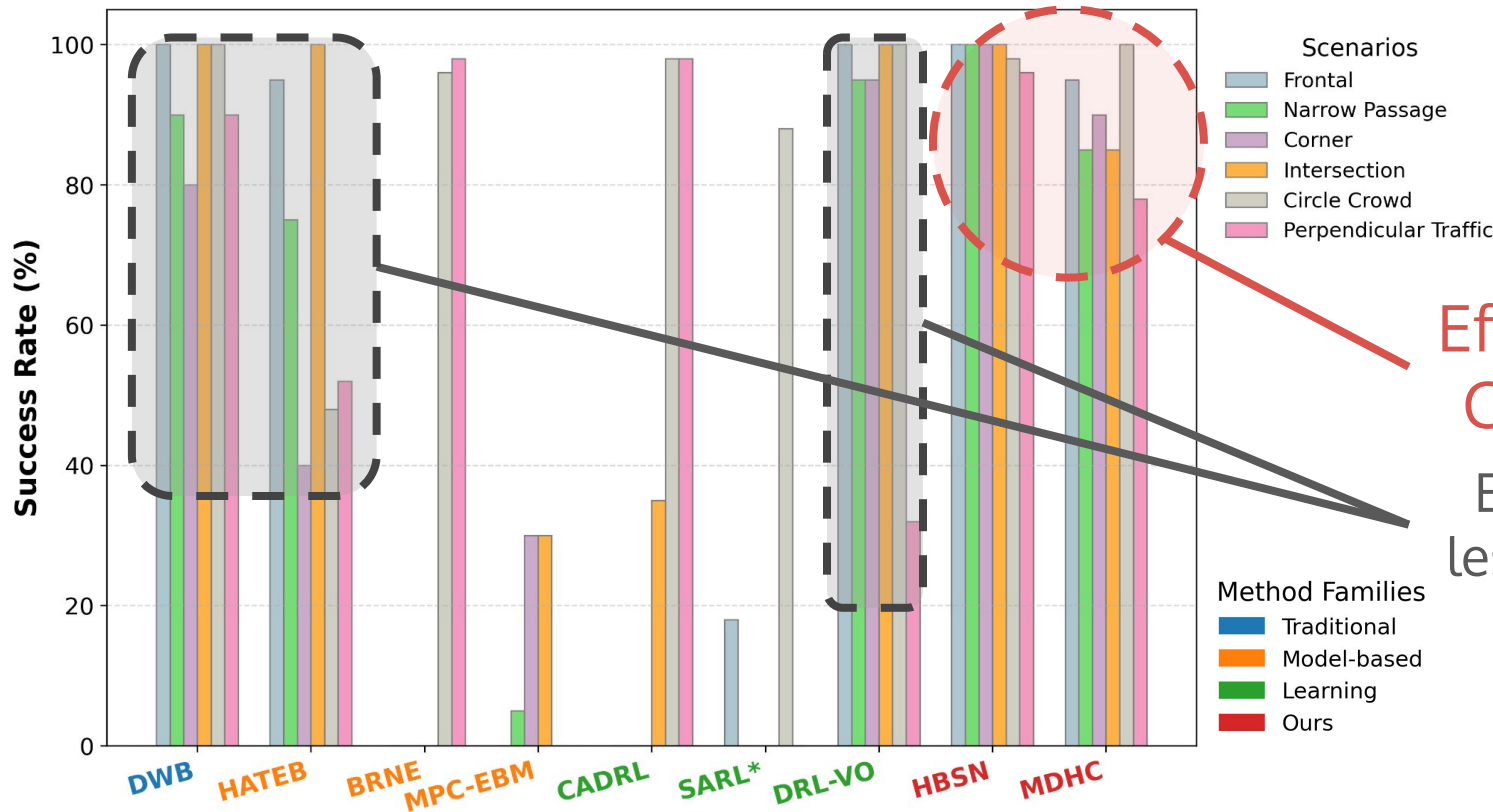
Success Rate (%) Across Methods and Scenarios





Performance Metrics: Success Rate

Success Rate (%) Across Methods and Scenarios

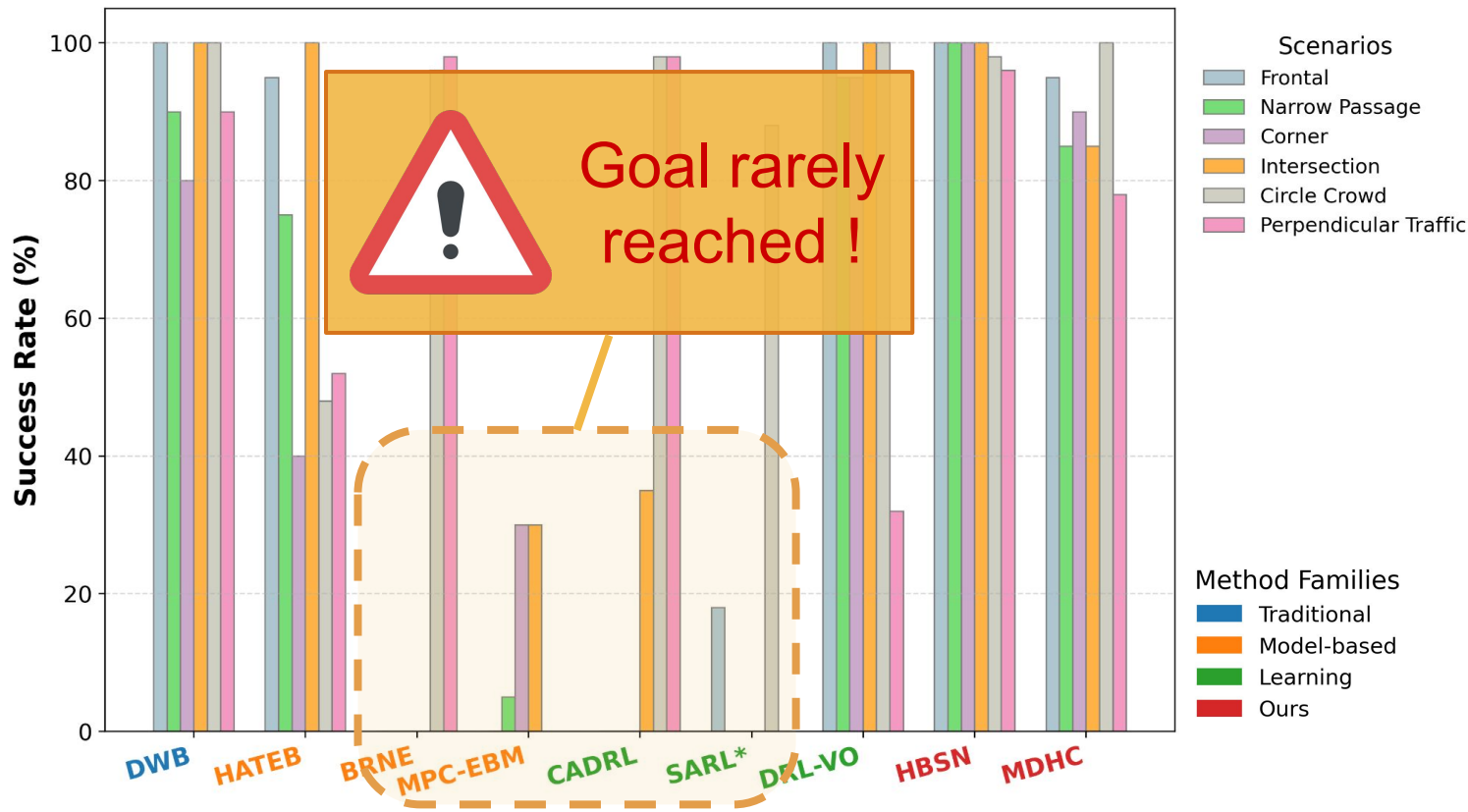


Effective and Consistent.

Effective but less consistent

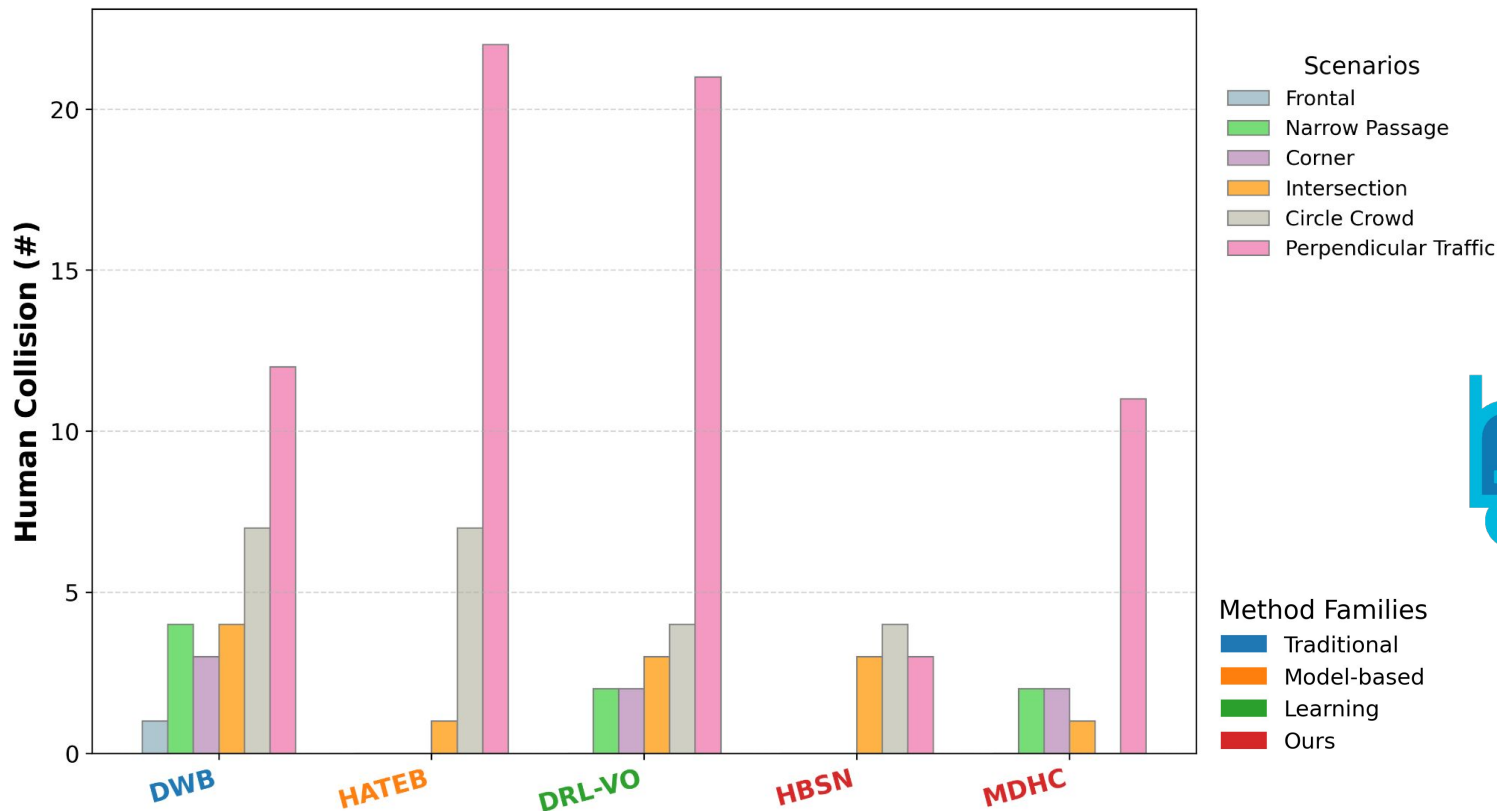
Performance Metrics: Success Rate

Success Rate (%) Across Methods and Scenarios



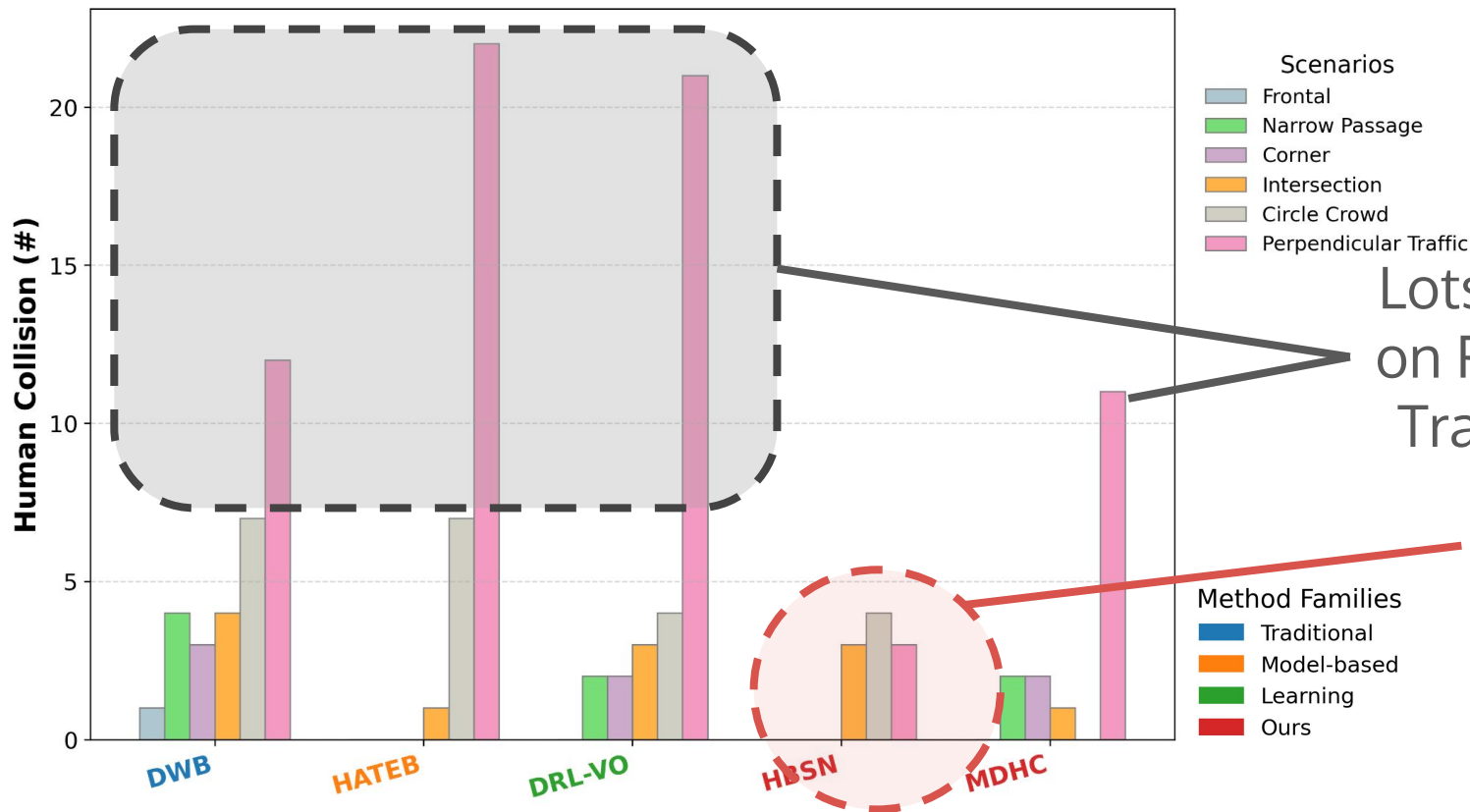
Social Metrics: Human collision

Human Collision (#) Across Methods and Scenarios



Social Metrics: Human collision

Human Collision (#) Across Methods and Scenarios



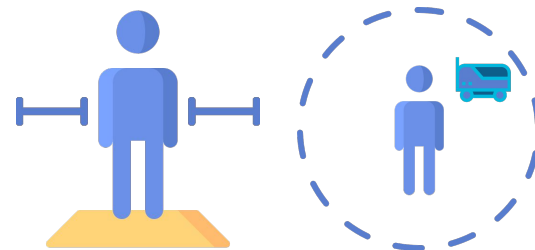
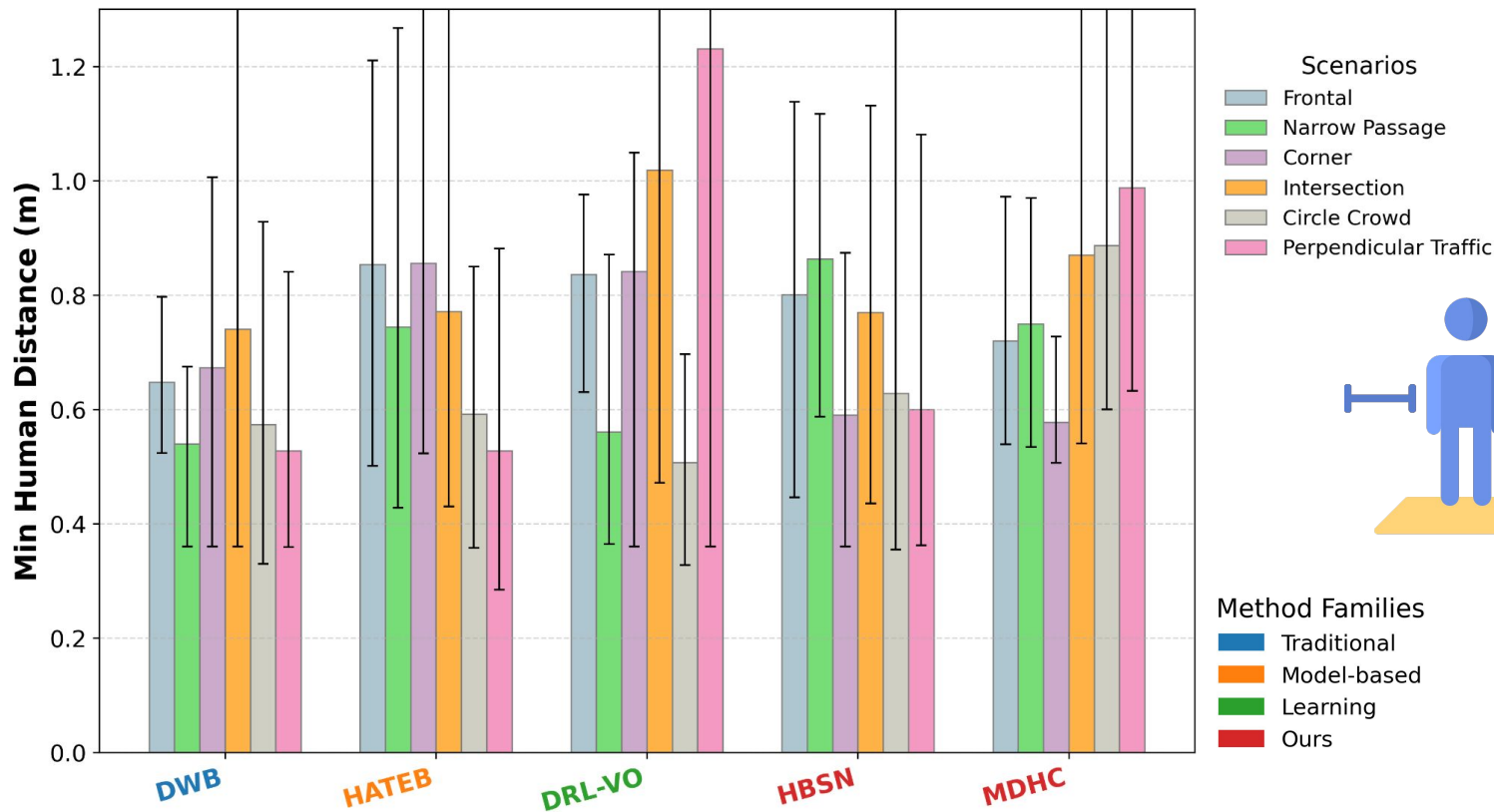
Lots of collisions
on Perpendicular
Traffic scenario

Only few
collisions

Social Metrics: Distance to Human



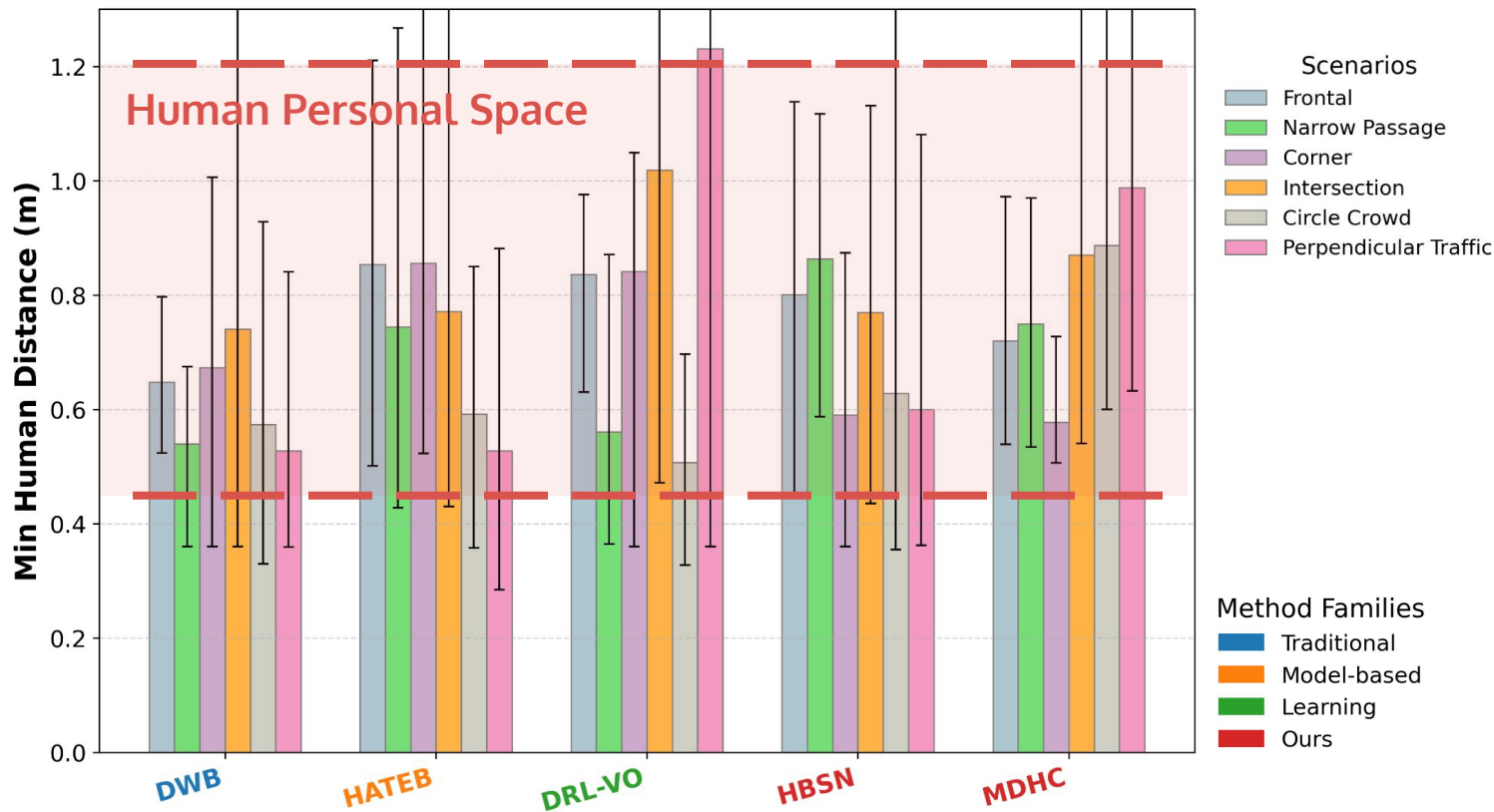
Min Human Distance (m) Across Methods and Scenarios



Social Metrics: Distance to Human



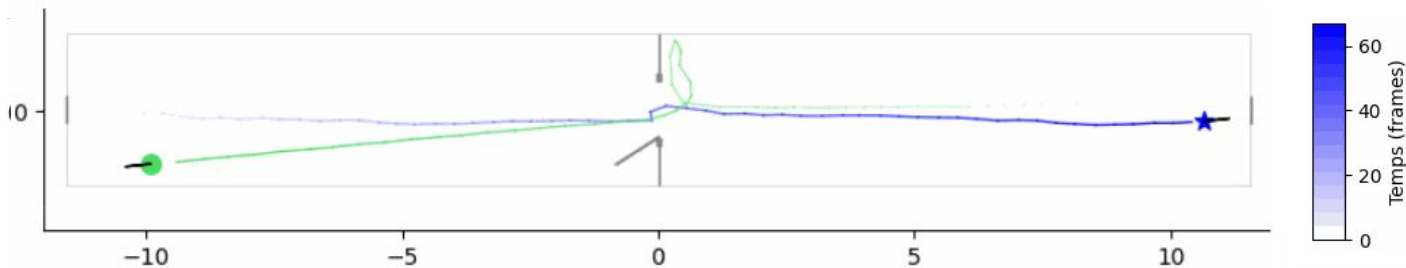
Min Human Distance (m) Across Methods and Scenarios



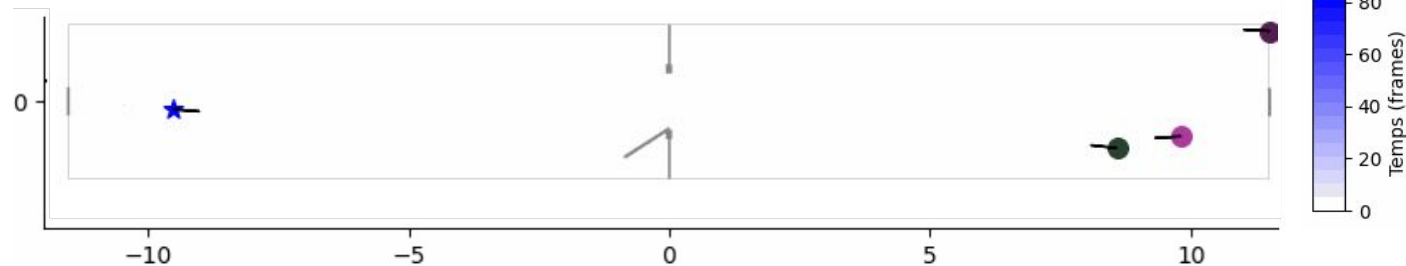
Narrow Passage Example



DRL-VO
(the best off-the-shelf
baseline)



HBSN



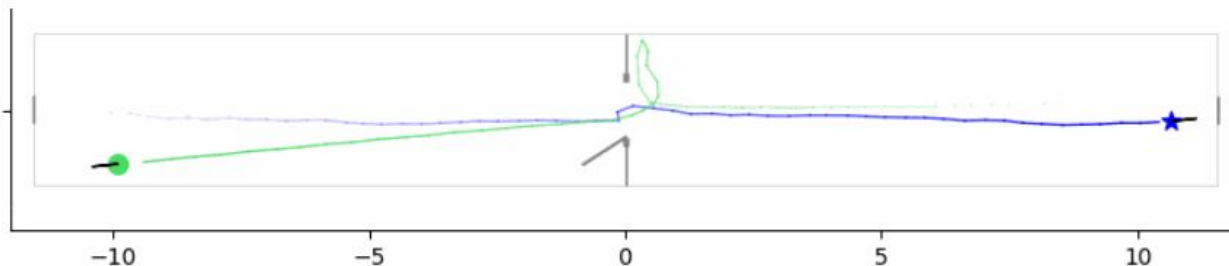
Robot with
orientation

Human with
orientation

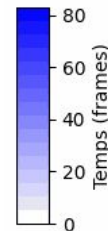
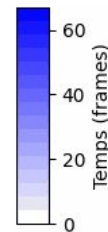
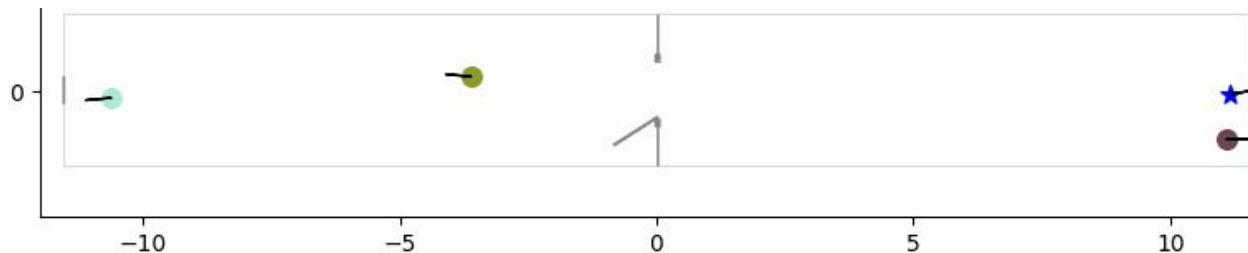
Narrow Passage Example



DRL-VO
(the best off-the-shelf
baseline)



HBSN



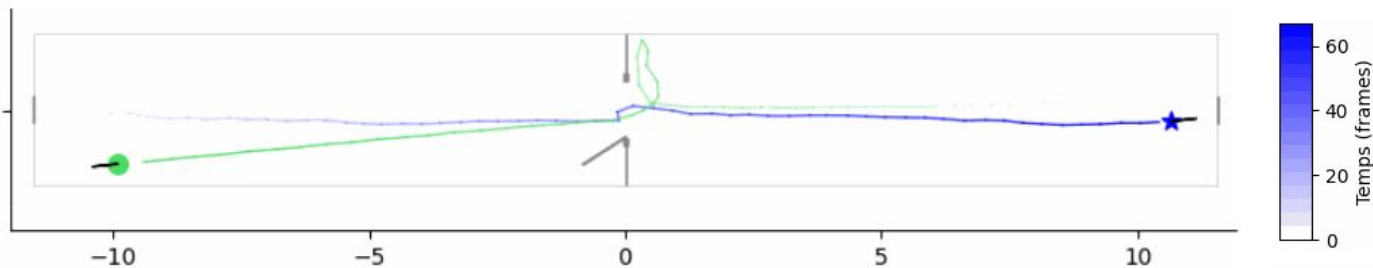
Robot with
orientation

Human with
orientation

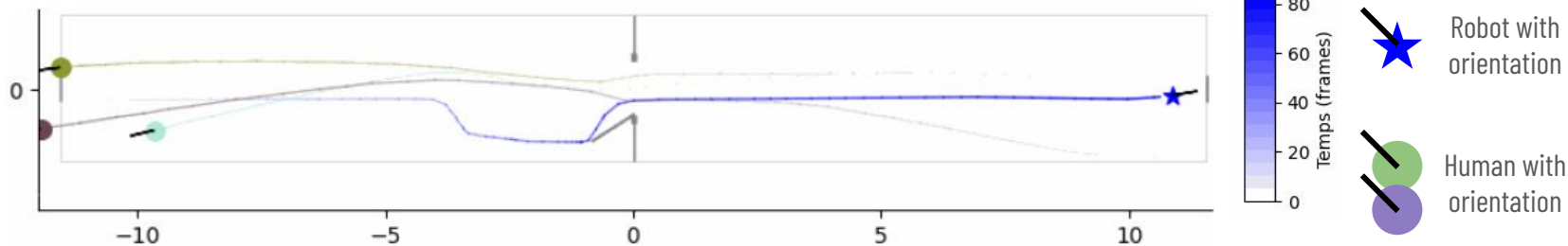
Narrow Passage Example



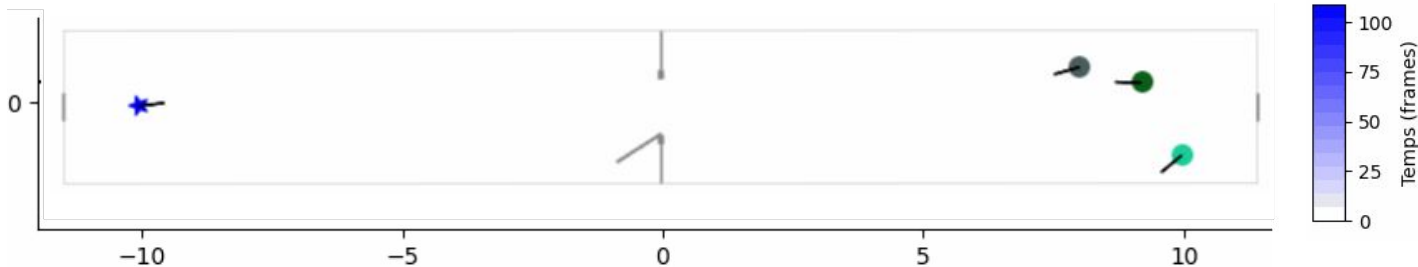
DRL-VO
(the best off-the-shelf
baseline)



HBSN



MDHC



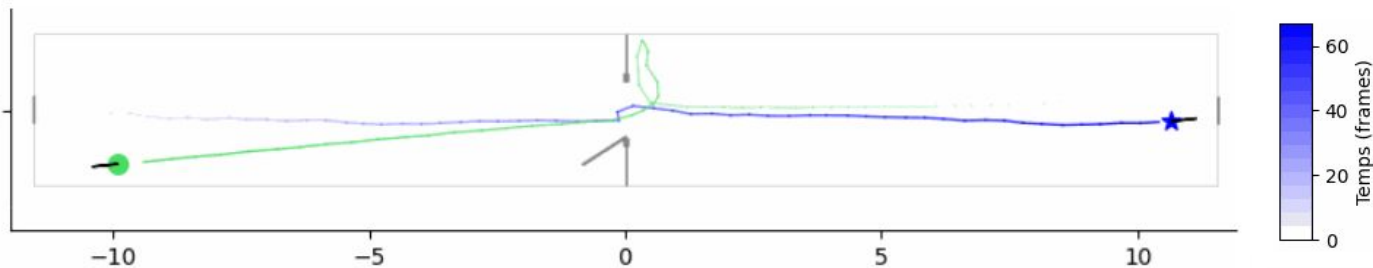
Robot with
orientation

Human with
orientation

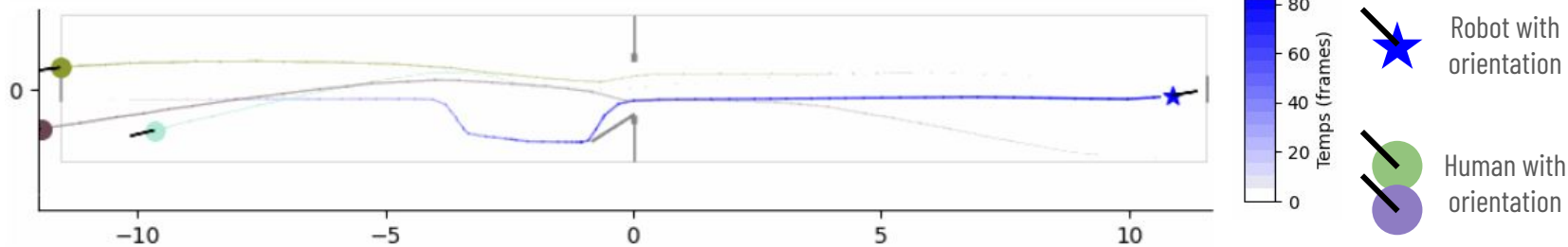
Narrow Passage Example



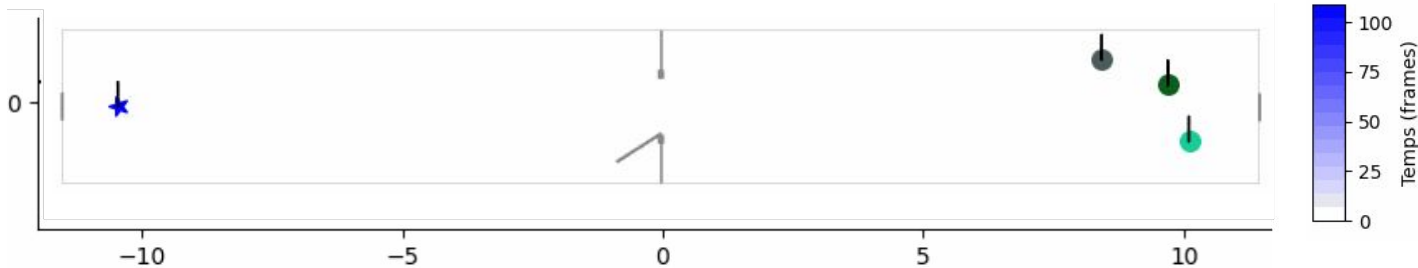
DRL-VO
(the best off-the-shelf
baseline)



HBSN



MDHC



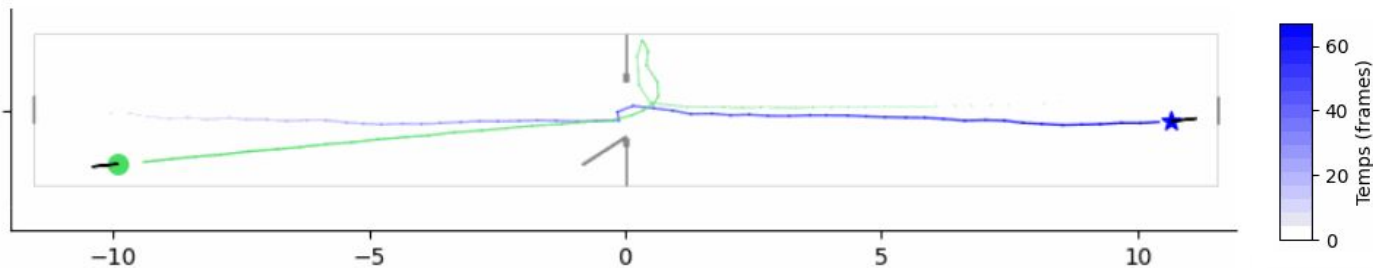
Robot with
orientation

Human with
orientation

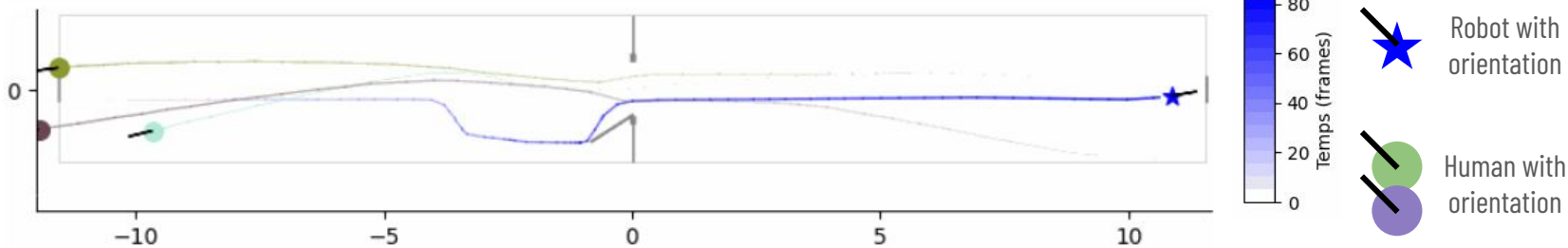
Narrow Passage Example



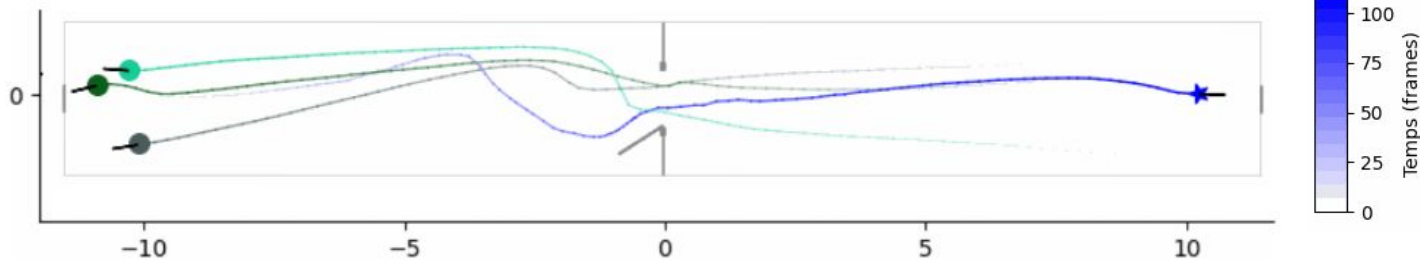
DRL-VO
(the best off-the-shelf
baseline)



HBSN



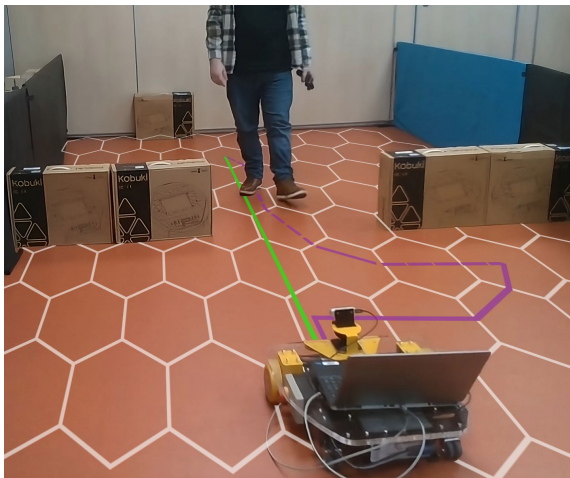
MDHC



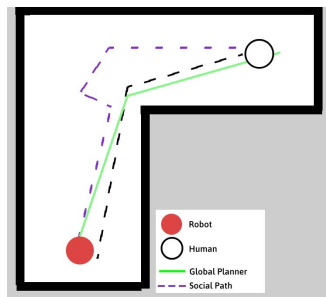
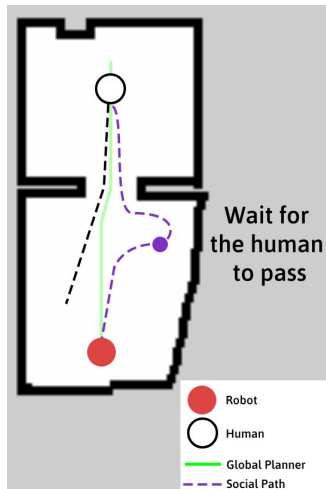
Real-World Experiments



Real-World Scenario



Final Paths



Real Case



Results



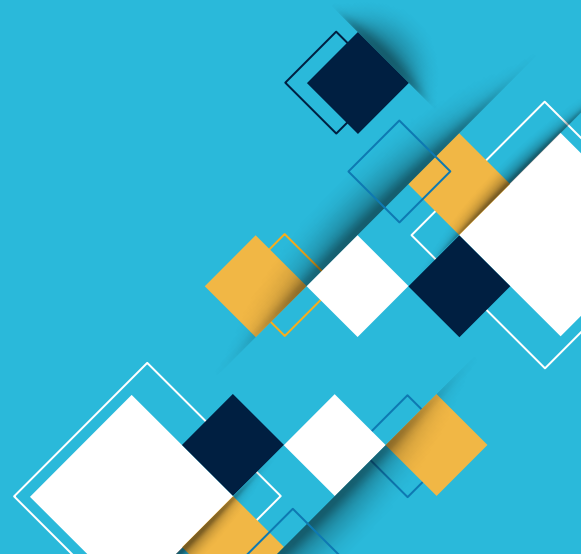
Discussion

- **HBSN** is better than the set of implemented methods, but the social rules and given weights are defined manually.
- **MDHC** is the most versatile machine learning solution.

No agent is truly aware that waiting can resolve blocking situations (narrow passage).

- Correlated but degraded results in real setup :
 - Sensor Uncertainty / Robot dynamics
 - Human Model vs Real Human

Conclusion



Conclusion



Contributions

- A new tool, named **RobotSNAP**, for evaluating social navigation methods has been developed : Open Source / Realistic / Reproducible / Sim2Real / Map Dataset
- Two main social navigation solutions have been proposed:
 - one based on heuristics (**HBSN**).
 - one based on reinforcement learning (**MDHC**).
- A comprehensive benchmark of **9 solutions** was carried out across the entire thesis.

Future Work



Short Term:

- Integrate (Progressive) **Multi-Layer Cross-Modal** Learning in **MDHC**.
- **User studies** for social evaluation: validate socially-aware behaviors of our solutions with **real humans**.

Long Term:

- An improved version of **RobotSNAP** to have a complete evaluation from simulation to the real world.
- Richer Structured Observations: integrating **contextual** knowledge.
- Multi-agents: managing robot blocking situations with **intention communication**.



Thank you for your attention

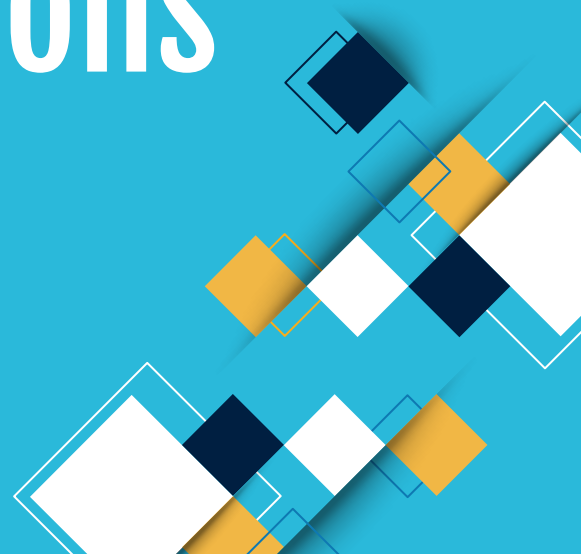


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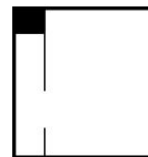
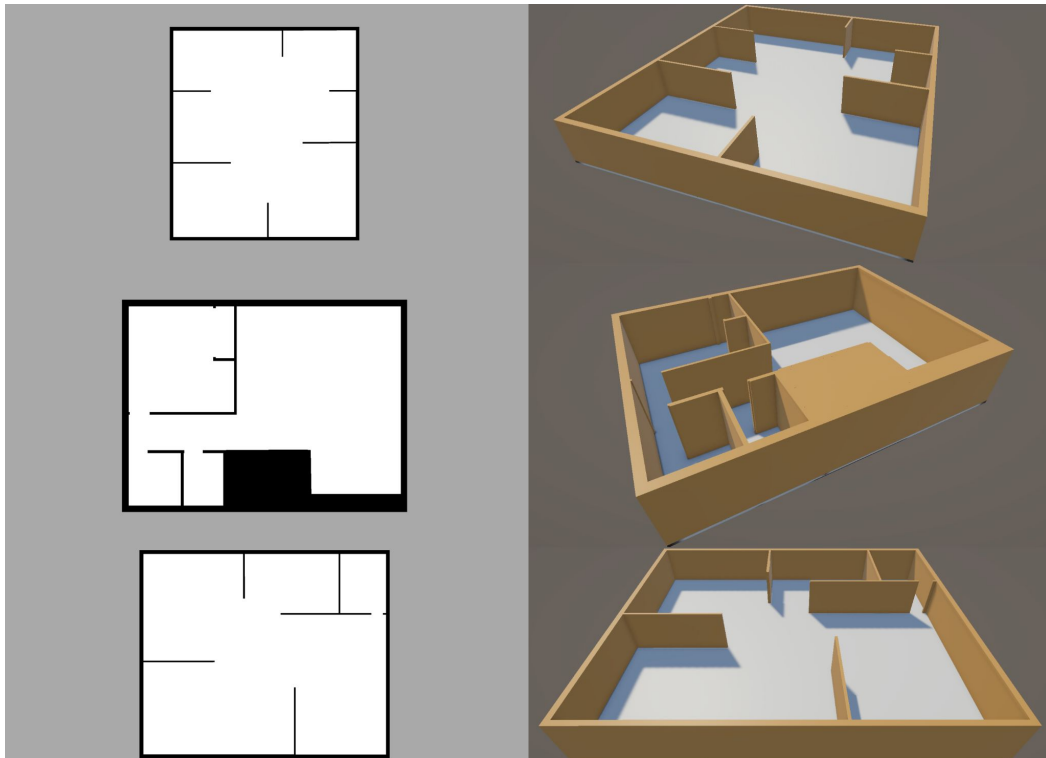
adam.gouguet@imt-nord-europe.fr

Appendix: Contributions

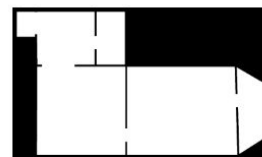




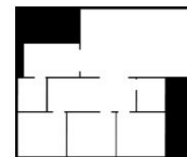
RobotSNAP HouseExpo



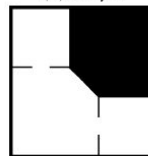
(a) Easy 1



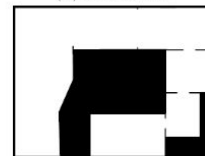
(b) Medium 1



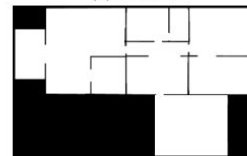
(c) Hard 1



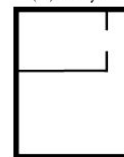
(d) Easy 2



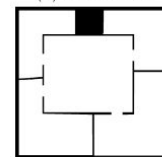
(e) Medium 2



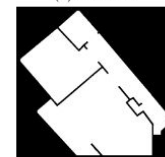
(f) Hard 2



(g) Easy 3



(h) Medium 3

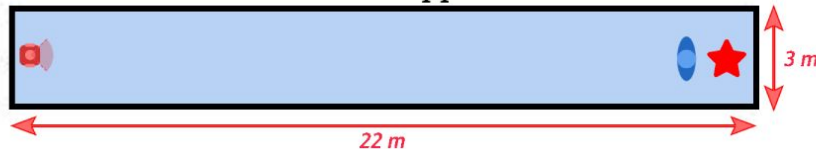


(i) Hard 3

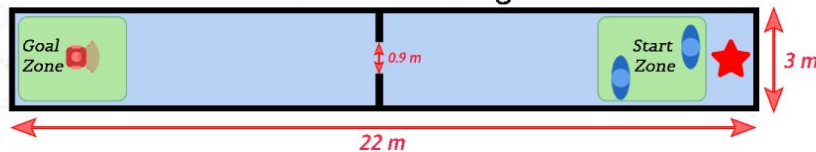
RobotSNAP Base Scenarios



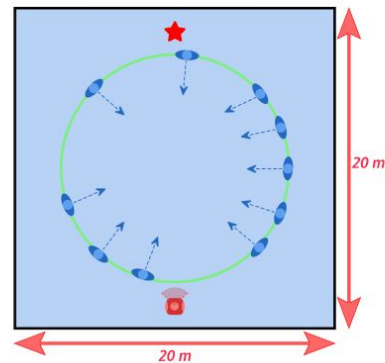
Frontal Approach



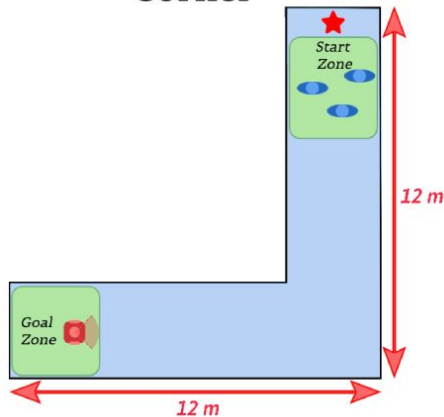
Narrow Passage



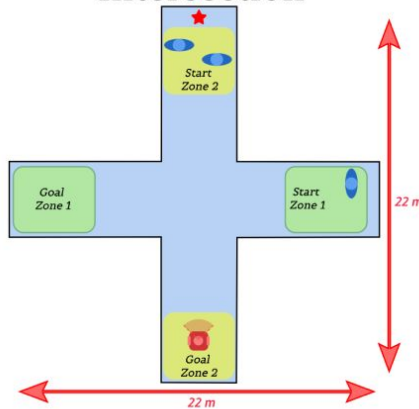
Crowd Circle



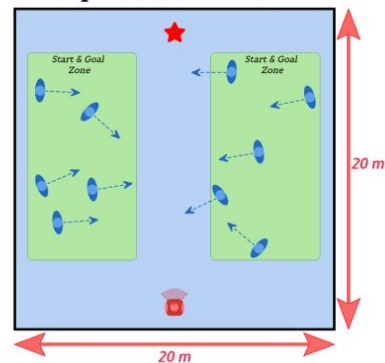
Corner



Intersection

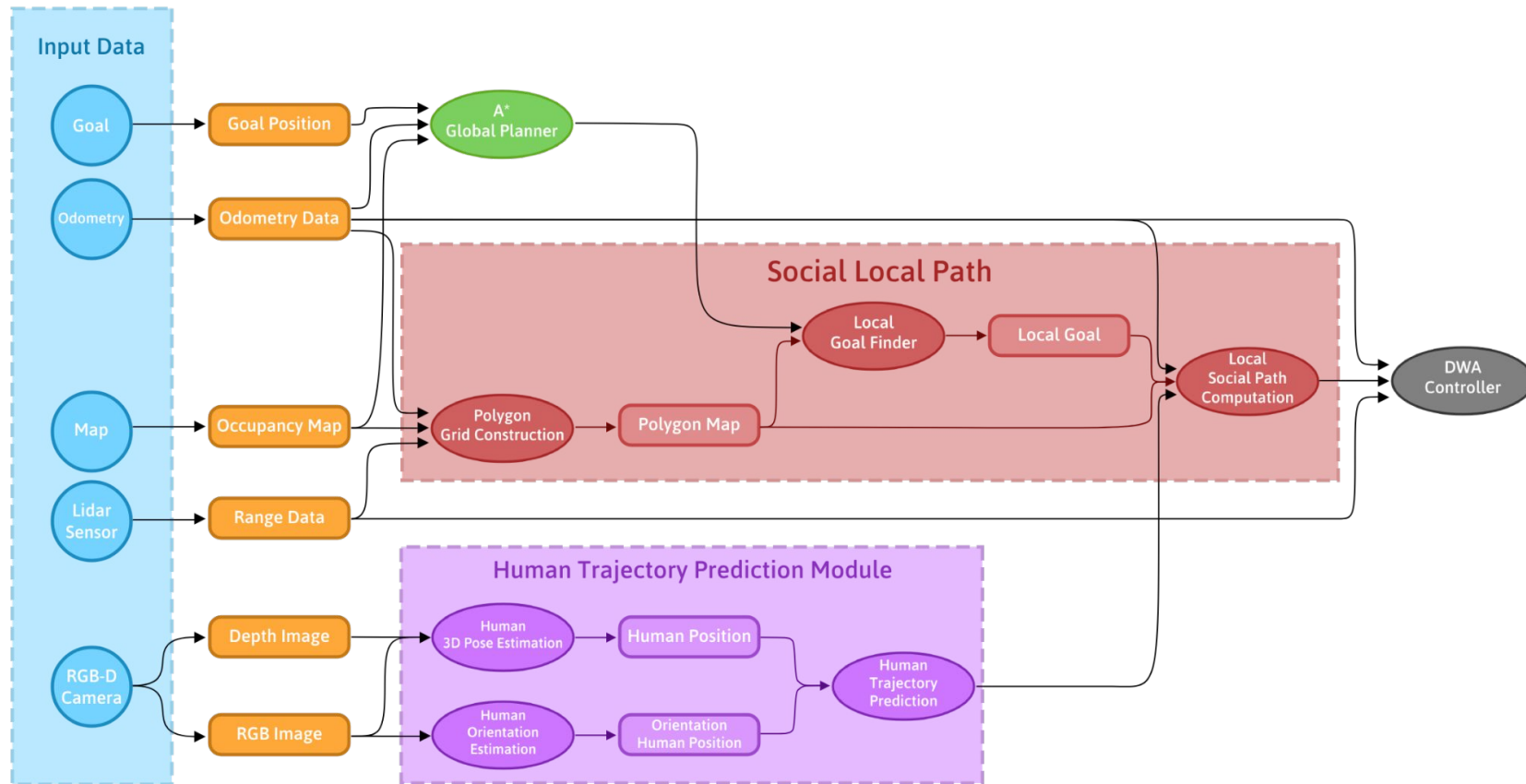


Perpendicular Traffic





HBSN: Full Stack



HBSN: Sub-Scores



- Movement: reducing movement effort.

$$score_{move}(a) = 1 - \frac{d_{m_a} - \min(d_m)}{\max(d_m) - \min(d_m)}$$

$$d_m = \{d_{m_a} \mid \forall a \in A^t\}$$

where d_{m_a} is the distance to action a for the robot.

- Goal: achieving the goal.

$$score_{goal}(a) = 1 - \frac{d_{g_a} - \min(d_g)}{\max(d_g) - \min(d_g)}$$

$$d_g = \{d_{g_a} \mid \forall a \in A^t\}$$

where d_{g_a} is the distance to the goal for action a .

- Near: respecting social space with Human Personal Space (HPS) equals to 1.0 meter.

$$score_{near}(a) = \begin{cases} \frac{d_{n_a}}{HPS} * \exp^{-\alpha(HPS-d_{n_a})^2}, & \text{if } d_{n_a} < HPS \\ 1, & \text{otherwise} \end{cases}$$

$$d_{n_a} = \min_{n \in \{1, \dots, N\}} \min_{t \in [T_0, T_f]} \|p_r^{(a)}(t) - p_n(t)\|_2$$

where d_{n_a} is the distance to the nearest agent and their future positions for action a .

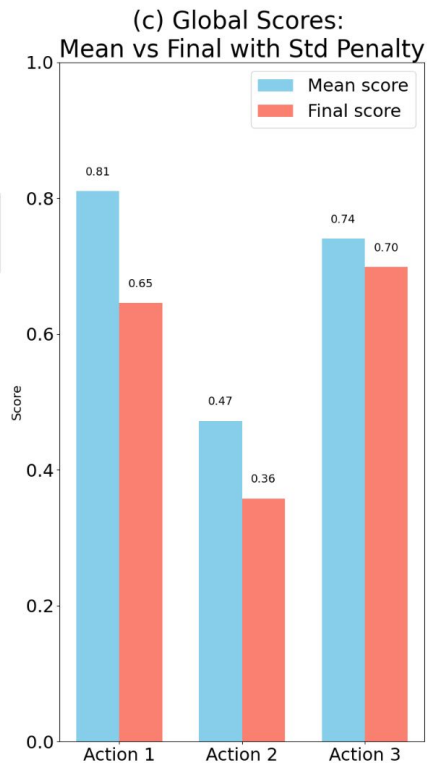
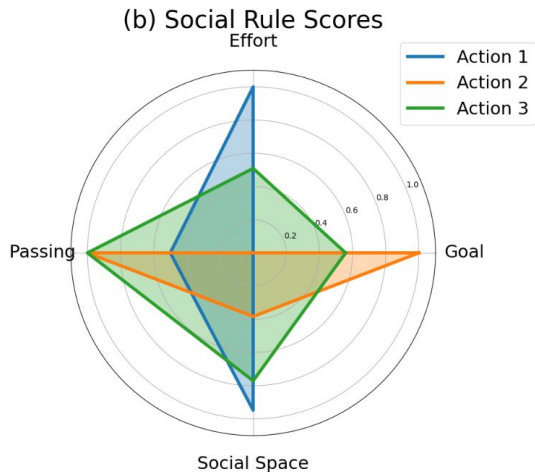
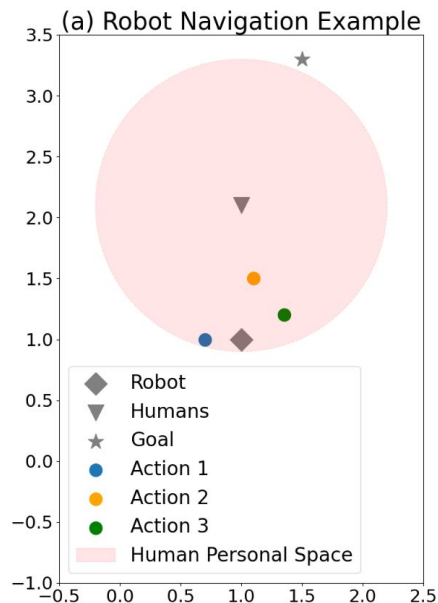
- Side: prioritize moving to the right or left in symmetrical situations.

$$score_{side}(a) = \begin{cases} 0, & \text{if } \vec{RA} \times \vec{RH} = 0 \\ 0.99, & \text{if } \vec{RA} \times \vec{RH} < 0 \\ 1, & \text{if } \vec{RA} \times \vec{RH} > 0 \end{cases}$$

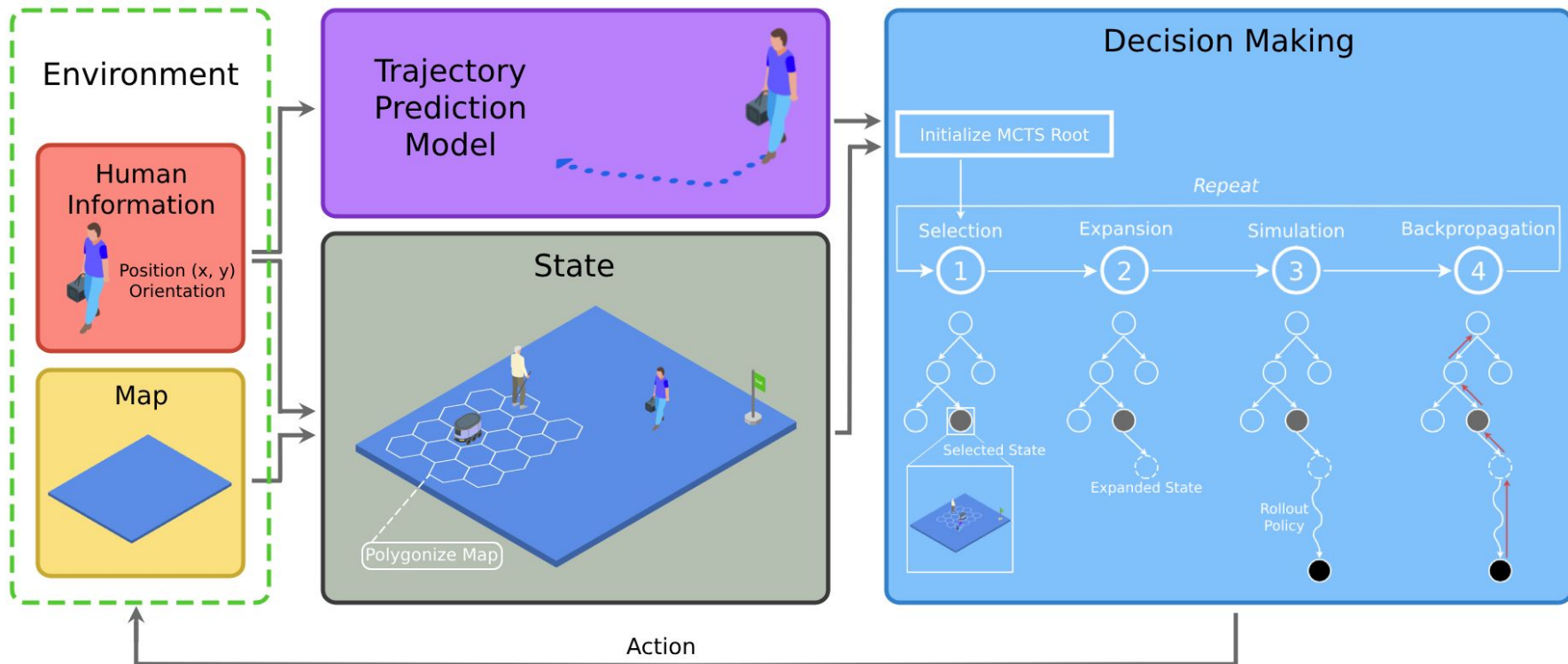
The score depends on the cross product of vectors $\vec{RA}$ (Robot-Action) and $\vec{RH}$ (Robot-Human).

This heuristic scores each social or performance-based rule from 0 (non-compliance) to 1 (full compliance).

HBSN: Action Selection



MBSN: Monte-Carlo Tree Search





MDHC: MDP Details

State:	$[v, w, \theta_{\text{local}}]$	$\{s^1, s^2, \dots, s^N\}$	$[d_x, d_y, v_x, v_y] \forall h \in \text{humans}$
	<i>Robot</i>	<i>Scan</i>	<i>Humans</i>

Reward Function:

$$R_t = R_{\text{goal}}(t) + R_{\text{collision}}(t) + R_{\text{heading}}(t) + R_{\text{human}}(t), \quad (1)$$

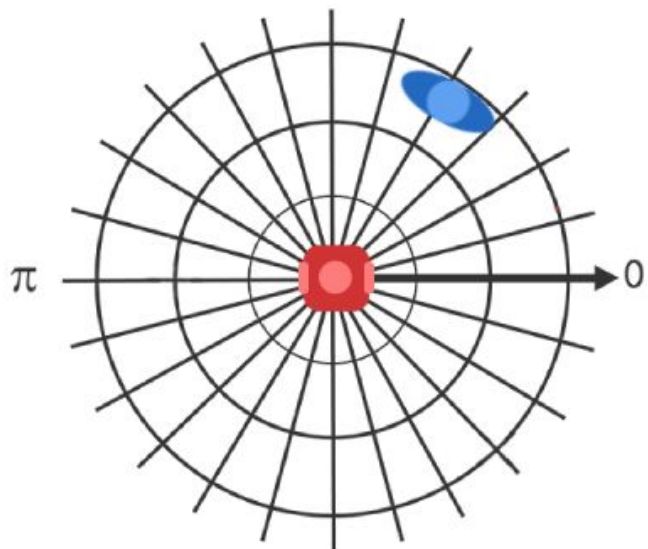
$$R_{\text{goal}}(t) = \begin{cases} r_{\text{goal}}, & \text{if } d_t \leq r_{\text{threshold}}, \\ -r_{\text{goal}}, & \text{if } t \geq T_{\text{max}}, \\ r_{\text{progress}} \cdot (d_{t-1} - d_t), & \text{otherwise,} \end{cases}$$

$$R_{\text{collision}}(t) = \begin{cases} r_{\text{collision}}, & \delta_t \leq r_{\text{robot}}, \\ r_{\text{scan}} \cdot (\alpha r_{\text{robot}} - \delta_t), & \delta_t < \alpha r_{\text{robot}}, \\ 0, & \text{otherwise,} \end{cases}$$

$$R_{\text{heading}}(t) = r_{\text{angle}} \cdot \max(0, \theta_{\text{thresh}} - |\theta_t^* - \theta_t|),$$

$$R_{\text{human}}(t) = - \sum_j r_{\text{human}} \cdot \max(0, d_{\text{safe}} - d_t^j),$$

MDHC: Social Angular Field (SAF)



Bin	Occupancy	Distance	TTC
1	occ_1	d_1	ttc_1
2	occ_2	d_2	ttc_2
3	occ_3	d_3	ttc_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	occ_n	d_n	ttc_n



Proximal Policy Optimization (PPO)

$$L^{PPO}(\theta) = \mathbb{E}_t \left[L_t^{CLIP}(\theta) + c_1 L_t^V(\theta) - c_2 L_t^{ENT}(\theta) \right]$$

Safe Update
Policy

Value Error

Exploration
Bonus

$$L_t^{CLIP}(\theta) = \min \left(\underbrace{r_t(\theta) A_t}_{\text{normal update}}, \underbrace{\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t}_{\text{limited update}} \right)$$

$$r_t(\theta) = \frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

$$L^{VF}(\theta) = (V_{\theta}(s_t) - R_t)^2$$

$$L^{ENT}(\theta) = \mathcal{H}[\pi_{\theta}(\cdot \mid s_t)]$$

CL Levels



Level	Dataset	Min Humans	Max Humans
1	basic	0	0
2	basic	0	1
3	basic	0	3
4	basic	0	5
5	easy	0	0
6	easy	0	1
7	easy	0	2
8	easy	1	2
9	medium	0	0
10	medium	0	2
11	medium	0	4
12	medium	2	5
13	hard	0	5
14	hard	2	5
15	hard	5	9
16	hard	10	15

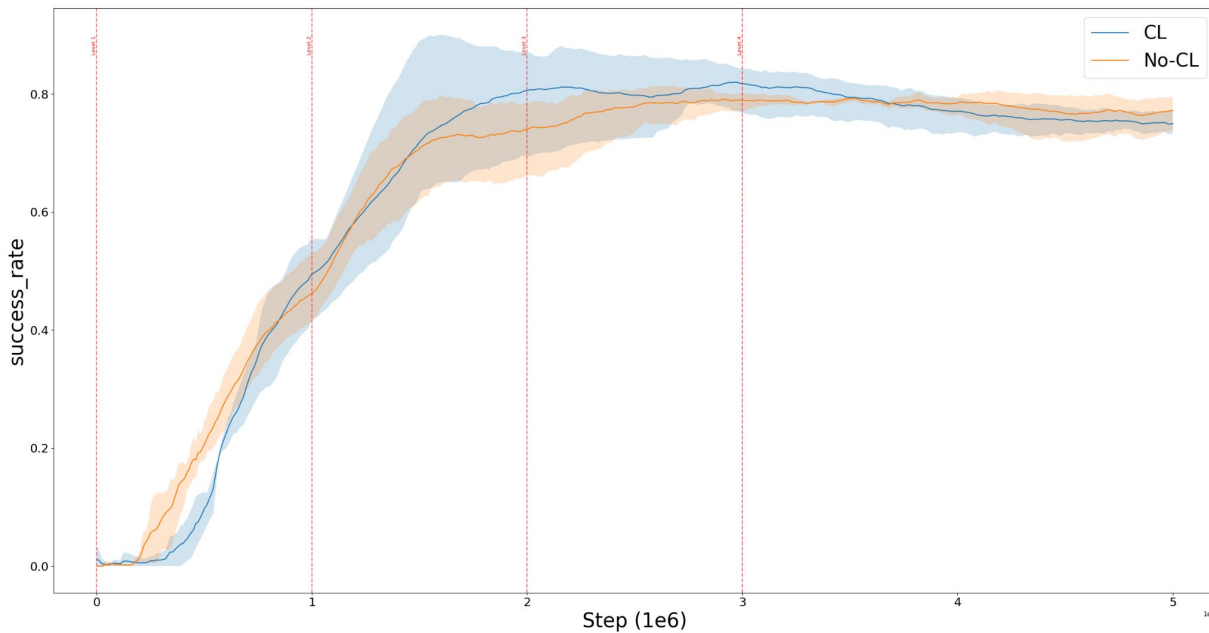
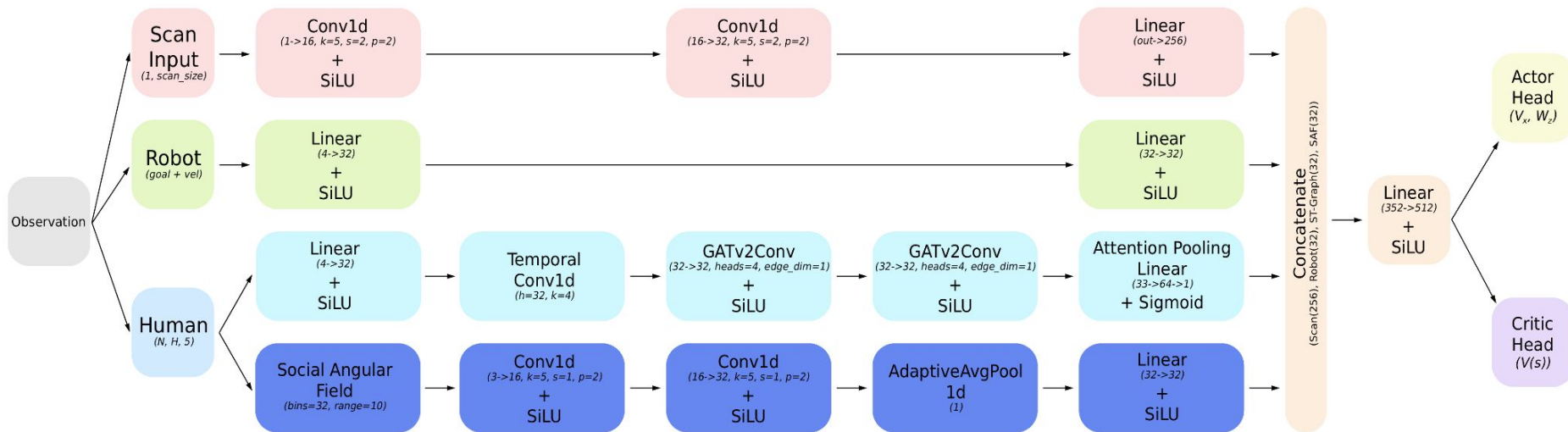
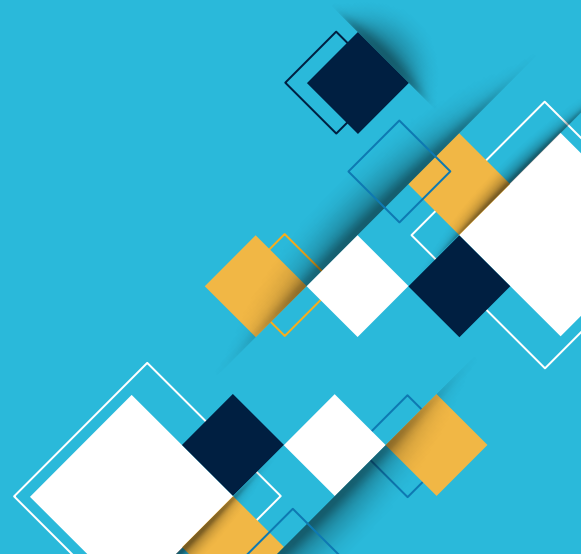


Table B.2 – Curriculum learning schedule configuration.

MDHC: Network

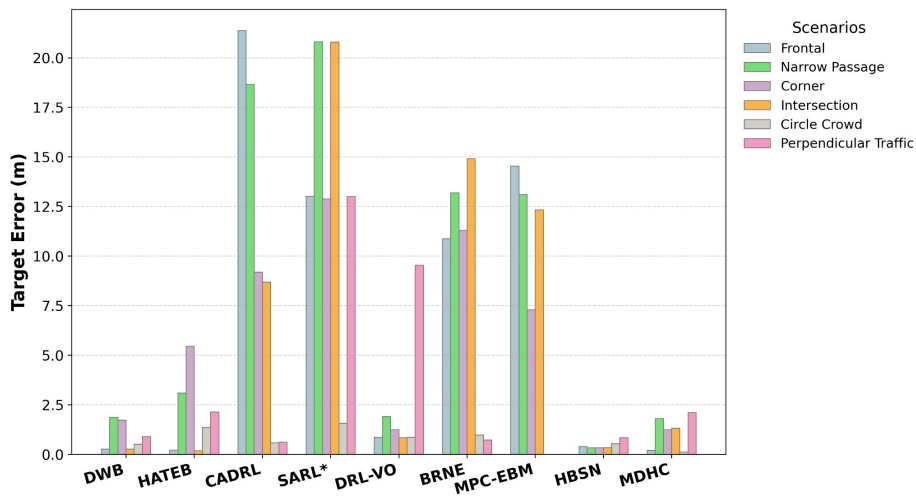


Appendix: Results

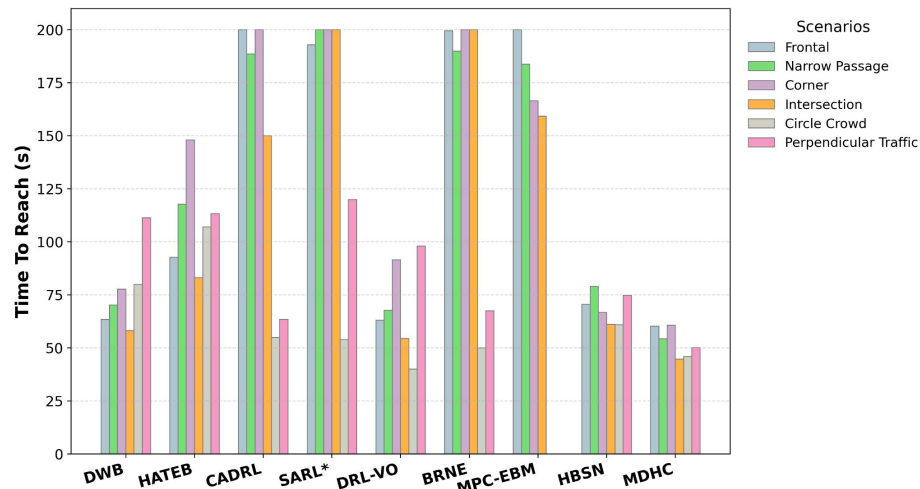


Results Performance

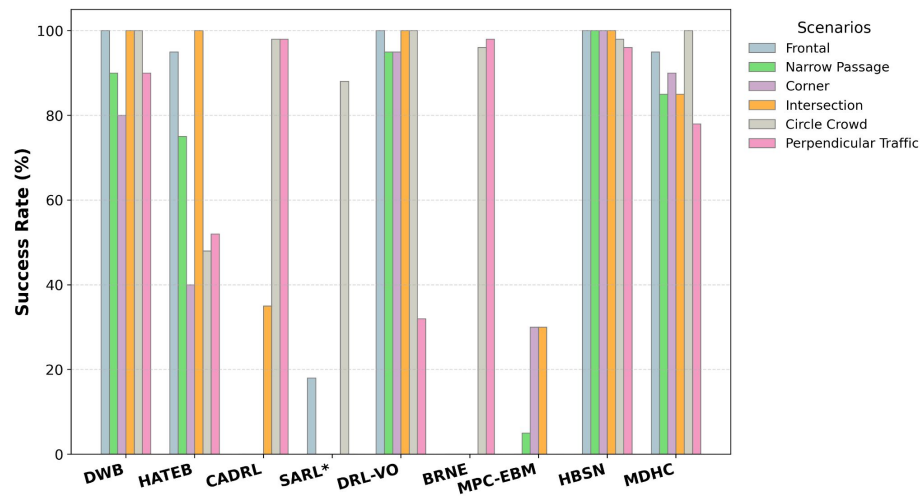
Target Error (m) Across Methods and Scenarios



Time To Reach (s) Across Methods and Scenarios



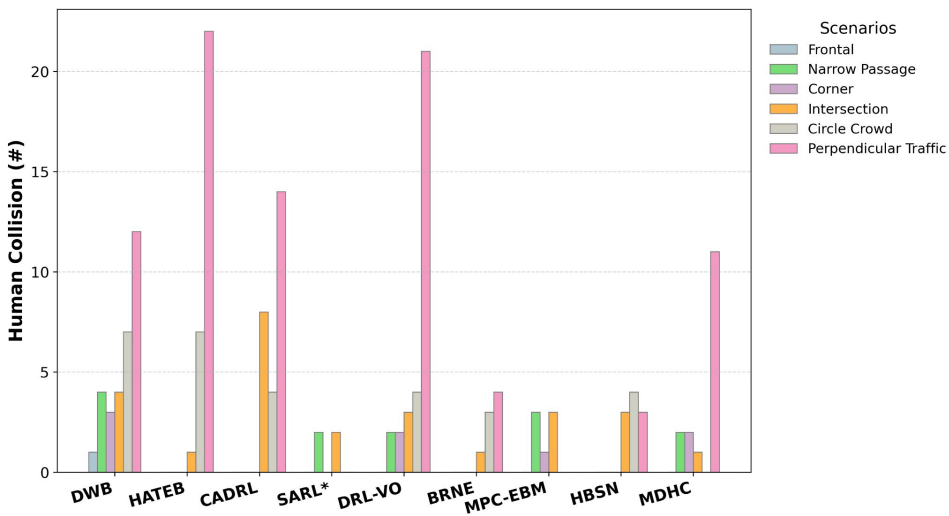
Success Rate (%) Across Methods and Scenarios



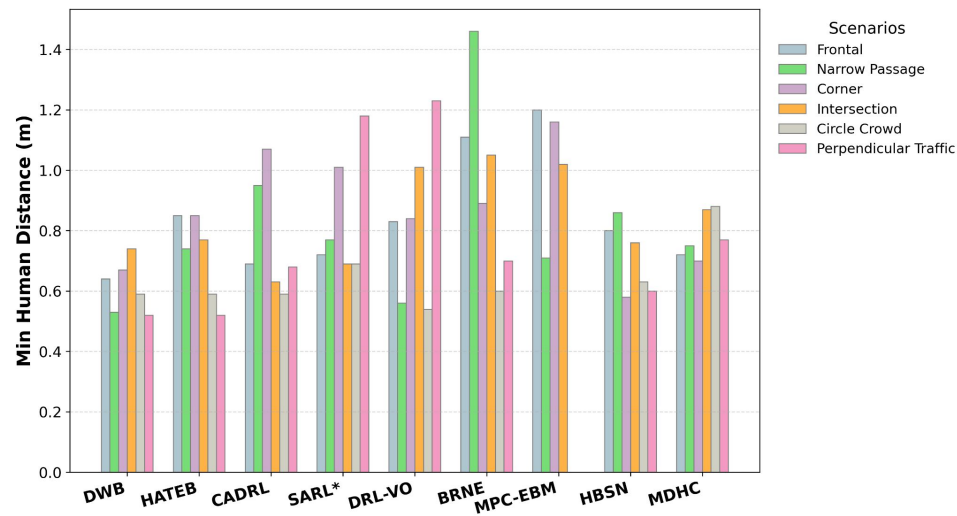
Results Social



Human Collision (#) Across Methods and Scenarios



Min Human Distance (m) Across Methods and Scenarios





MBSN: Results Performance

	Static Humans				Overtaking				Crossing				Door Passing			
	DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*
Goal Achieved	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Path Length (m)	21.75	21.80	22.05	-	21.87	21.55	22.12	28.35	22.73	21.87	21.78	26.38	21.98	22.46	24.86	56.87
Time to Reach Goal (s)	190.0	194.0	191.0	-	169.0	201.0	188.0	345.0	189.0	201.0	178.0	250.0	206.0	240.0	212.0	633.0
Error Distance to Target (m)	0.24	0.19	0.18	14.46	0.24	0.59	0.28	0.67	0.24	0.36	0.17	0.85	0.24	0.17	0.18	0.54
Total Rotation (degree)	22.69	31.02	235.73	218.09	208.09	160.99	235.83	2839.79	294.52	117.12	89.98	3274.92	250.12	150.21	279.16	8432.11
Time Not Moving (s)	0.0	2.0	5.0	-	1.0	9.0	1.0	253.0	3.0	6.0	1.0	111.0	35.0	33.0	13.0	354.0
Collision with Static Obstacle	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

Performance and Social Metrics of Four Different Methods on Proposed Scenarios

MBSN: Results Social



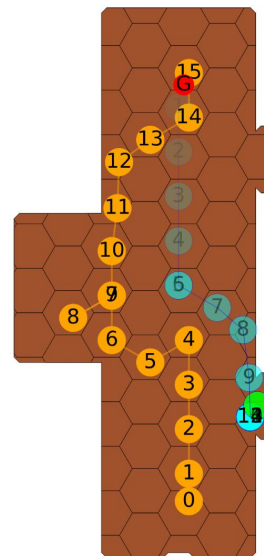
		Static Humans				Overtaking				Crossing				Door Passing			
		DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*	DWA	TEB	HATEB	SARL*
Social Metrics	Violation of Human Personal Space	3	3	0	-	1	1	0	1	2	0	0	0	0	0	0	1
	Violation of Human Personal Space While the Robot is Stationary	0	0	0	-	0	0	0	0	1	0	0	0	1	1	1	1
	Time of Robot Violate Human Personal Space (s)	47.53	47.79	0.0	-	22.71	24.23	0.0	1.16	26.02	0.0	0.0	0.0	28.44	17.87	7.56	61.28
	Minimum Distance Robot-Human (m)	0.92	0.99	1.36	-	0.82	1.06	1.53	1.38	0.97	1.50	3.56	3.75	0.97	1.02	1.19	0.88
	Collision with Human	0	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0

Performance and Social Metrics of Four Different Methods on Proposed Scenarios

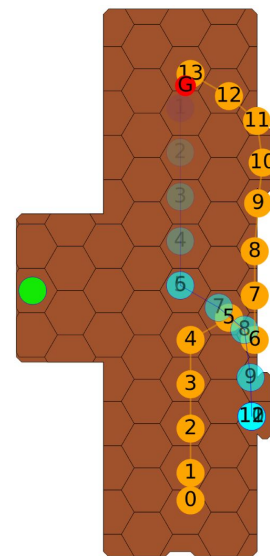
MBSN-MCTS: Experiments



	Action Restriction-Based		Score-Based	
	Local Prediction	Goal Prediction	Local Prediction	Goal Prediction
Heuristics Only				
Using MCTS with Heuristics				



(a) Result when human path predicted by the robot is correct



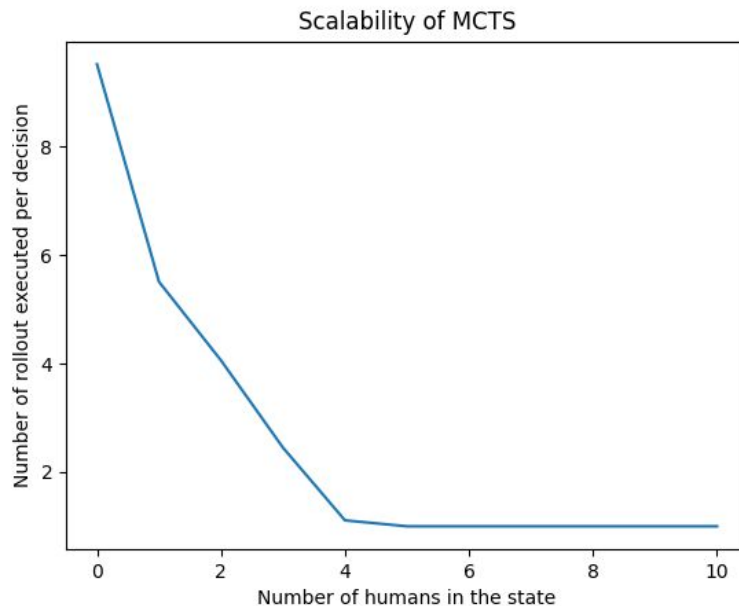
(b) Result when human path predicted by the robot is incorrect



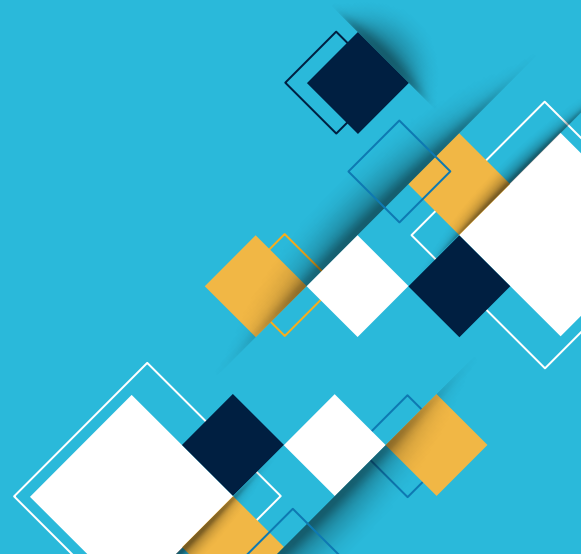
MBSN-MCTS: Scalability

	h=3	h=5	h=10	h=20
Success Rate (%)	100	100	100	86
Average Minimal Distance Human-Robot (m)	2.81	2.36	1.22	0.84
Number of collisions	0	0	1	9
Proportion HPS (% of room)	2.7	4.5	9.0	18.0

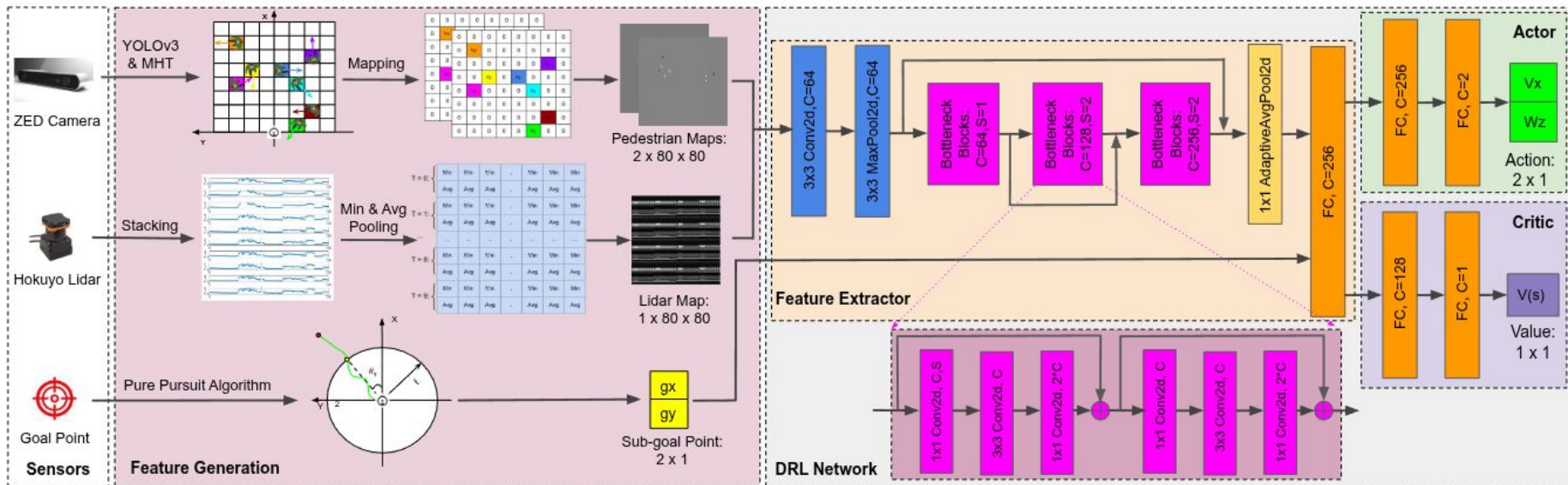
Table 3.1 – The results per scenario, launched 50 times for each, where h is the number of humans present in the scenario.



Appendix: State-of-the-art



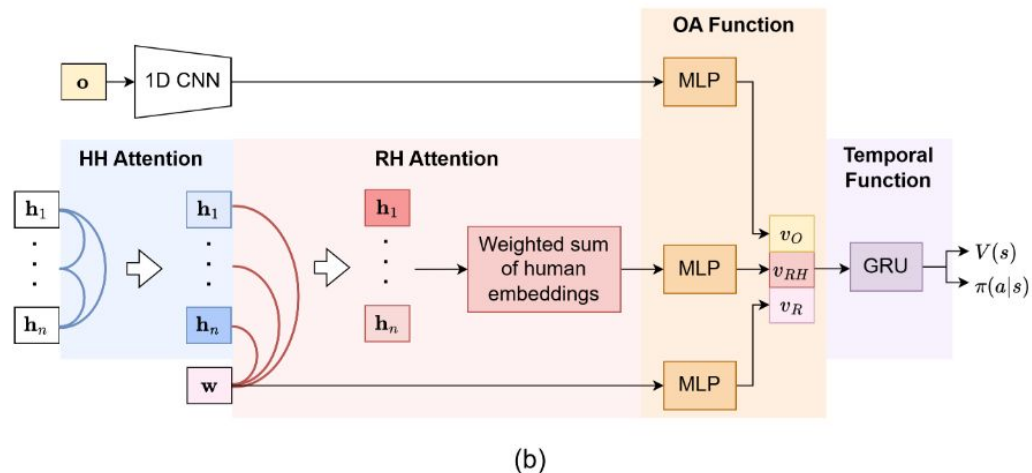
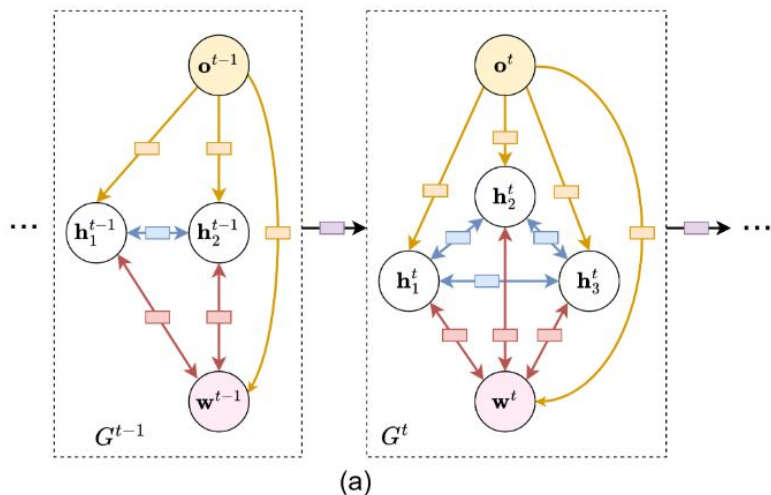
DRL-VO Architecture



CrowdHEIGHT Architecture



$\mathbf{o}$: obstacles state, $\mathbf{h}_i$: state of the i -th human, $\mathbf{w}$: robot state, $\mathbf{a}$: robot action



BRNE

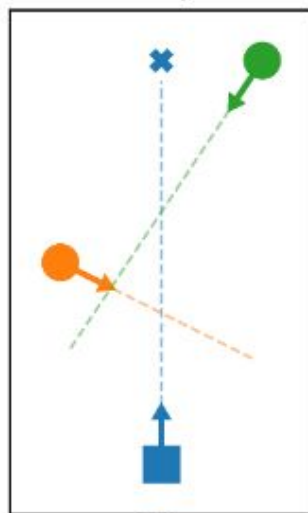


Take measurements



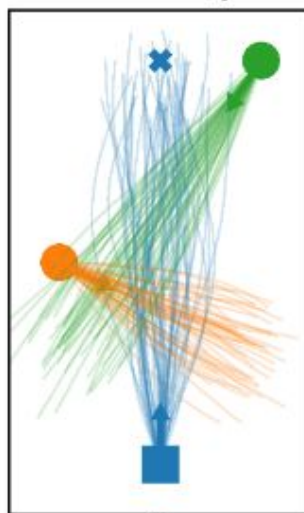
(a)

Generate nominal trajectories



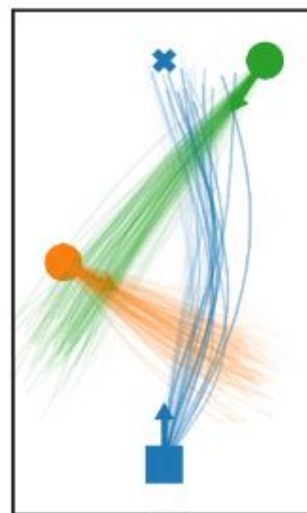
(b)

Generate nominal mixed strategies



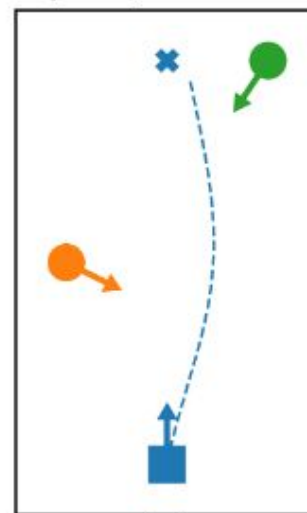
(c)

Weight update



(d)

Generate planned trajectory for the robot



(e)